

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION
C	0000813234	PRODUCTION RELEASED



K84 MLB SCHEMATIC
PROD OK2FAB 11/01/2009

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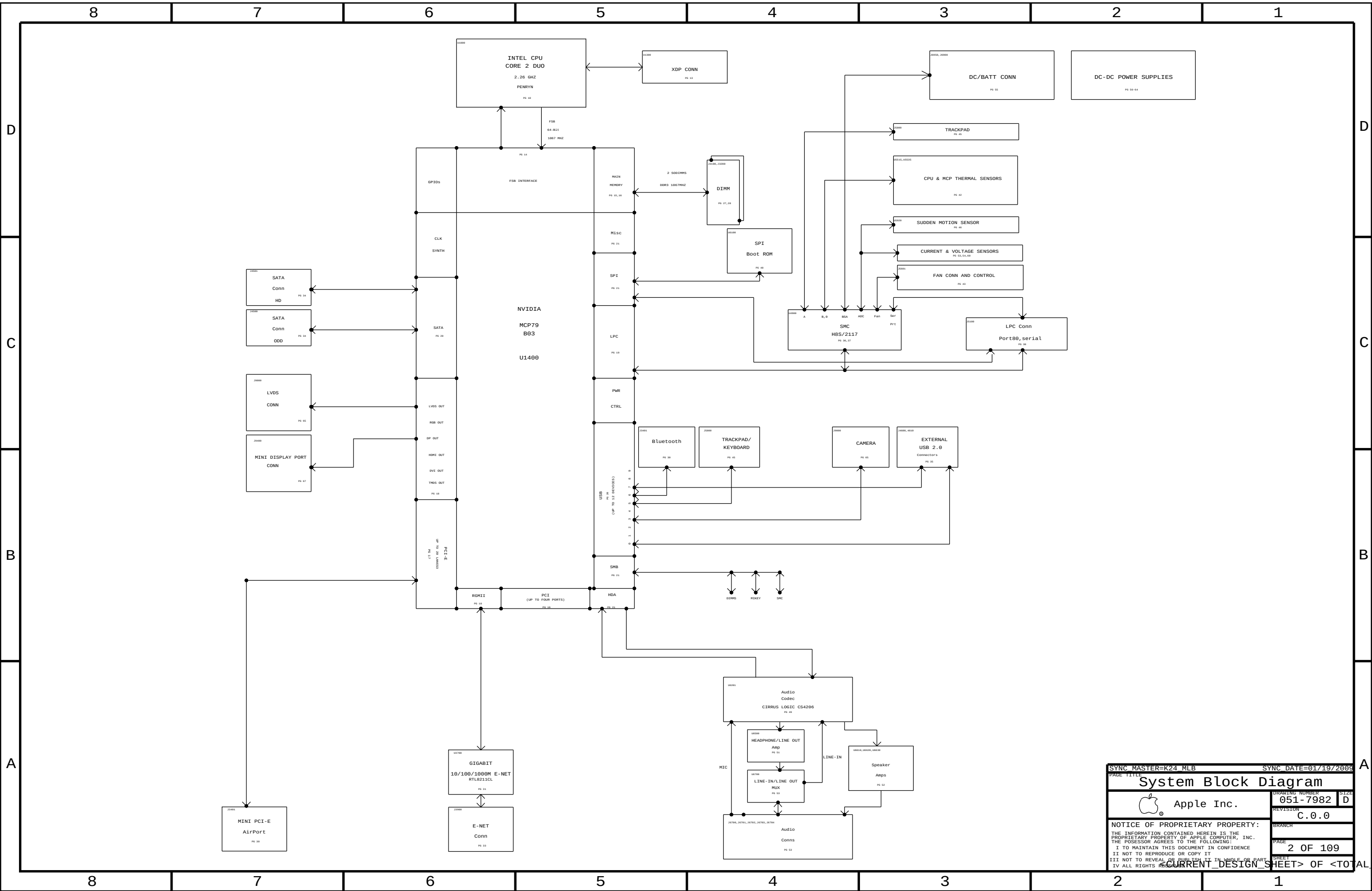
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Schematic / PCB #'s

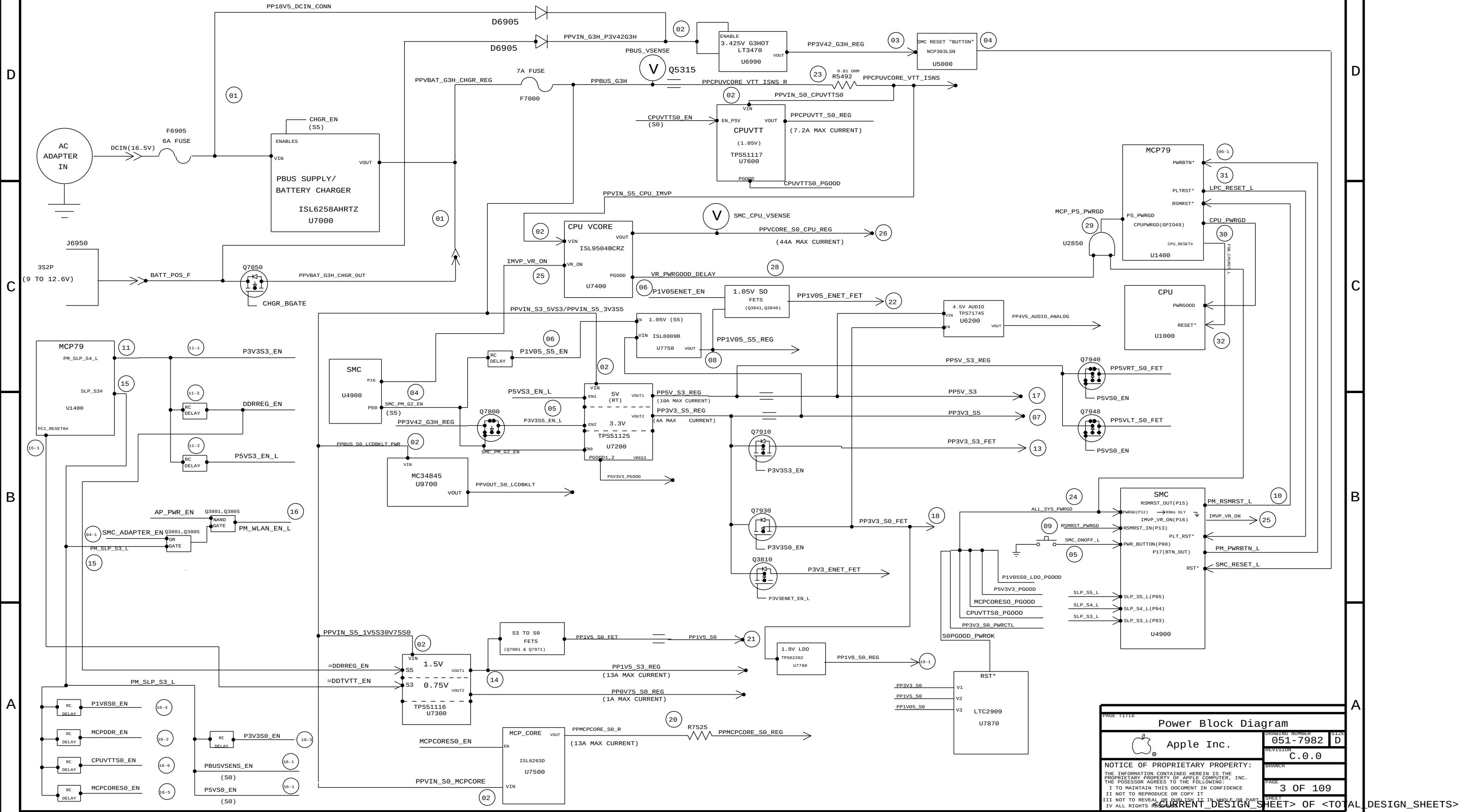
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-7982	1	SCHEM,MLB,K84	SCH	CRITICAL	
820-2567	1	PCBF,MLB,K84	PCB	CRITICAL	

DRAWING TITLE		SCHEM, MLB, K84	
	Apple Inc.		DRAWING NUMBER 051-7982
			REVISION C.0.0
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SYNC MASTER=K24 MLB		SYNC DATE=01/19/2009	
PAGE TITLE			
System Block Diagram			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-7982	D
		REVISION	
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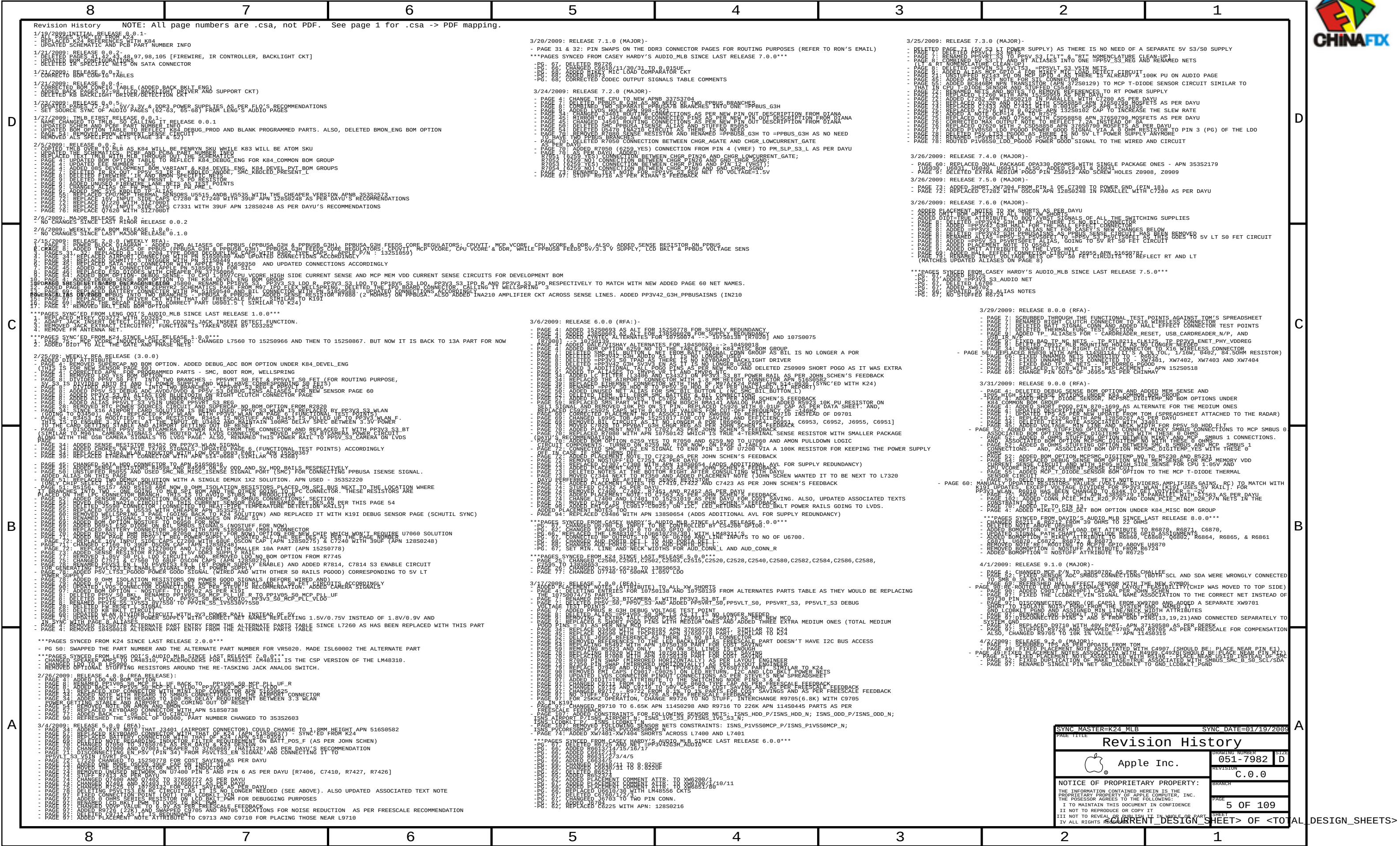
K84 POWER SYSTEM ARCHITECTURE

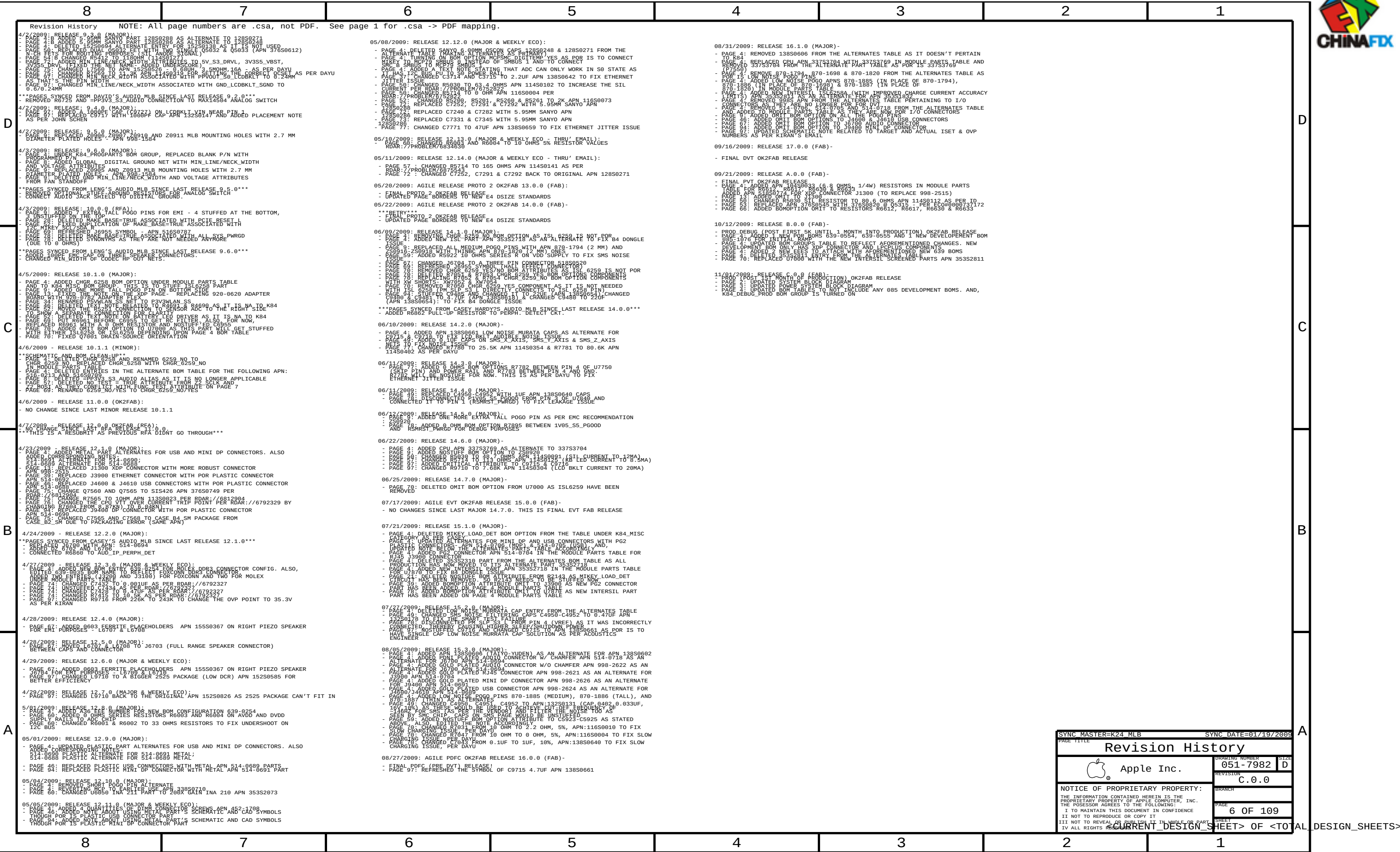


Power Block Diagram			
Apple Inc.	DRAWING NUMBER	051-7982	SIZE D
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SHEET OF TOTAL DESIGN SHEETS			



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	<table><tr><th>BOM NUMBER</th><th>BOM NAME</th><th>BOM OPTIONS</th></tr><tr><td>639-0035</td><td>PCBA,MLB,FOX DDR CONN,K84</td><td>K84_COMMON,CPU_2_0GHZ,FOX_DDR_CONN,EEE_BCG</td></tr><tr><td>639-0254</td><td>PCBA,MLB,MLX DDR CONN,K84</td><td>K84_COMMON,CPU_2_0GHZ,MLX_DDR_CONN,EEE_A36</td></tr><tr><td>085-0748</td><td>K84 MLB DEVELOPMENT BOM</td><td>K84_DEVEL_ENG</td></tr><tr><td>639-0554</td><td>PCBA,MLB,FOX DDR CONN,PVT K84</td><td>K84_COMMON_PVT,CPU_2_0GHZ,FOX_DDR_CONN,EEE_CXR</td></tr><tr><td>639-0555</td><td>PCBA,MLB,MLX DDR CONN,PVT K84</td><td>K84_COMMON_PVT,CPU_2_0GHZ,MLX_DDR_CONN,EEE_CY1</td></tr><tr><td>085-1076</td><td>K84 MLB DEVELOPMENT PVT</td><td>K84_DEVEL_PVT</td></tr></table>									BOM NUMBER	BOM NAME	BOM OPTIONS	639-0035	PCBA,MLB,FOX DDR CONN,K84	K84_COMMON,CPU_2_0GHZ,FOX_DDR_CONN,EEE_BCG	639-0254	PCBA,MLB,MLX DDR CONN,K84	K84_COMMON,CPU_2_0GHZ,MLX_DDR_CONN,EEE_A36	085-0748	K84 MLB DEVELOPMENT BOM	K84_DEVEL_ENG	639-0554	PCBA,MLB,FOX DDR CONN,PVT K84	K84_COMMON_PVT,CPU_2_0GHZ,FOX_DDR_CONN,EEE_CXR	639-0555	PCBA,MLB,MLX DDR CONN,PVT K84	K84_COMMON_PVT,CPU_2_0GHZ,MLX_DDR_CONN,EEE_CY1	085-1076	K84 MLB DEVELOPMENT PVT	K84_DEVEL_PVT																																																																																																		
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Module Parts																																																																																																																																
<table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>33753769</td><td>1</td><td>PWC,SILV1,Z-20,25M,1000,90,30,80A,P7500</td><td>U1000</td><td>CRITICAL</td><td>CPU_2_0GHZ</td></tr><tr><td>33850710</td><td>1</td><td>IC,GNCP,MCP79,35X35MM,BGA1437,B03</td><td>U1400</td><td>CRITICAL</td><td>MCP_B03</td></tr><tr><td>51650706</td><td>1</td><td>CONN,204P,5001MM,SOCKET,DDR3,RAM,BGA</td><td>J3200</td><td>CRITICAL</td><td>FOX_DDR_CONN</td></tr><tr><td>516-0201</td><td>1</td><td>CONN,204P,5001MM,P=0.6MM</td><td>J3100</td><td>CRITICAL</td><td>FOX_DDR_CONN</td></tr><tr><td>51650790</td><td>1</td><td>CONN,204P,5001MM,SOCKET,DDR3,RAM,NON/SC</td><td>J3200</td><td>CRITICAL</td><td>MLX_DDR_CONN</td></tr><tr><td>516-0213</td><td>1</td><td>CONN,204P,5001MM,P=0.6MM,HF</td><td>J3100</td><td>CRITICAL</td><td>MLX_DDR_CONN</td></tr><tr><td>452-1708</td><td>4</td><td>SCR,R1,6X0.35X0.9,D4,HD,3,BLK,M07</td><td>SCREW1,SCREW2,SCREW3,SCREW4</td><td>CRITICAL</td><td></td></tr><tr><td>514-0704</td><td>1</td><td>CONN,RCPT,RJ45,PLASTIC,HF,K83/K84</td><td>J3900</td><td>CRITICAL</td><td></td></tr><tr><td>514-0705</td><td>2</td><td>CONN,RCPT,USB,4P,PLASTIC,HF,K83/K84</td><td>J4000,J4010</td><td>CRITICAL</td><td></td></tr><tr><td>514-0706</td><td>1</td><td>CONN,RCPT,MCP,20P,PLASTIC,HF,K83/K84</td><td>J9400</td><td>CRITICAL</td><td></td></tr><tr><td>514-0718</td><td>1</td><td>CONN,RCPT,S/PDIF,TX,HF,CFR,K83/K84</td><td>J6700</td><td>CRITICAL</td><td></td></tr><tr><td>35352718</td><td>1</td><td>IC,15L88042,4X V MONTN,Z-70/2-80V,TQFN8</td><td>U7870</td><td>CRITICAL</td><td></td></tr><tr><td>870-1885</td><td>4</td><td>POSD P1N,RED,NOISE-IMPROVED,K84</td><td>Z50900,Z50901,Z50902,Z50903</td><td>CRITICAL</td><td></td></tr><tr><td>870-1885</td><td>3</td><td>POSD P1N,RED,NOISE-IMPROVED,K84</td><td>Z50900,Z50909,Z50911</td><td>CRITICAL</td><td></td></tr><tr><td>870-1886</td><td>5</td><td>POSD P1N,TALL,NOISE-IMPROVED,K84</td><td>Z50904,Z50905,Z50906,Z50907,Z50910</td><td>CRITICAL</td><td></td></tr><tr><td>870-1886</td><td>5</td><td>POSD P1N,TALL,NOISE-IMPROVED,K84</td><td>Z50912,Z50913,Z50914,Z50915,Z50919</td><td>CRITICAL</td><td></td></tr><tr><td>870-1887</td><td>3</td><td>POSD P1N,THIN,NOISE-IMPROVED,K84</td><td>Z50917,Z50918,Z50916</td><td>CRITICAL</td><td></td></tr><tr><td>10450033</td><td>4</td><td>RES,HF,1/4W,6.800M,5%,0805,SMD</td><td>R6612,R6617,R6630,R6633</td><td>CRITICAL</td><td></td></tr><tr><td>51850774</td><td>1</td><td>CONN,RCPT,60P,P=0.4,STK HT 1.0</td><td>J1300</td><td>CRITICAL</td><td>XDP_CONN</td></tr></table>								PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	33753769	1	PWC,SILV1,Z-20,25M,1000,90,30,80A,P7500	U1000	CRITICAL	CPU_2_0GHZ	33850710	1	IC,GNCP,MCP79,35X35MM,BGA1437,B03	U1400	CRITICAL	MCP_B03	51650706	1	CONN,204P,5001MM,SOCKET,DDR3,RAM,BGA	J3200	CRITICAL	FOX_DDR_CONN	516-0201	1	CONN,204P,5001MM,P=0.6MM	J3100	CRITICAL	FOX_DDR_CONN	51650790	1	CONN,204P,5001MM,SOCKET,DDR3,RAM,NON/SC	J3200	CRITICAL	MLX_DDR_CONN	516-0213	1	CONN,204P,5001MM,P=0.6MM,HF	J3100	CRITICAL	MLX_DDR_CONN	452-1708	4	SCR,R1,6X0.35X0.9,D4,HD,3,BLK,M07	SCREW1,SCREW2,SCREW3,SCREW4	CRITICAL		514-0704	1	CONN,RCPT,RJ45,PLASTIC,HF,K83/K84	J3900	CRITICAL		514-0705	2	CONN,RCPT,USB,4P,PLASTIC,HF,K83/K84	J4000,J4010	CRITICAL		514-0706	1	CONN,RCPT,MCP,20P,PLASTIC,HF,K83/K84	J9400	CRITICAL		514-0718	1	CONN,RCPT,S/PDIF,TX,HF,CFR,K83/K84	J6700	CRITICAL		35352718	1	IC,15L88042,4X V MONTN,Z-70/2-80V,TQFN8	U7870	CRITICAL		870-1885	4	POSD P1N,RED,NOISE-IMPROVED,K84	Z50900,Z50901,Z50902,Z50903	CRITICAL		870-1885	3	POSD P1N,RED,NOISE-IMPROVED,K84	Z50900,Z50909,Z50911	CRITICAL		870-1886	5	POSD P1N,TALL,NOISE-IMPROVED,K84	Z50904,Z50905,Z50906,Z50907,Z50910	CRITICAL		870-1886	5	POSD P1N,TALL,NOISE-IMPROVED,K84	Z50912,Z50913,Z50914,Z50915,Z50919	CRITICAL		870-1887	3	POSD P1N,THIN,NOISE-IMPROVED,K84	Z50917,Z50918,Z50916	CRITICAL		10450033	4	RES,HF,1/4W,6.800M,5%,0805,SMD	R6612,R6617,R6630,R6633	CRITICAL		51850774	1	CONN,RCPT,60P,P=0.4,STK HT 1.0	J1300	CRITICAL	XDP_CONN	
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33850710	1	IC,GNCP,MCP79,35X35MM,BGA1437,B03	U1400	CRITICAL	MCP_B03																																																																																																																											
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516-0201	1	CONN,204P,5001MM,P=0.6MM	J3100	CRITICAL	FOX_DDR_CONN																																																																																																																											
51650790	1	CONN,204P,5001MM,SOCKET,DDR3,RAM,NON/SC	J3200	CRITICAL	MLX_DDR_CONN																																																																																																																											
516-0213	1	CONN,204P,5001MM,P=0.6MM,HF	J3100	CRITICAL	MLX_DDR_CONN																																																																																																																											
452-1708	4	SCR,R1,6X0.35X0.9,D4,HD,3,BLK,M07	SCREW1,SCREW2,SCREW3,SCREW4	CRITICAL																																																																																																																												
514-0704	1	CONN,RCPT,RJ45,PLASTIC,HF,K83/K84	J3900	CRITICAL																																																																																																																												
514-0705	2	CONN,RCPT,USB,4P,PLASTIC,HF,K83/K84	J4000,J4010	CRITICAL																																																																																																																												
514-0706	1	CONN,RCPT,MCP,20P,PLASTIC,HF,K83/K84	J9400	CRITICAL																																																																																																																												
514-0718	1	CONN,RCPT,S/PDIF,TX,HF,CFR,K83/K84	J6700	CRITICAL																																																																																																																												
35352718	1	IC,15L88042,4X V MONTN,Z-70/2-80V,TQFN8	U7870	CRITICAL																																																																																																																												
870-1885	4	POSD P1N,RED,NOISE-IMPROVED,K84	Z50900,Z50901,Z50902,Z50903	CRITICAL																																																																																																																												
870-1885	3	POSD P1N,RED,NOISE-IMPROVED,K84	Z50900,Z50909,Z50911	CRITICAL																																																																																																																												
870-1886	5	POSD P1N,TALL,NOISE-IMPROVED,K84	Z50904,Z50905,Z50906,Z50907,Z50910	CRITICAL																																																																																																																												
870-1886	5	POSD P1N,TALL,NOISE-IMPROVED,K84	Z50912,Z50913,Z50914,Z50915,Z50919	CRITICAL																																																																																																																												
870-1887	3	POSD P1N,THIN,NOISE-IMPROVED,K84	Z50917,Z50918,Z50916	CRITICAL																																																																																																																												
10450033	4	RES,HF,1/4W,6.800M,5%,0805,SMD	R6612,R6617,R6630,R6633	CRITICAL																																																																																																																												
51850774	1	CONN,RCPT,60P,P=0.4,STK HT 1.0	J1300	CRITICAL	XDP_CONN																																																																																																																											
C	35352718 IS NEW INTERSIL PART FOR FIXING B4 DONGLE ISSUE 514-0704 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0692 PART FOR RJ45 CONNECTOR 514-0705 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0689 PART FOR USB CONNECTORS 514-0706 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0691 PART FOR MINI DP CONNECTOR 514-0718 IS CLOUD GREY 4/LB3 PLASTIC W/PDNI PLATING VERSION OF 514-0694 PART FOR AUDIO CONNECTOR								C																																																																																																																							
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	Programmable Parts																																																																																																																															
	<table><tr><td>33850563</td><td>1</td><td>IC,SMC,H5B/2117,9X9MM,TLP,HF</td><td>U4900</td><td>CRITICAL</td><td>SMC_BLANK</td></tr><tr><td>34152485</td><td>1</td><td>IC,SMC,K84</td><td>U4900</td><td>CRITICAL</td><td>SMC_PROD</td></tr><tr><td>33550610</td><td>1</td><td>IC,FLASH,SPT,32MBIT,3.3V,80Mhz,8-SOP</td><td>U6100</td><td>CRITICAL</td><td>BOOTROM_BLANK</td></tr><tr><td>34152487</td><td>1</td><td>IC,PROGR,EF1 BOOTROM_UNLOCK,K84</td><td>U6100</td><td>CRITICAL</td><td>BOOTROM_PROD</td></tr><tr><td>33752983</td><td>1</td><td>IC,PSOC+ M/ USB,56 P1N,MLF,CY8C24704</td><td>U5701</td><td>CRITICAL</td><td>WELLSPRING_BLANK</td></tr><tr><td>34152491</td><td>1</td><td>IC,WELLSPRING_CONTROLLER,K84</td><td>U5701</td><td>CRITICAL</td><td>WELLSPRING_PROD</td></tr></table>									33850563	1	IC,SMC,H5B/2117,9X9MM,TLP,HF	U4900	CRITICAL	SMC_BLANK	34152485	1	IC,SMC,K84	U4900	CRITICAL	SMC_PROD	33550610	1	IC,FLASH,SPT,32MBIT,3.3V,80Mhz,8-SOP	U6100	CRITICAL	BOOTROM_BLANK	34152487	1	IC,PROGR,EF1 BOOTROM_UNLOCK,K84	U6100	CRITICAL	BOOTROM_PROD	33752983	1	IC,PSOC+ M/ USB,56 P1N,MLF,CY8C24704	U5701	CRITICAL	WELLSPRING_BLANK	34152491	1	IC,WELLSPRING_CONTROLLER,K84	U5701	CRITICAL	WELLSPRING_PROD																																																																																			
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Functional Test Points

FAN CONNECTORS FUNC_TEST

PP5V	TRUE	PP5VRT_S0	7 8
PP5V	TRUE	FAN_RT_PWM	43
PP5V	TRUE	FAN_RT_TACH	43
(NEED TO ADD 1 GND TP)			

MIC FUNC_TEST

PP5V	TRUE	BI_MIC_LO	53 54
PP5V	TRUE	BI_MIC_HI	53 54
PP5V	TRUE	BI_MIC_SHIELD	53 54

SPEAKER FUNC_TEST

PP5V	TRUE	SPKRAMP_L_N_OUT	52 53
PP5V	TRUE	SPKRAMP_L_P_OUT	52 53
PP5V	TRUE	SPKRAMP_R_N_OUT	52 53
PP5V	TRUE	SPKRAMP_R_P_OUT	52 53
PP5V	TRUE	SPKRAMP_SUB_N_OUT	52 53
PP5V	TRUE	SPKRAMP_SUB_P_OUT	52 53

LVDS FUNC_TEST

PP5V	TRUE	PP3V3_LCDVDD_SW_F	7 65 (NEED 2 TP)
PP5V	TRUE	PP3V3_S0_LCD_F	65
PP5V	TRUE	PPVOUT_S0_LCDBKLT	7 47 65 68 (NEED 2 TP)
PP5V	TRUE	LVDS_IG_DDC_CLK	18 65
PP5V	TRUE	LVDS_IG_DDC_DATA	18 65
PP5V	TRUE	LVDS_IG_A_DATA_N<0>	18 65 72
PP5V	TRUE	LVDS_IG_A_DATA_P<0>	18 65 72
PP5V	TRUE	LVDS_IG_A_DATA_N<1>	18 65 72
PP5V	TRUE	LVDS_IG_A_DATA_P<1>	18 65 72
PP5V	TRUE	LVDS_IG_A_DATA_N<2>	18 65 72
PP5V	TRUE	LVDS_IG_A_DATA_P<2>	18 65 72
PP5V	TRUE	LVDS_IG_A_CLK_F_N	65 72
PP5V	TRUE	LVDS_IG_A_CLK_F_P	65 72
PP5V	TRUE	LED_RETURN_1	65 68
PP5V	TRUE	LED_RETURN_2	65 68
PP5V	TRUE	LED_RETURN_3	65 68
PP5V	TRUE	LED_RETURN_4	65 68
PP5V	TRUE	LED_RETURN_5	65 68
PP5V	TRUE	LED_RETURN_6	65 68
PP5V	TRUE	PP5V_S3_CAMERA_F	7 65
PP5V	TRUE	USB_CAMERA_CONN_P	65 73
PP5V	TRUE	USB_CAMERA_CONN_N	65 73
(NEED TO ADD 5 GND TP)			

SATA ODD CONN FUNC_TEST

PP5V	TRUE	PP5V_SW_ODD	(NEED 2 TP) 7 34 47
PP5V	TRUE	SMC_ODD_DETECT	34 36
PP5V	TRUE	SATA_ODD_D2R_C_P	34 72
PP5V	TRUE	SATA_ODD_D2R_C_N	34 72
PP5V	TRUE	SATA_ODD_R2D_P	34 72
PP5V	TRUE	SATA_ODD_R2D_N	34 72
(NEED TO ADD 2 GND TP)			

SATA HDD/SIL FUNC_TEST

PP5V	TRUE	PP5V_S0_HDD_FLT	(NEED 2 TP) 7 34
PP5V	TRUE	SATA_HDD_R2D_P	34 72
PP5V	TRUE	SATA_HDD_R2D_N	34 72
PP5V	TRUE	SATA_HDD_D2R_C_P	34 72
PP5V	TRUE	SATA_HDD_D2R_C_N	34 72
PP5V	TRUE	SYS_LED_ANODE_R	34
(NEED TO ADD 3 GND TP)			

BATT POWER CONN FUNC_TEST

PP5V	TRUE	SMBUS_SMC_BSA_SCL	39 75
PP5V	TRUE	SMBUS_SMC_BSA_SDA	39 75
PP5V	TRUE	SYS_DETECT_L	55
PP5V	TRUE	BATT_POS_F	55 56
(NEED TO ADD 2 GND TP) (NEED 2 TP)			

HALL EFFECT CONNECTOR FUNC_TEST

PP5V	TRUE	PP3V42_G3H	7 8
PP5V	TRUE	SMC_LID_R	55

X16 WIRELESS CONN FUNC_TEST

PP5V	TRUE	PP3V3_S3_BT_F	39
PP5V	TRUE	CONN_PCIE_MINI_D2R_P	39 72
PP5V	TRUE	CONN_PCIE_MINI_D2R_N	39 72
PP5V	TRUE	CONN_PCIE_MINI_R2D_P	39 72
PP5V	TRUE	CONN_PCIE_MINI_R2D_N	39 72
PP5V	TRUE	PCIE_CLK100M_MINI_CONN_P	39 72
PP5V	TRUE	PCIE_CLK100M_MINI_CONN_N	39 72
PP5V	TRUE	PP3V3_WLAN	7 39 (NEED 2 TP)
PP5V	TRUE	PCIE_WAKE_L	17 39
PP5V	TRUE	CONN_USB2_BT_P	39 73
PP5V	TRUE	CONN_USB2_BT_N	39 73
PP5V	TRUE	MINI_CLKREQ_Q_L	39
PP5V	TRUE	MINI_RESET_CONN_L	39
(NEED TO ADD 2 GND TP)			

IPD_FLEX_CONN FUNC_TEST

PP5V	TRUE	PP3V3_S3_LDO	7 45
PP5V	TRUE	PP18V5_S3	7 45
PP5V	TRUE	Z2_CS_L	44 45
PP5V	TRUE	Z2_DEBUG3	44 45
PP5V	TRUE	Z2_MOSI	44 45
PP5V	TRUE	Z2_MISO	44 45
PP5V	TRUE	Z2_SCLK	44 45
PP5V	TRUE	Z2_BOOTST_EN	45
PP5V	TRUE	Z2_HOST_INTN	44 45
PP5V	TRUE	Z2_CLKIN	44 45
PP5V	TRUE	Z2_KEY_ACT_L	44 45
PP5V	TRUE	Z2_RESET	44 45
PP5V	TRUE	PSOC_MISO	44 45
PP5V	TRUE	PSOC_MOSI	44 45
PP5V	TRUE	PSOC_SCLK	44 45
PP5V	TRUE	SMBUS_SMC_A_S3_SDA	39 75
PP5V	TRUE	SMBUS_SMC_A_S3_SCL	39 75
PP5V	TRUE	PSOC_F_CS_L	44 45
PP5V	TRUE	PICKB_L	44 45
(NEED TO ADD 2 GND TP)			

KEYBOARD CONN FUNC_TEST

PP5V	TRUE	PP3V3_S3	7 8
PP5V	TRUE	PP3V42_G3H	7 8
PP5V	TRUE	WS_KBD1	44
PP5V	TRUE	WS_KBD2	44
PP5V	TRUE	WS_KBD3	44
PP5V	TRUE	WS_KBD4	44
PP5V	TRUE	WS_KBD5	44
PP5V	TRUE	WS_KBD6	44
PP5V	TRUE	WS_KBD7	44
PP5V	TRUE	WS_KBD8	44
PP5V	TRUE	WS_KBD9	44
PP5V	TRUE	WS_KBD10	44
PP5V	TRUE	WS_KBD11	44
PP5V	TRUE	WS_KBD12	44
PP5V	TRUE	WS_KBD13	44
PP5V	TRUE	WS_KBD14	44
PP5V	TRUE	WS_KBD15_CAP	44
PP5V	TRUE	WS_KBD16_NUM	44
PP5V	TRUE	WS_KBD17	44
PP5V	TRUE	WS_KBD18	44
PP5V	TRUE	WS_KBD19	44
PP5V	TRUE	WS_KBD20	44
PP5V	TRUE	WS_KBD21	44
PP5V	TRUE	WS_KBD22	44
PP5V	TRUE	WS_KBD23	44
PP5V	TRUE	WS_KBD_ONOFF_L	44
PP5V	TRUE	WS_LEFT_SHIFT_KBD	44
PP5V	TRUE	WS_LEFT_OPTION_KBD	44
PP5V	TRUE	WS_CONTROL_KBD	44
(NEED TO ADD 1 GND TP)			

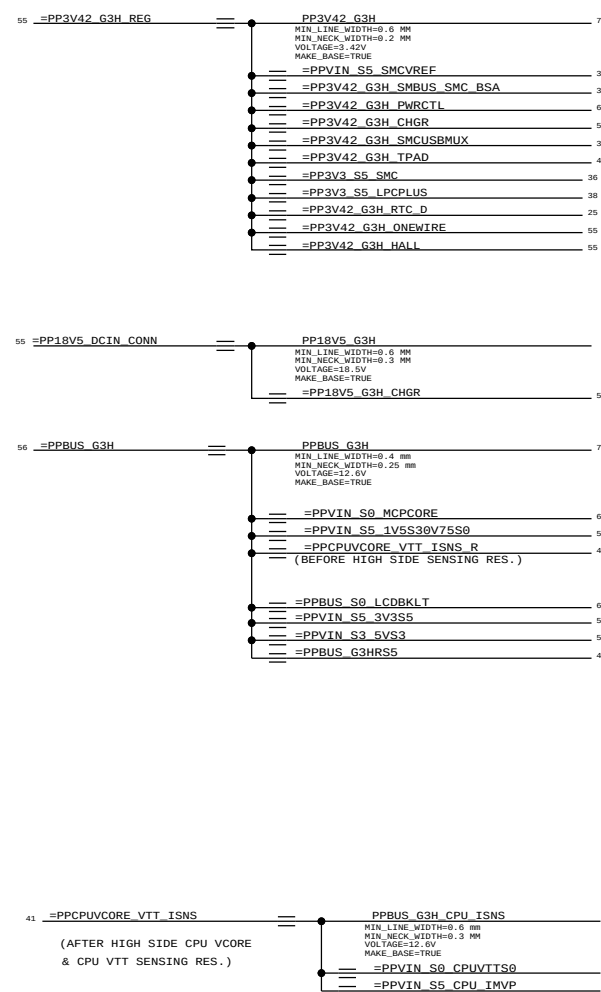
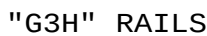
POWER NETS FUNC_TEST

PP5V	TRUE	PPVCORE_S0_CPU	8
PP5V	TRUE	PPVCORE_S0_MCP	8
PP5V	TRUE	PP0V75_S0	8
PP5V	TRUE	PP1V05_S0	8
PP5V	TRUE	PP1V5_S0	8
PP5V	TRUE	PP1V8_S0	8
PP5V	TRUE	PP5VLT_S0	8
PP5V	TRUE	PP5VRT_S0	7 8
PP5V	TRUE	PP3V3_S0	8
PP5V	TRUE	PP1V5_S3	8
PP5V	TRUE	PP3V3_S3	7 8
PP5V	TRUE	PP5V_S3	8
PP5V	TRUE	PP1V1R1V05_S5	8
PP5V	TRUE	PP3V3_S5	8
PP5V	TRUE	PP3V42_G3H	7 8
PP5V	TRUE	PPBUS_G3H	8
PP5V	TRUE	PP3V3_ENET_PHY	8
PP5V	TRUE	PP1V2R1V05_ENET	8
PP5V	TRUE	PP3V3_G3_RTC	21 22 25
PP5V	TRUE	PP3V3_WLAN	7 39
PP5V	TRUE	PP5V_SW_ODD	7 34 47
PP5V	TRUE	PP5V_S0_HDD_FLT	7 34
PP5V	TRUE	PP3V3_S5_AVREF_SMC	36 37
PP5V	TRUE	PP18V5_S3	7 45
PP5V	TRUE	PP3V3_S3_LDO	7 45
PP5V	TRUE	PP3V3_LCDVDD_SW_F	7 65
PP5V	TRUE	PPVOUT_S0_LCDBKLT	7 47 65 68
PP5V	TRUE	PP4V5_AUDIO_ANALOG	49
PP5V	TRUE	SMC_PM_G2_EN	36 57 63
PP5V	TRUE	PM_SLP_S4_L	21 36 37 63
PP5V	TRUE	PM_SLP_S3_L	21 32 36 63 67
PP5V	TRUE	PP5V_S3_CAMERA_F	7 65
(NEED TO ADD 1 GND TP)			

DC POWER CONN FUNC_TEST

PP5V	TRUE	PP18V5_DCIN_FUSE (NEED 2 TP)	55
PP5V	TRUE	ADAPTER_SENSE	55
(NEED TO ADD 2 GND TP)			

SYNC MASTER=K24 MLB		SYNC DATE=02/04/2009	
PAGE TITLE			
FUNC TEST			
		DRAWING NUMBER	SIZE
		051-7982	D
		REVISION	
		C.0.0	
		BRANCH	
		PAGE	7 OF 109
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"ENET" RAILS

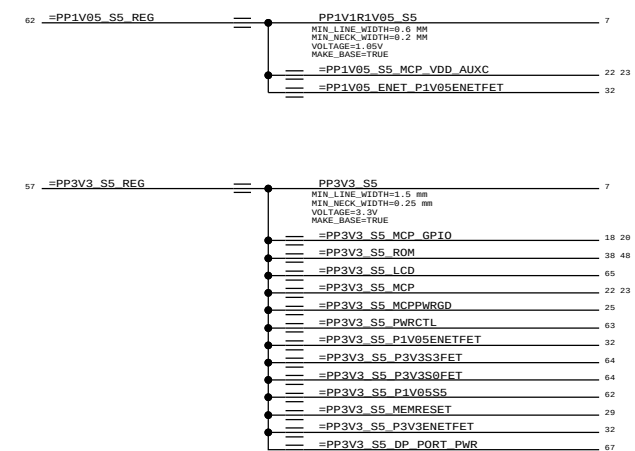
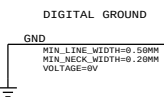


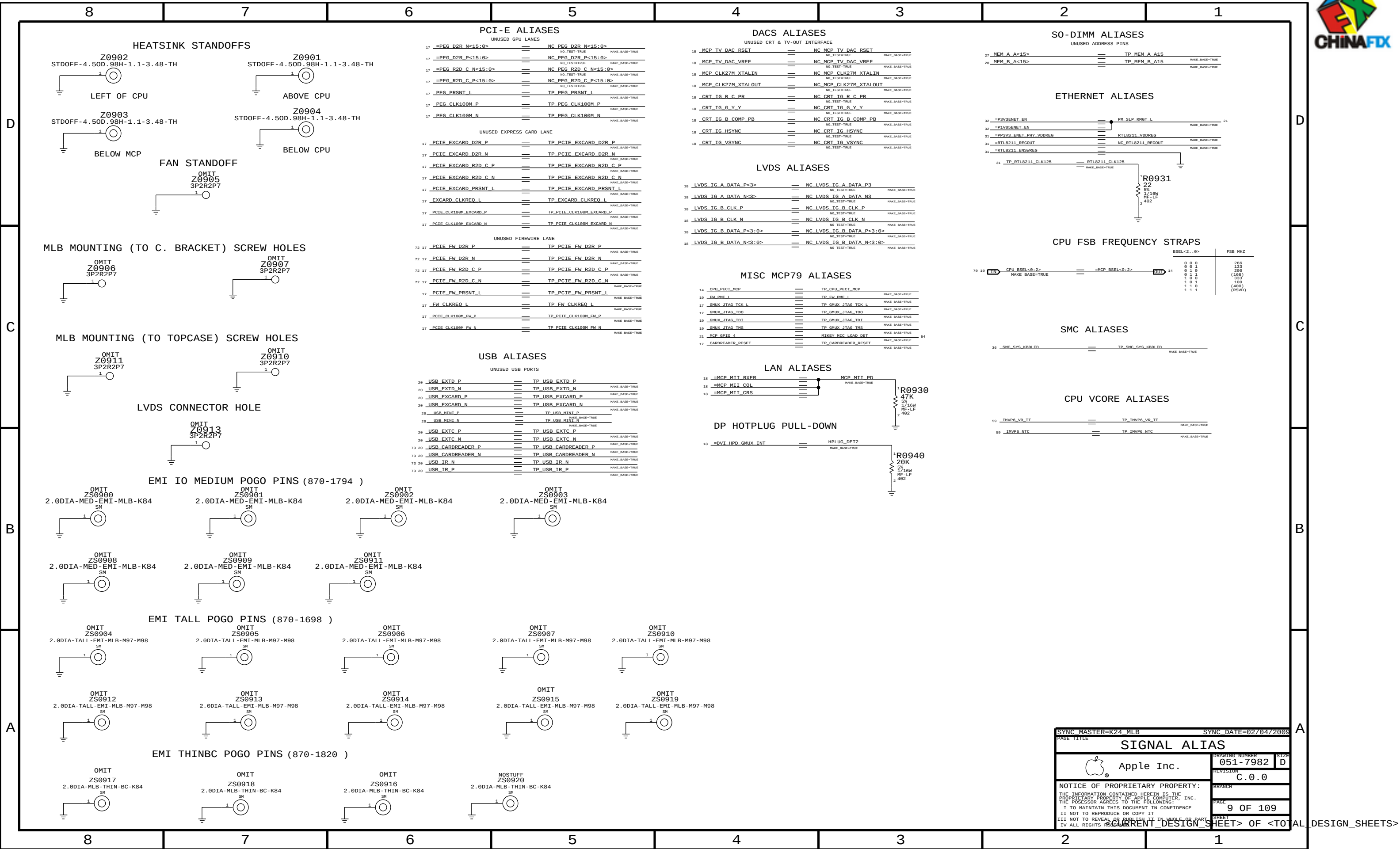
Diagram illustrating the power distribution network for the PP1V05_S0_MCP_PEX and PP1V05_S0_MCP_SATA components. The diagram shows connections to various power rails and components, including:

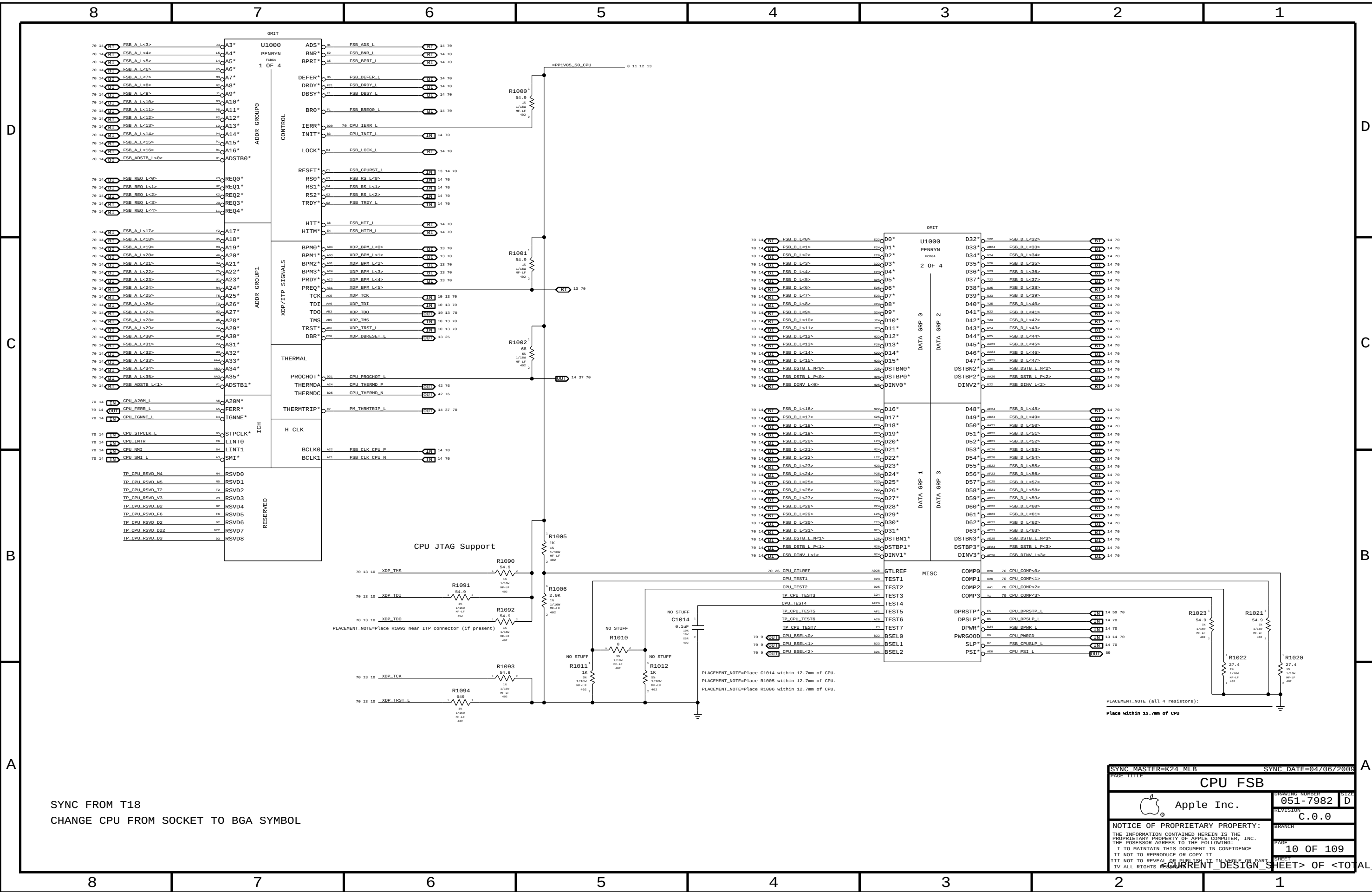
- PP1V05_S0_MCP_PEX_AVDD** (200 mA (AB1)) connected to **PP1V05_S0_MCP_PEX_AVDD0** (17) and **PP1V05_S0_MCP_PEX_AVDD1** (17).
- PP1V05_S0_MCP_PEX_DVDD** (200 mA (AB1)) connected to **PP1V05_S0_MCP_PEX_BVDD0** (17) and **PP1V05_S0_MCP_PEX_BVDD1** (17).
- PP1V05_S0_MCP_SATA_AVDD** (127 mA (AB1)) connected to **PP1V05_S0_MCP_SATA_AVDD0** (20) and **PP1V05_S0_MCP_SATA_AVDD1** (20).
- PP1V05_S0_MCP_SATA_DVDD** (127 mA (AB1)) connected to **PP1V05_S0_MCP_SATA_DVDD0** (20) and **PP1V05_S0_MCP_SATA_DVDD1** (20).

The diagram also shows a ground connection at the bottom.



SYNC MASTER=K24 MLB		SYNC DATE=02/04/2009	
PAGE TITLE			
Power Aliases			
		DRAWING NUMBER 051-7982	
Apple Inc.		SIZE D	
		REVISION C.0.0	
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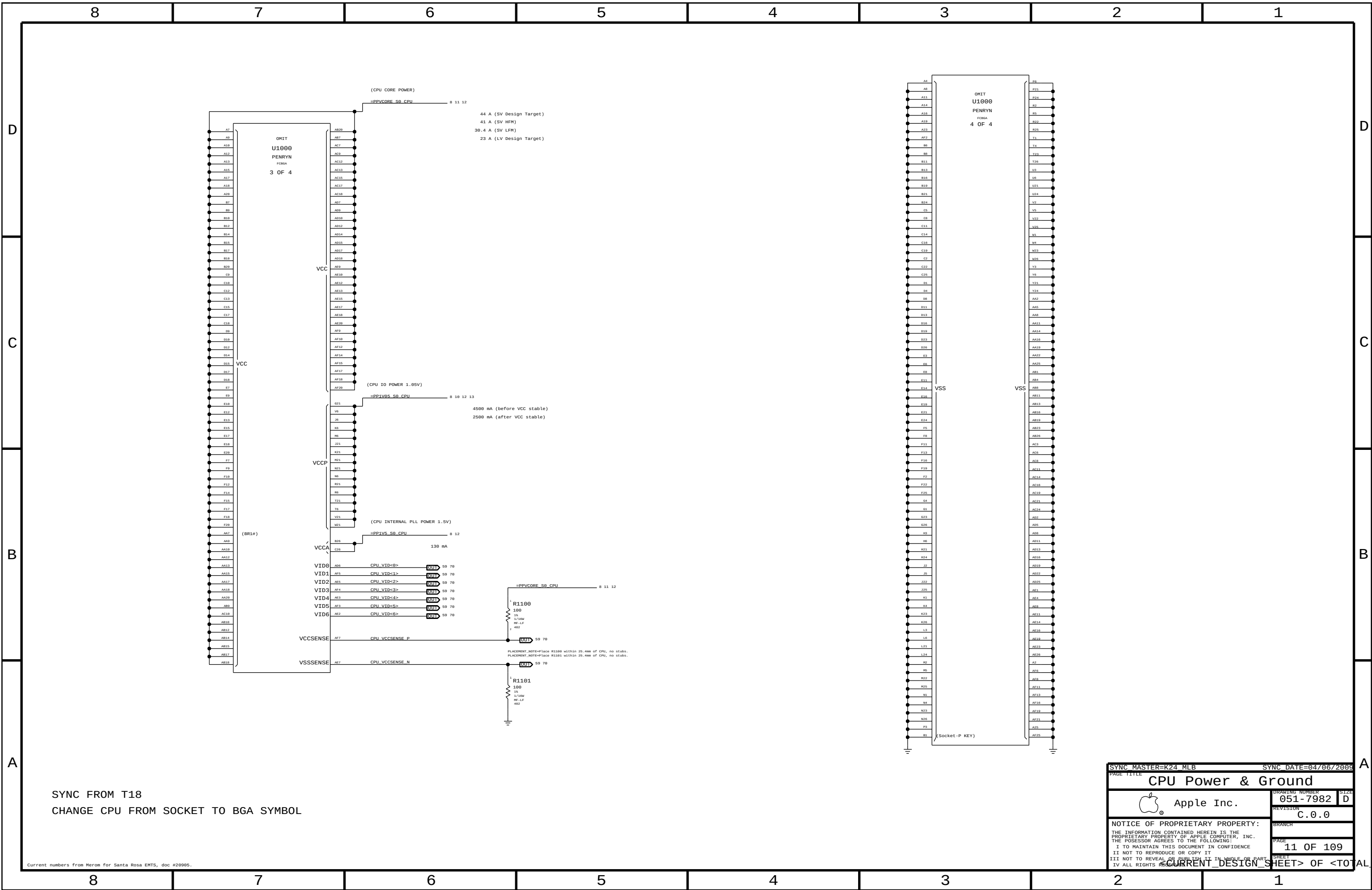




SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
CPU FSB			
		DRAWING NUMBER	SIZE
Apple Inc.		051-7982	D
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REGULATORY DESIGN SHEET OF <TO			

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SYNC FROM T18
CHANGE CPU FROM SOCKET TO BGA SYMBOL

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
CPU Power & Ground			
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		REVISION	
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		11 OF 109	
		SHEET> OF <TOTAL DESIGN SHEETS>	



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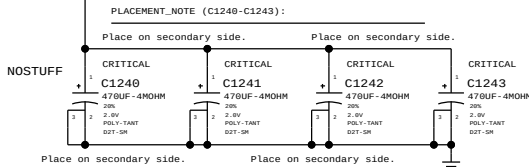
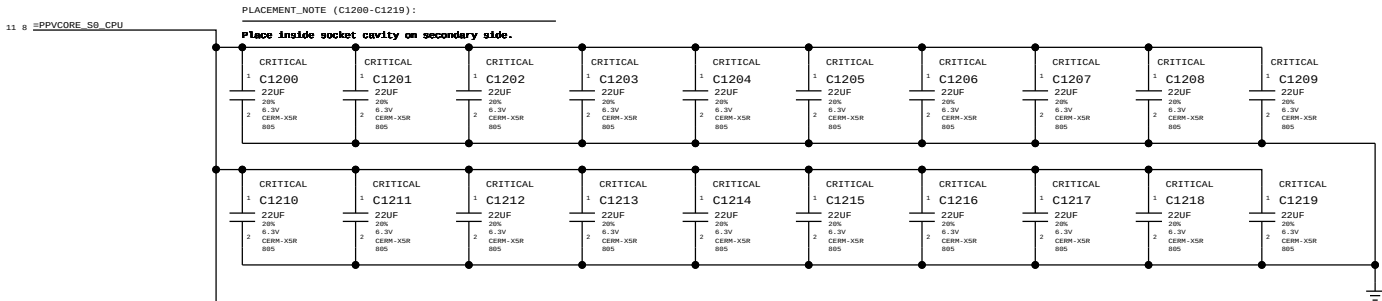
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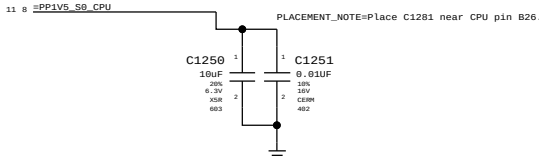
CPU VCore HF and Bulk Decoupling

4X 330UF, 20X 22UF 0805



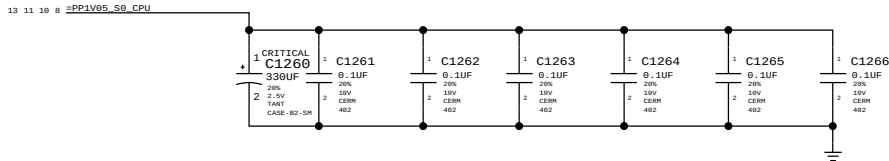
VCCA (CPU AVdd) DECOUPLING

1x 10uF, 1x 0.01uF



VCCP (CPU I/O) DECOUPLING

1x 330uF, 6x 0.1uF 0402



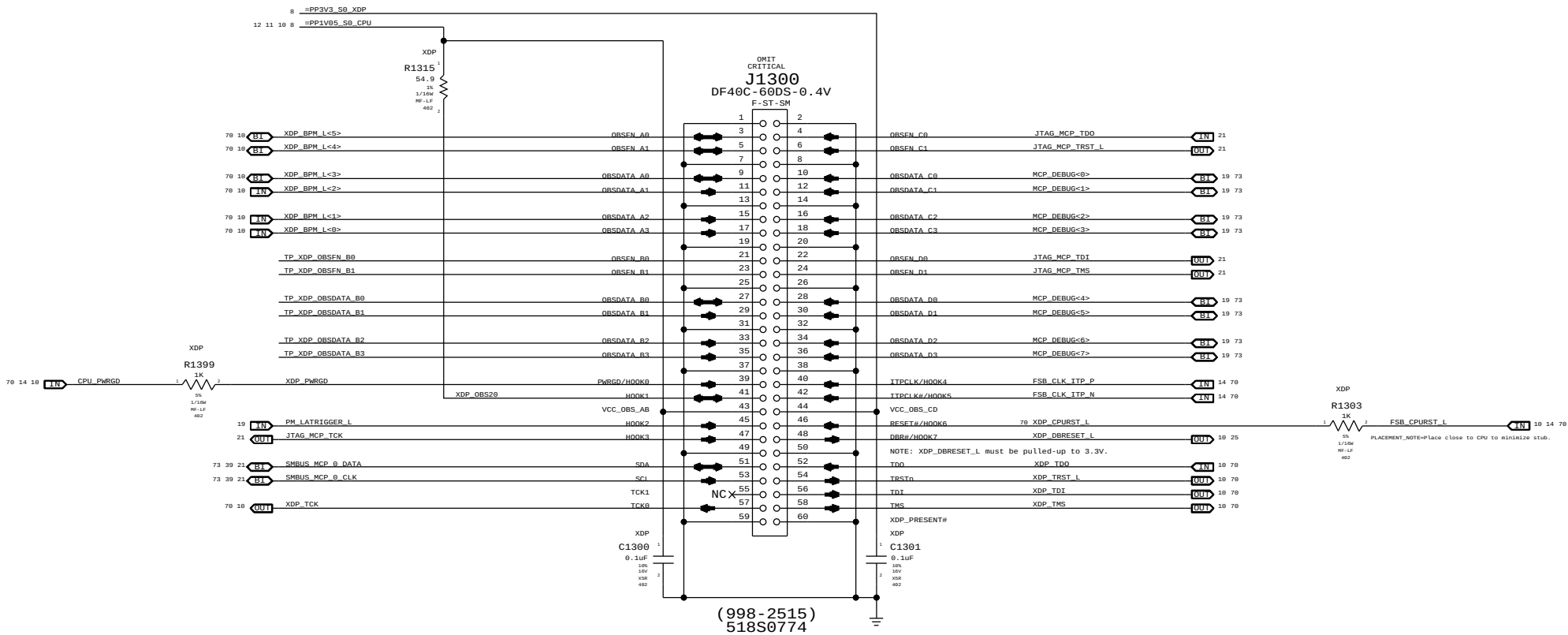
SYNC FROM T18
REMOVE NO STUFF CAPS C1220 TO C1231
REMOVE C1244 & C1245
CHANGE C1240-C1243 AND C1260 FROM 128S0241(9 MILLI-OHM) TO 128S0231(6 MILLI-OHM)

SYNC MASTER=K24 MLB		SYNC DATE=03/30/2009	
PAGE TITLE			
CPU Decoupling			
	DRAWING NUMBER		SIZE
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Apple Inc.	REVISION		PAGE
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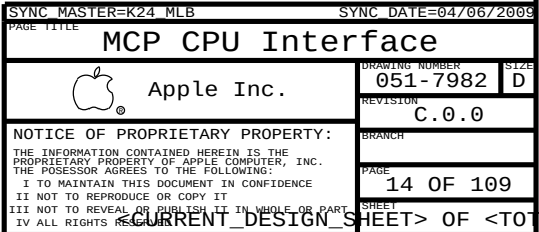
Mini-XDP Connector

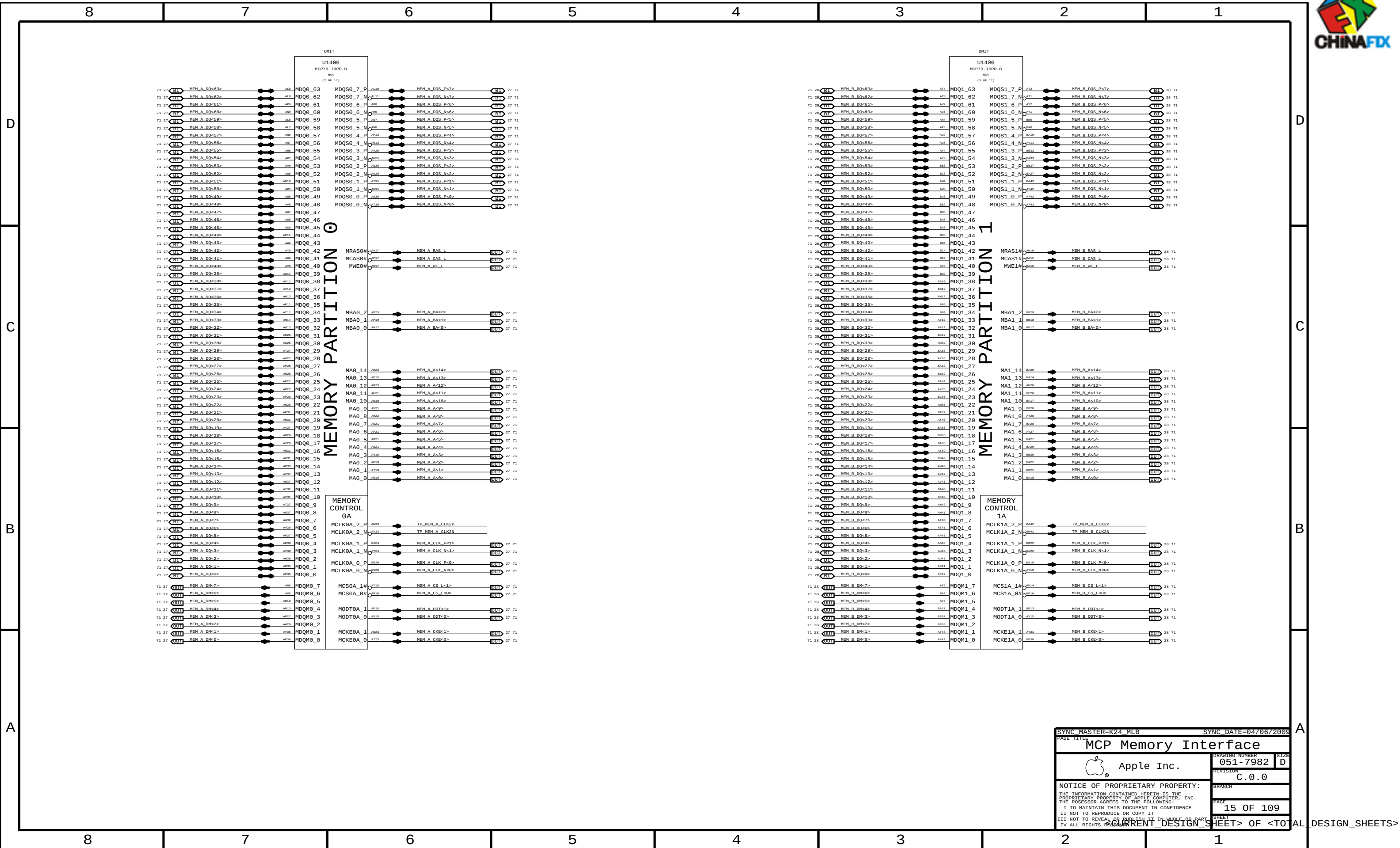
NOTE: This is not the standard XDP pinout.
USE WITH 928-0782 ADAPTER FLEX TO SUPPORT CPU, MCP DEBUGGING.

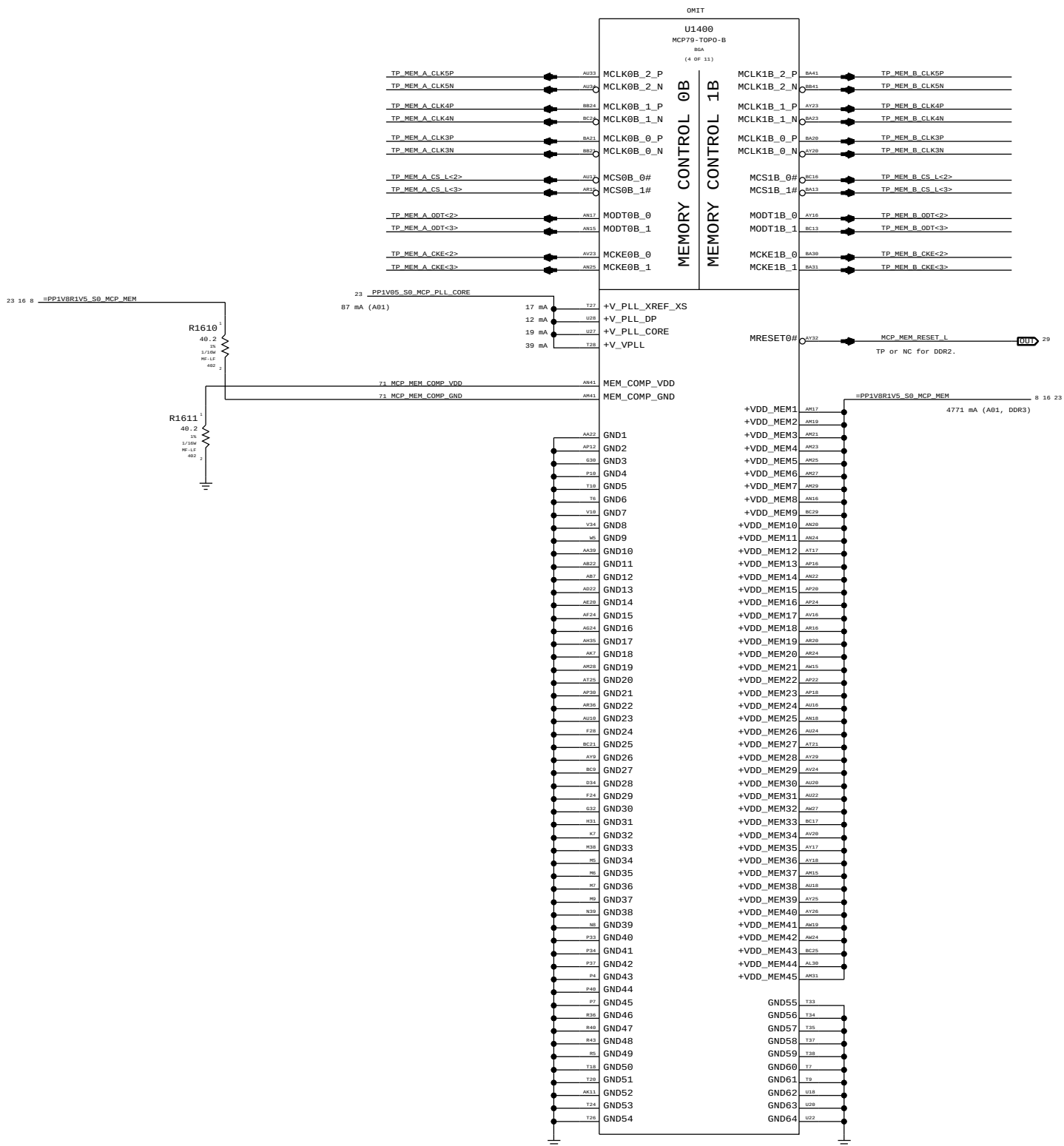
MCP79-specific pinout



Direction of XDP module
Please avoid any obstructions
ON 000-NUMBERED SIDE OF J1300

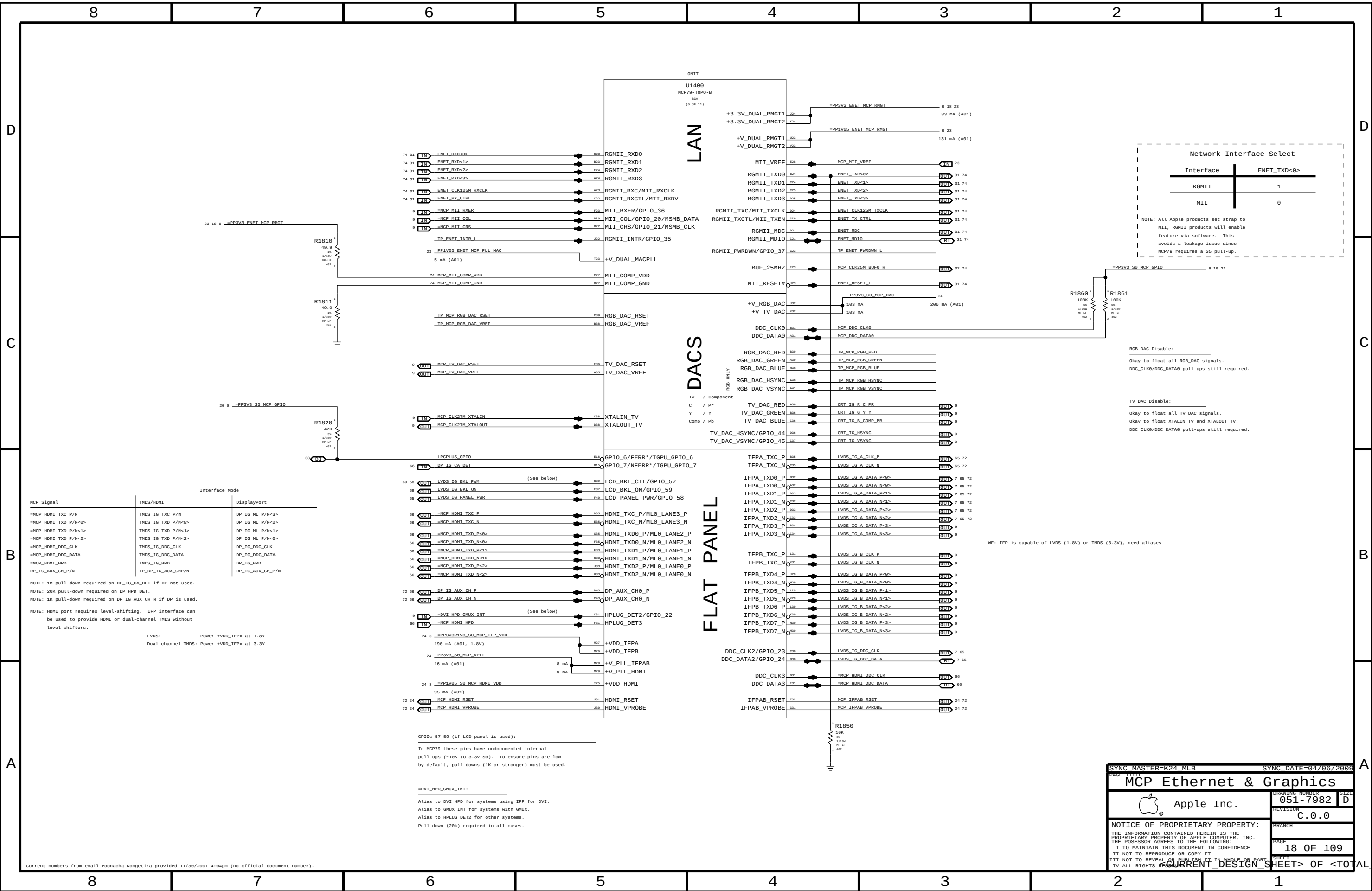




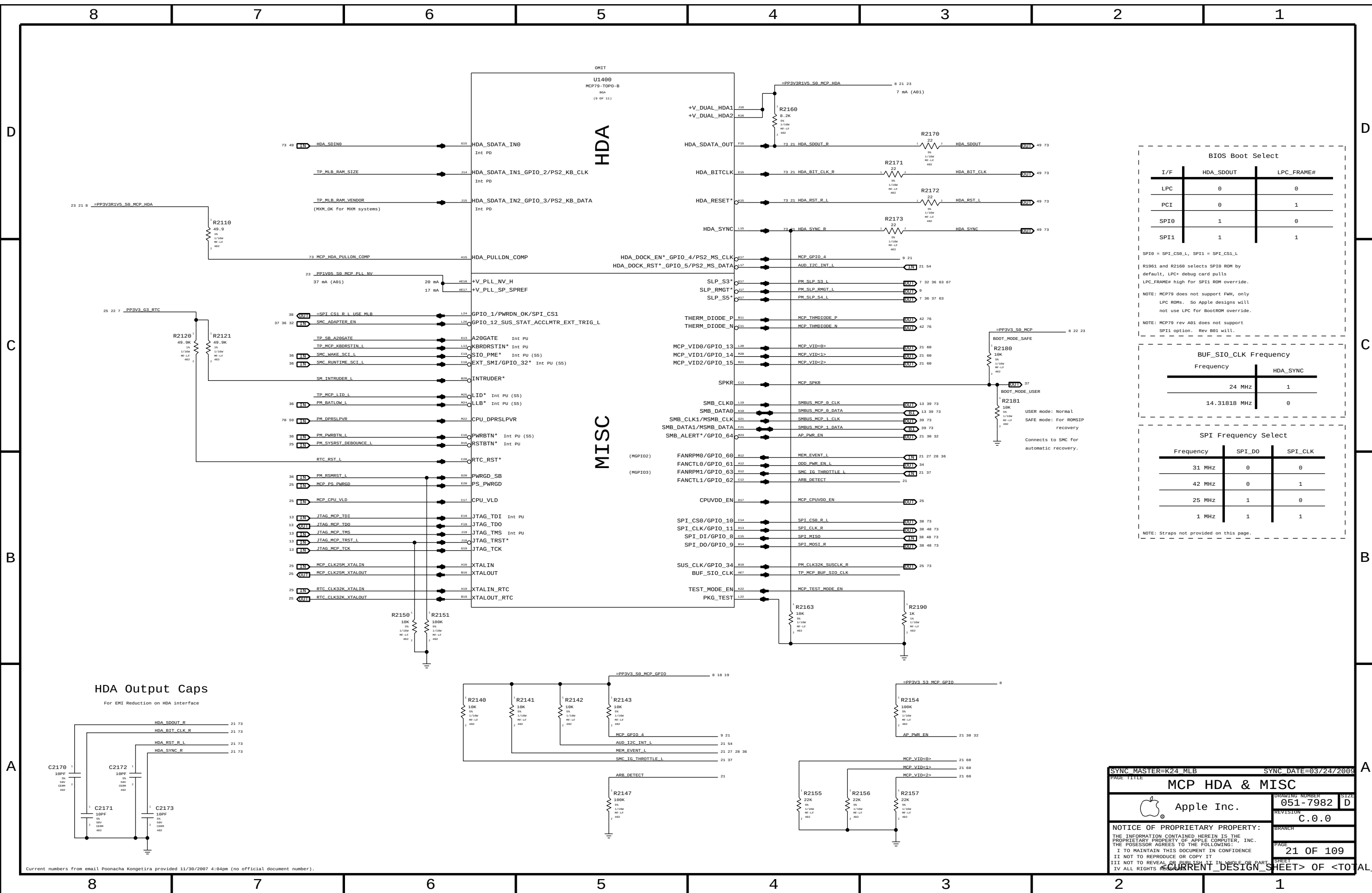


SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
MCP Memory Misc			
DRAWING NUMBER		SIZE	
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BIOS Boot Select

I/F	HDA_SDOUT	LPC_FRAME#
LPC	0	0
PCI	0	1
SPI0	1	0
SPI1	1	1

SPI0 = SPI_CS0_L, SPI1 = SPI_CS1_L

R1961 and R2160 selects SPI0 ROM by default, LPC+ debug card pulls LPC_FRAME# high for SPI1 ROM override.

NOTE: MCP79 does not support FWH, only LPC ROMs. So Apple designs will not use LPC for BootROM override.

NOTE: MCP79 rev A01 does not support SPI1 option. Rev B01 will.

BUF_SIO_CLK Frequency

Frequency	HDA_SYNC
24 MHz	1
14.31818 MHz	0

SPI Frequency Select

Frequency	SPI_D0	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

NOTE: Straps not provided on this page.

SYNC MASTER=K24 MLB SYNC DATE=03/24/2009

MCP HDA & MISC

DRAWING NUMBER: 051-7982 SIZE: D

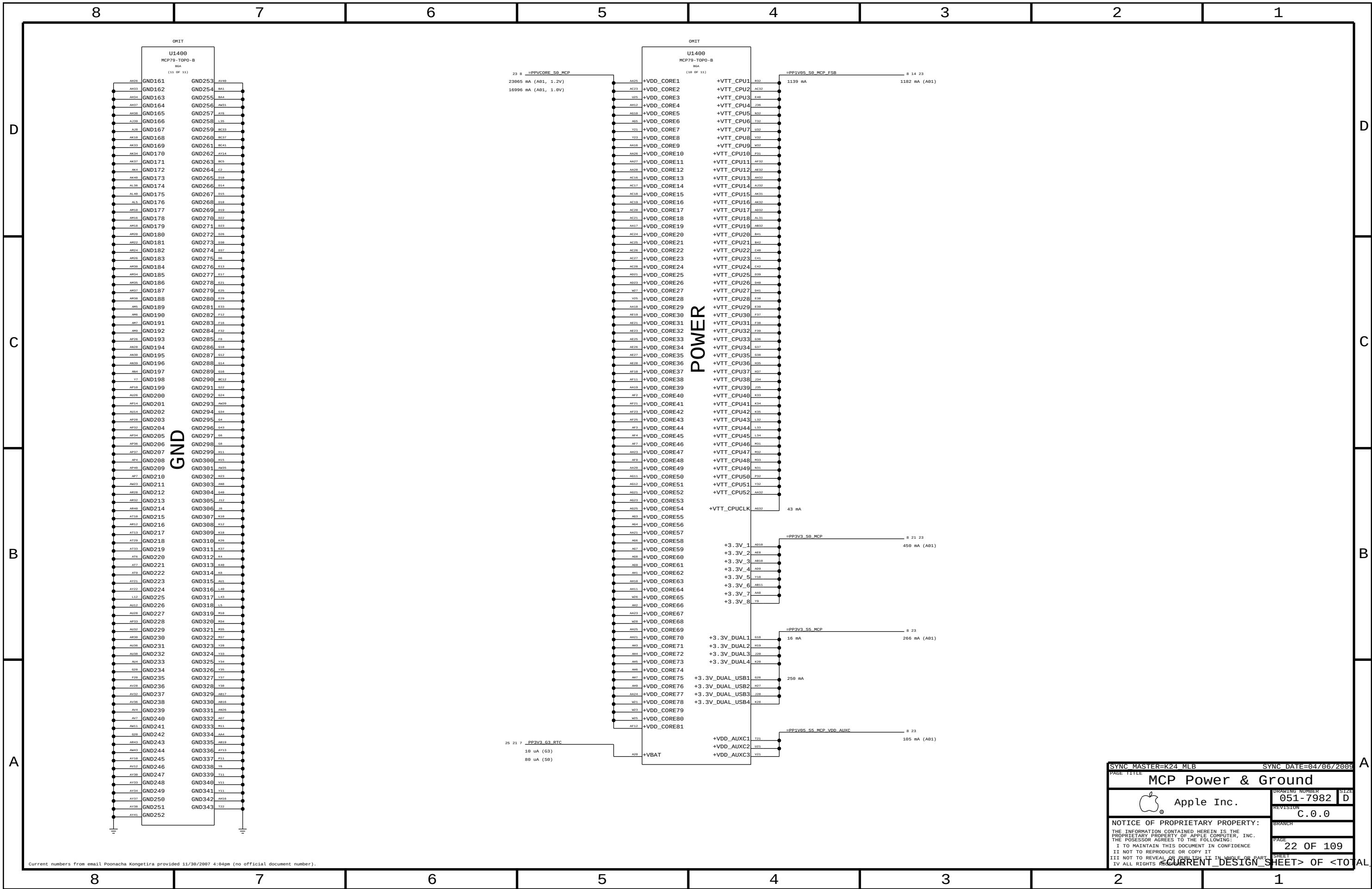
REVISION: C.0.0

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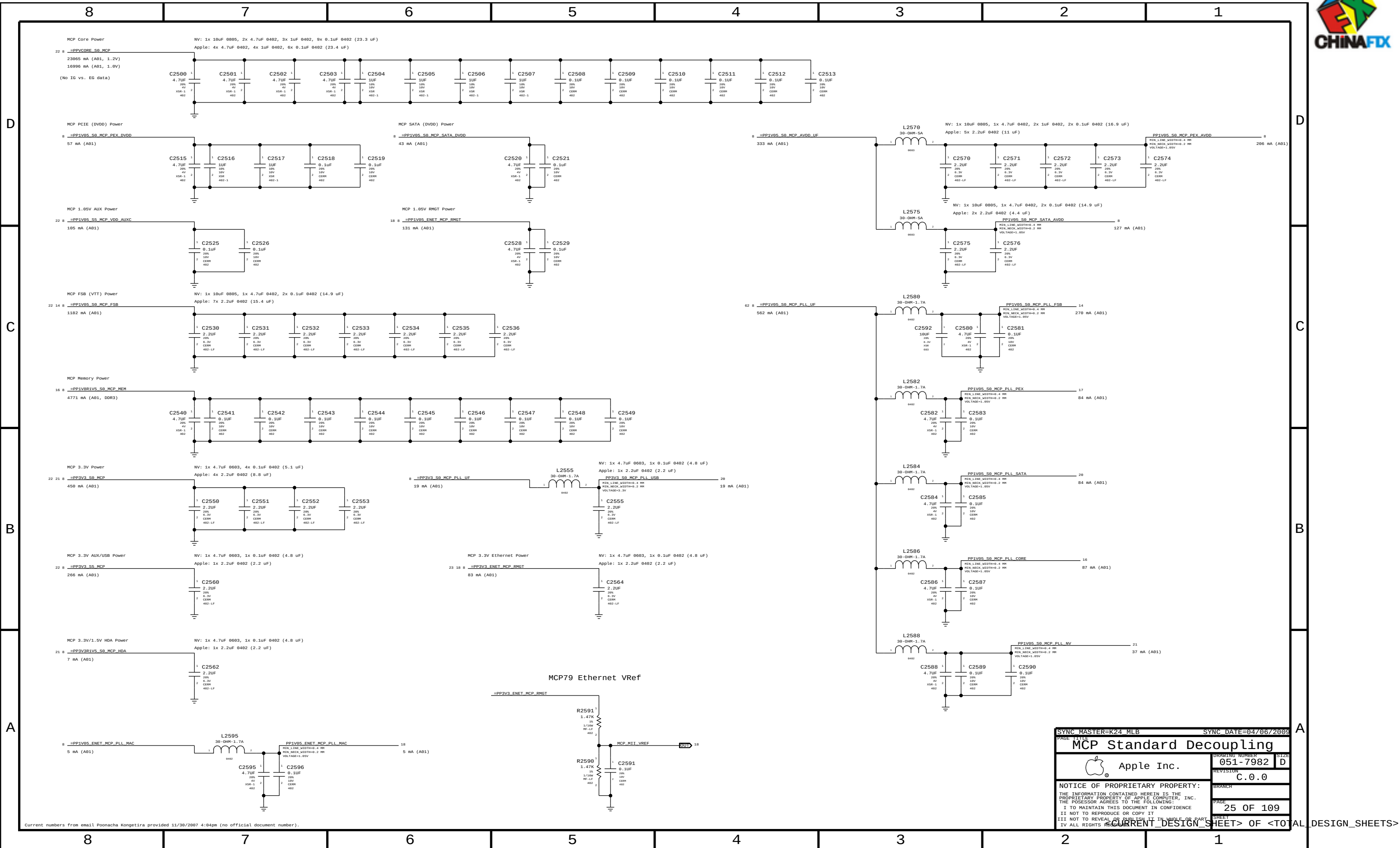
PAGE: 21 OF 109 SHEET:

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SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
MCP Power & Ground			
 Apple Inc.		DRAWING NUMBER	SIZE
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SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
MCP Standard Decoupling			
Apple Inc.		DRAWING NUMBER	051-7982
		REVISION	C.0.0
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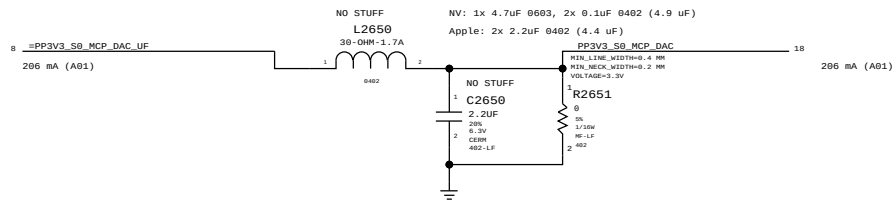
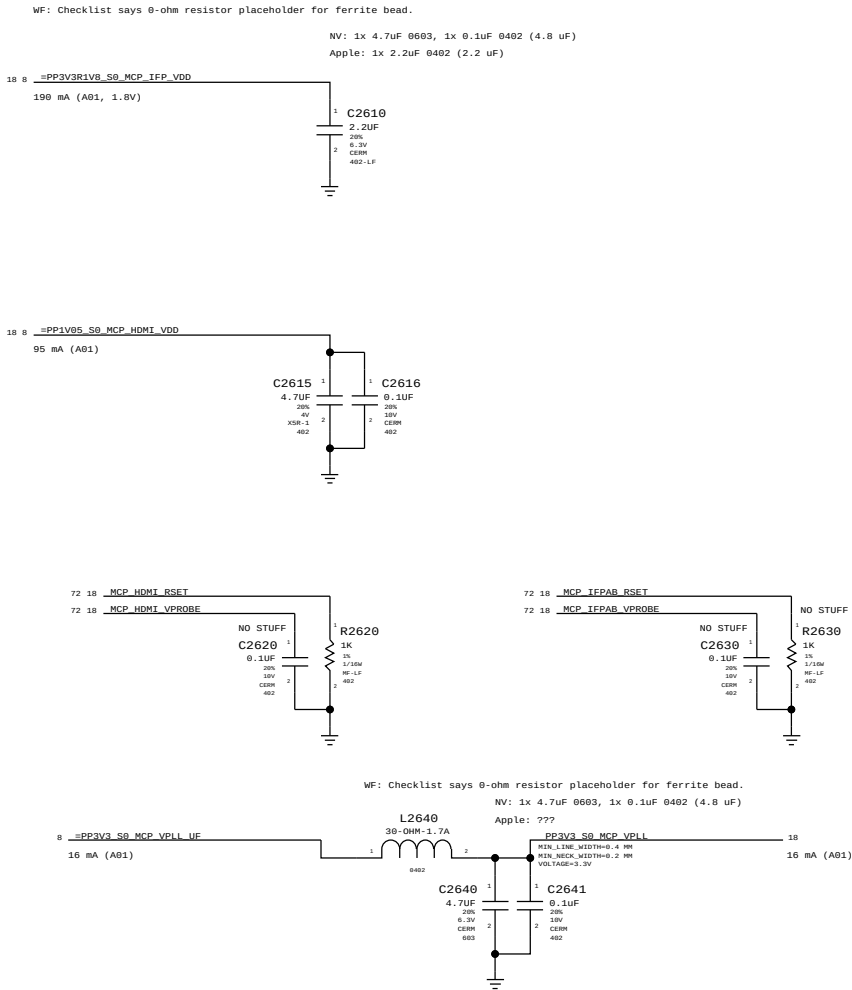
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SYNC FROM T18
REMOVE MCP 27MHZ CRYSTAL CRICUIT SINCE NOT SUPPORTING TV-OUT
REMOVE DAC TERMINATIONS R2665,C2665 AND R2670 TO R2672
NOSTUFF PP3V3_S0_MCP_DAC RAIL COMPONENTS (L2650 AND C2650)
CHANGE C2651 TO R2651 TO GND PP3V3_S0_MCP_DAC
REMOVE HDCP ROMS

SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE			
MCP Graphics Support			
 Apple Inc.		DRAWING NUMBER	SIZE
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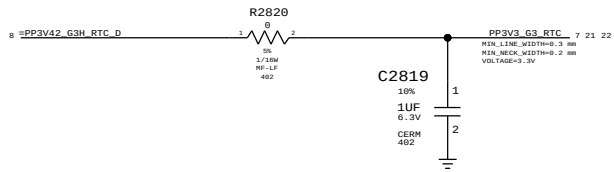
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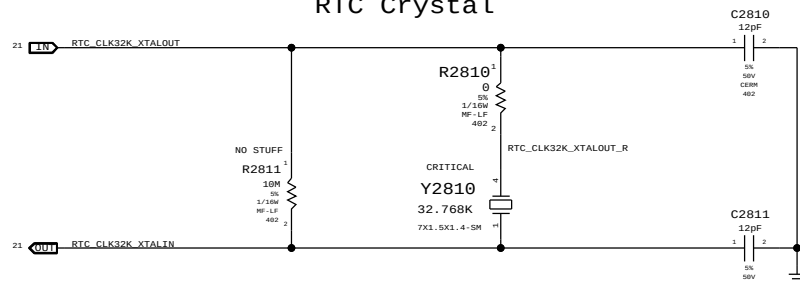
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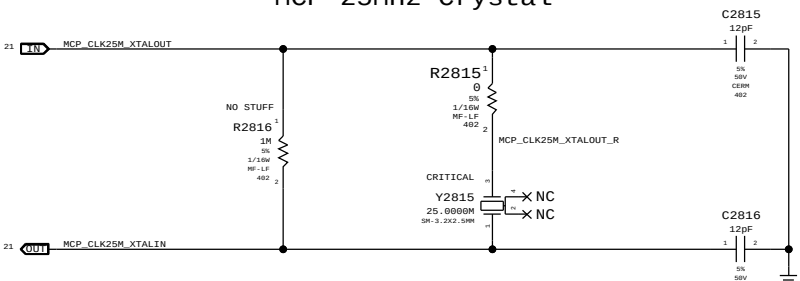


PLACEMENT_NOTE=PLACE C2819 CLOSE TO MCP79
PLACE C2819 CLOSE TO MCP79

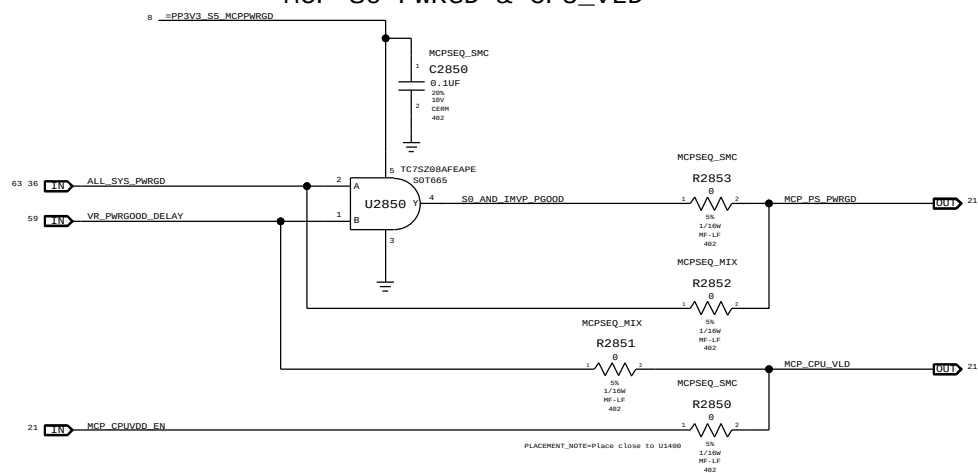
RTC Crystal



MCP 25MHz Crystal



MCP S0 PWRGD & CPU_VLD



MCPSEQ_SMC represents MCP79 'MLB' power sequencing connections, but results in MCP79 ROMSIP sequence happening after CPU powers up.

MCPSEQ_MIX is cross between MLB and internal power sequencing, which results in earlier ROMSIP and MCP FSB I/O interface initialization.

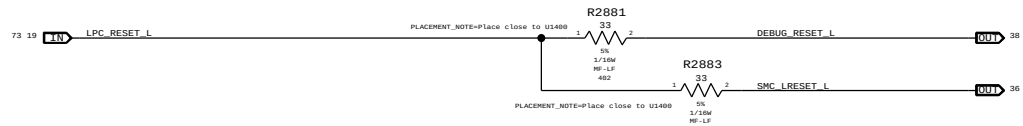
SMC 99ms delay from ALL_SYS_PWRGD to IMVP_VR_ON plus IMVP6 delay for VR_PWRGOOD_DELAY should guarantee CPU_VLD does not go high before CPUVDD_EN (which is 40-100ms after PS_PWRGD assertion).

NOTE: If CPU_VLD deasserts during S0 MCP79 will take system to S5 immediately.

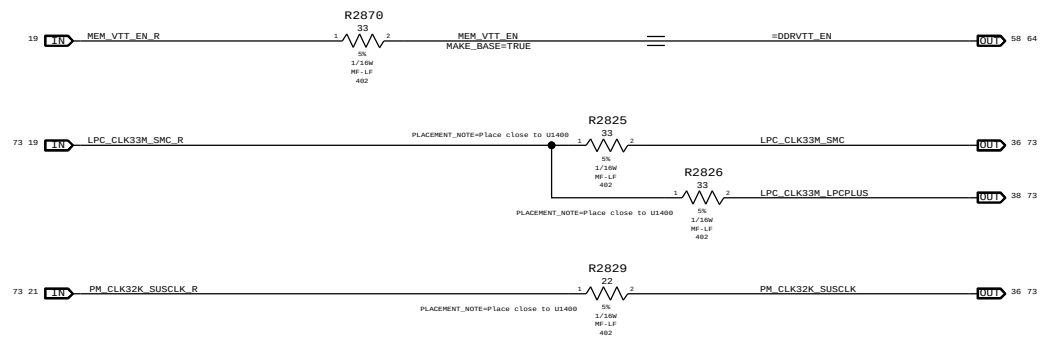
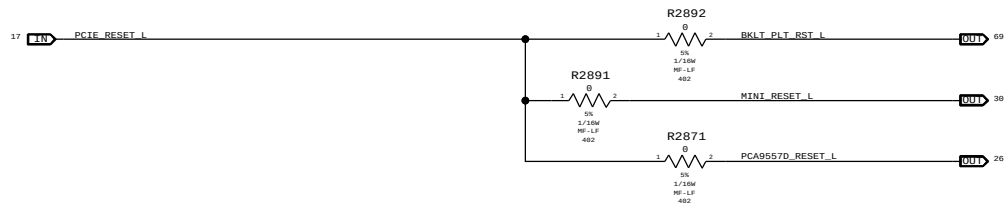
SYNC FROM T18
CHANGE RESET BUTTON TO RESET PADS
REMOVE UNUSED PCIE RESET SIGNALS
REMOVE R2824 AND NET PCI_CLK33M_SLOT_A
CHANGE RTC COIN CELL TO LDO & SUPERCAP
ALIAS MEM_VTT_EN TO =DDRVTT_EN
CHANGE Y2810 AND U2850 TO SMALLER PARTS

Platform Reset Connections

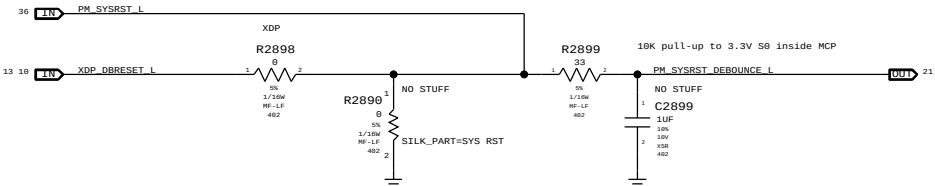
LPC Reset (Unbuffered)



PCIE Reset (Unbuffered)



Reset Button



SYNC MASTER=K24 MLB		SYNC DATE=02/15/2009	
PAGE TITLE			
SB Misc			
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Page Notes

Power aliases required by this page:

- PP1VS_S3_MEM_A
- PP1VS_S3_MEM_A
- PP1VS_S3_MEM_VTT_A
- PPSPD_S3_MEM_A (2.5 - 3.3V)

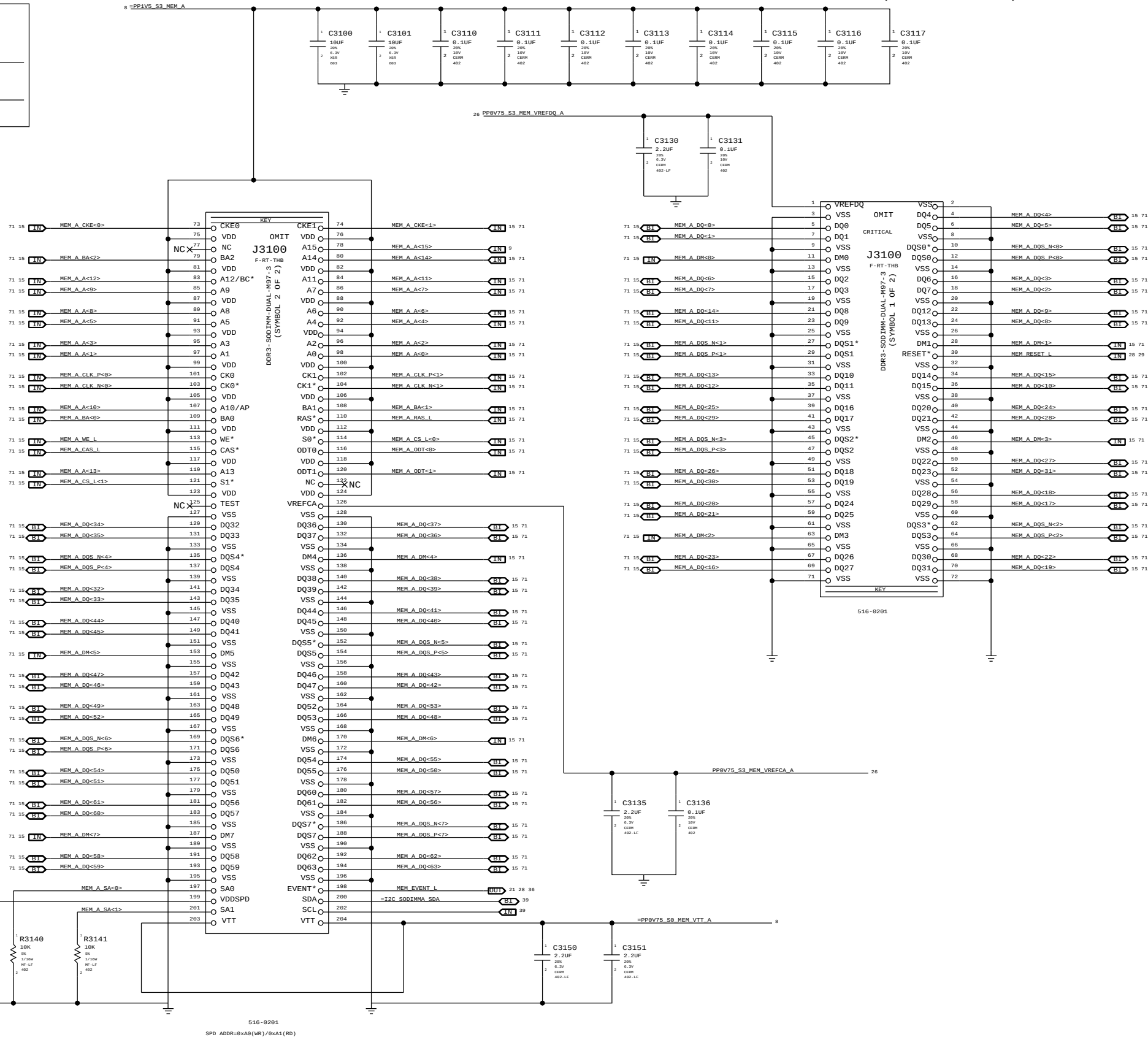
Signal aliases required by this page:

- I2C_S0D1MMA_SCL
- I2C_S0D1MMA_SDA

SOM options provided by this page:

(NONE)

DDR3 DECOUPLING AND GROUND RETURN CAPS (CONNECTOR SIDE)



"Factory" (top) slot

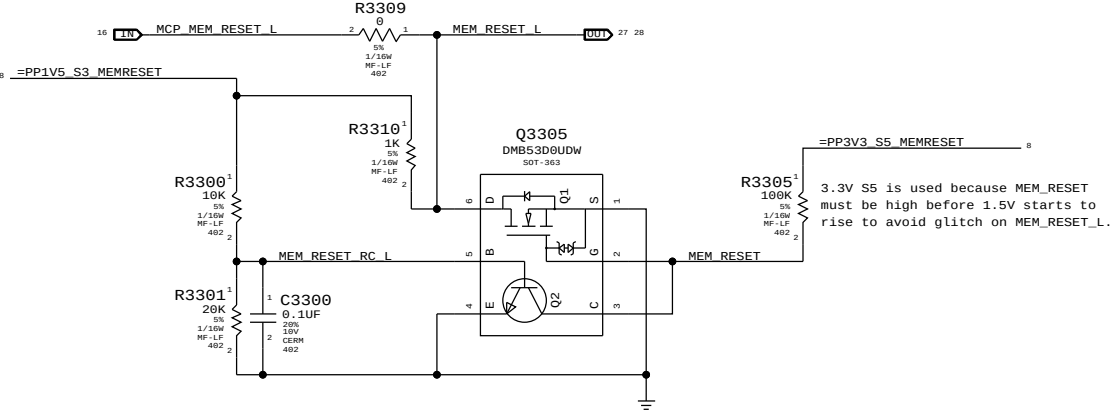
SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
PAGE TITLE			
DDR3 S0-DIMM Connector A			
DRAWING NUMBER		SIZE	
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DDR3 RESET Support

Required because MCP79 does not meet DDR3 spec power-up reset timing requirement.





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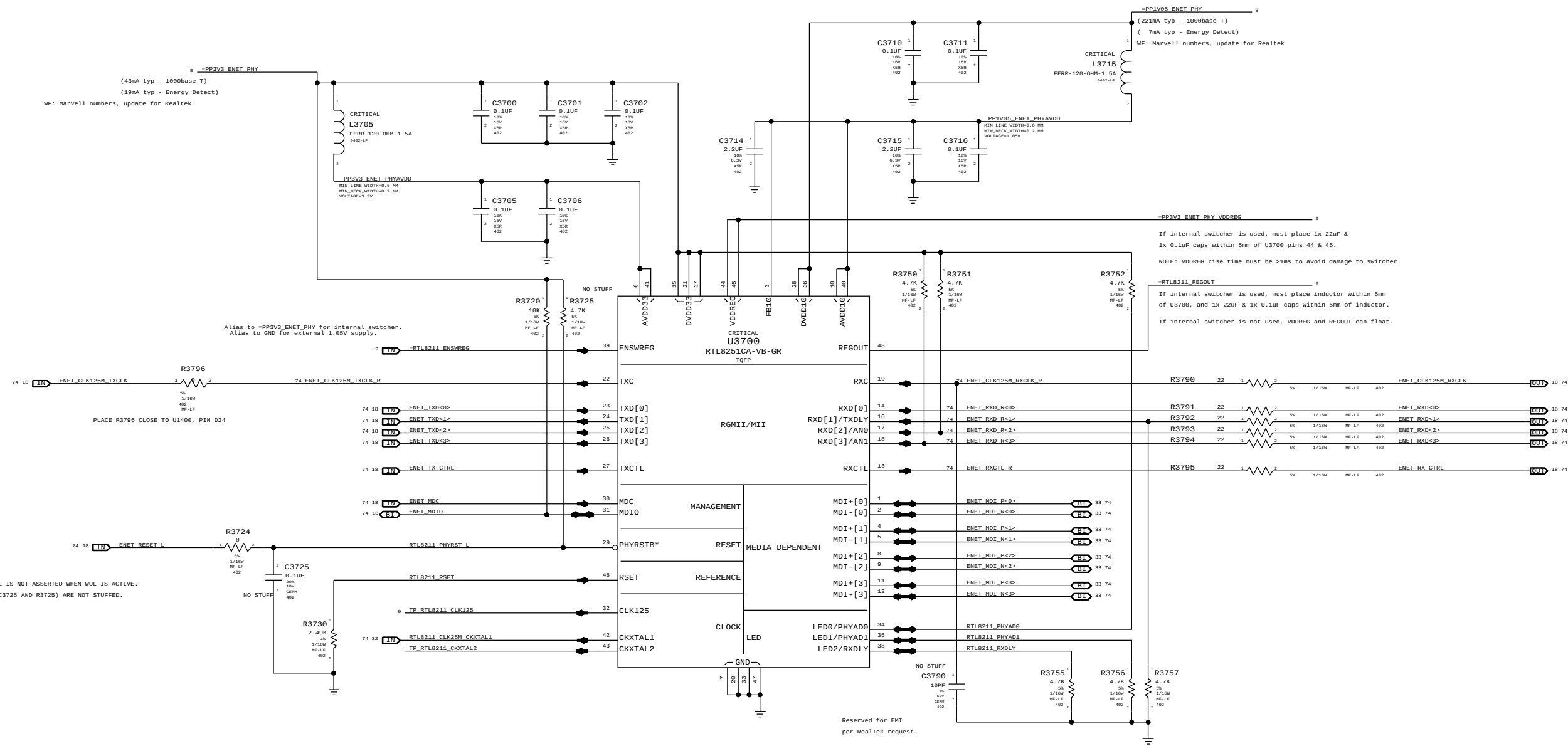
Apple Inc	DRAWING NUMBER 051-7982	SIZE D
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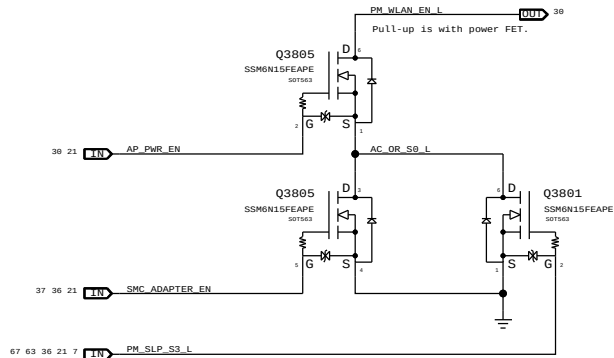




WLAN Enable Generation

"WLAN" = ("S3" && "AP_PWR_EN" && ("AC" || "S0"))

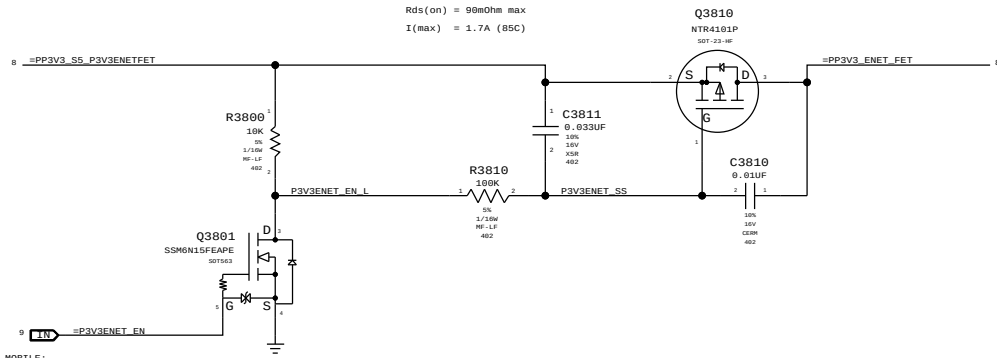
NOTE: S3 term is guaranteed by S3 pull-up on open-drain AP_PWR_EN signal.



3.3V ENET FET

@ 2.5V Vgs:
Rds(on) = 90mOhm max
I(max) = 1.7A (RSC)

CRITICAL



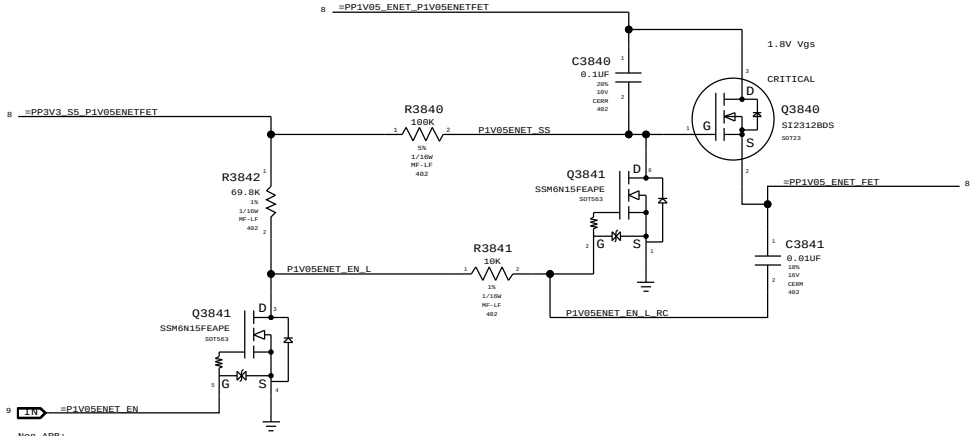
MOBILE:

Recommend aliasing PM_SLP_RMG_T_L and
=P3V3ENET_EN. Nets separated on
ARB for alternate power options.

1.05V ENET FET

1.8V Vgs

CRITICAL

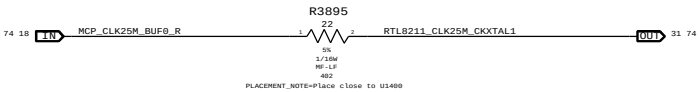


Non-ARB:

Recommend aliasing PM_SLP_RMG_T_L and
=P1V05ENET_EN. Nets separated on
ARB for alternate power options.

RTL8211 25MHz Clock

NOTE: MCP79 can provide 25MHz clock, but clock runs whenever RMGT rails are powered.
Designs must ensure PHY is powered whenever RMGT rails are, or use separate crystal.



SYNC MASTER=K24 MLB SYNC DATE=04/06/2009

Ethernet & AirPort Support

Apple Inc.

051-7982 D

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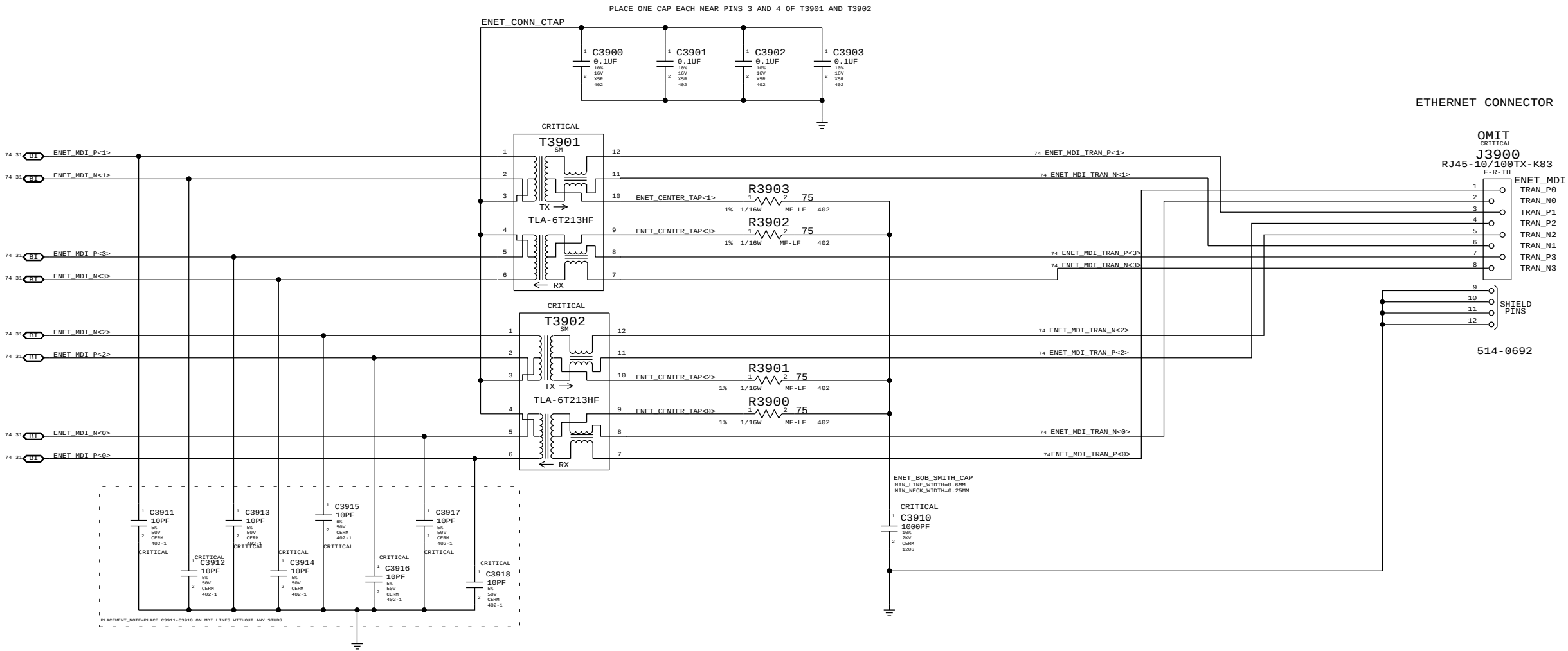
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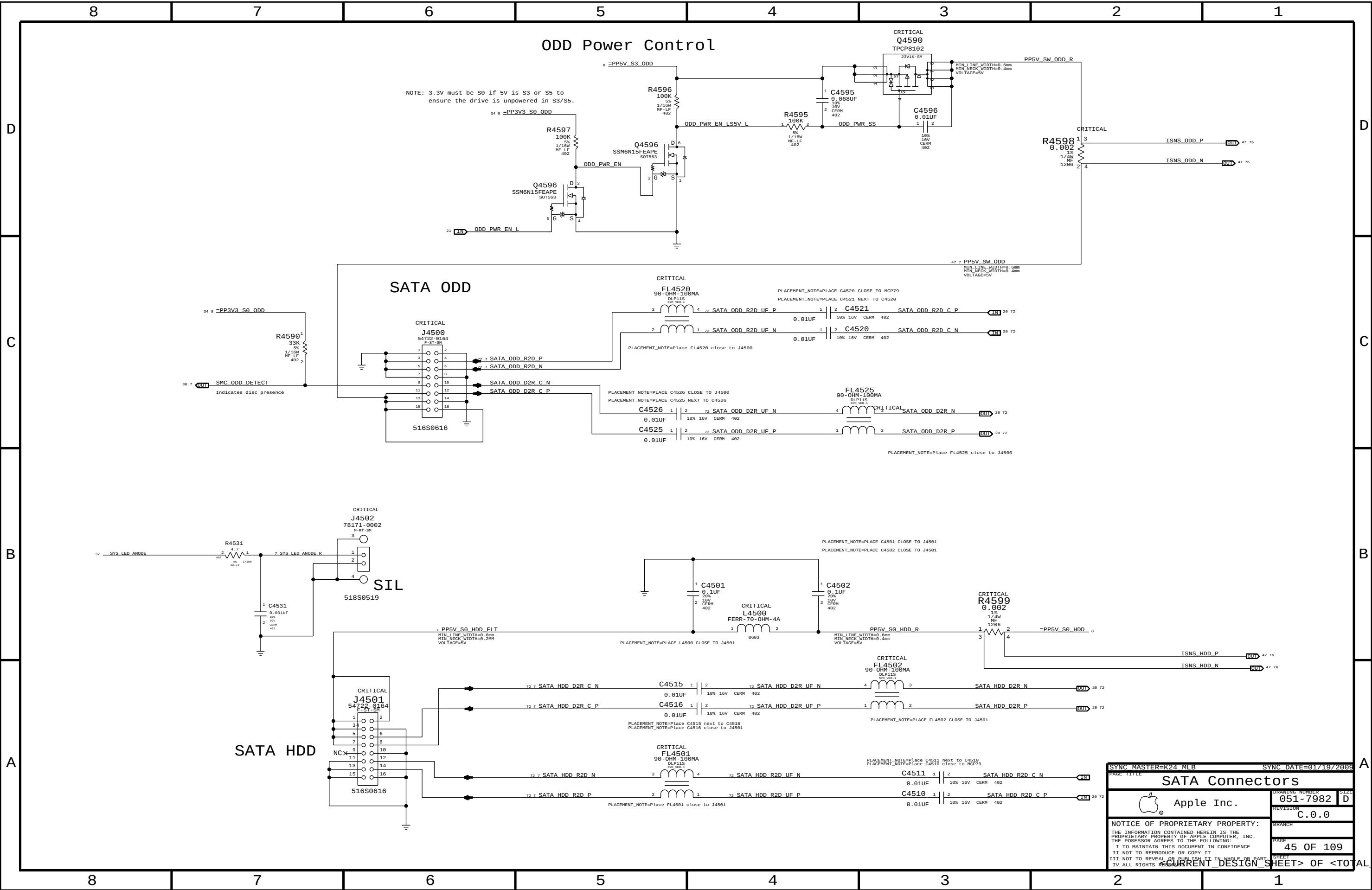
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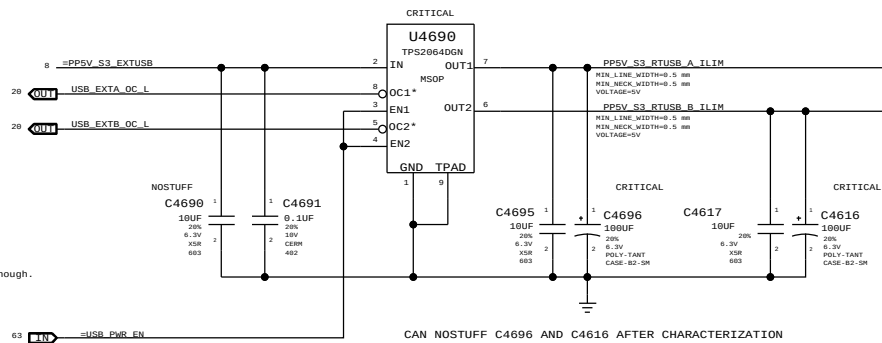


SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
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ETHERNET CONNECTOR			
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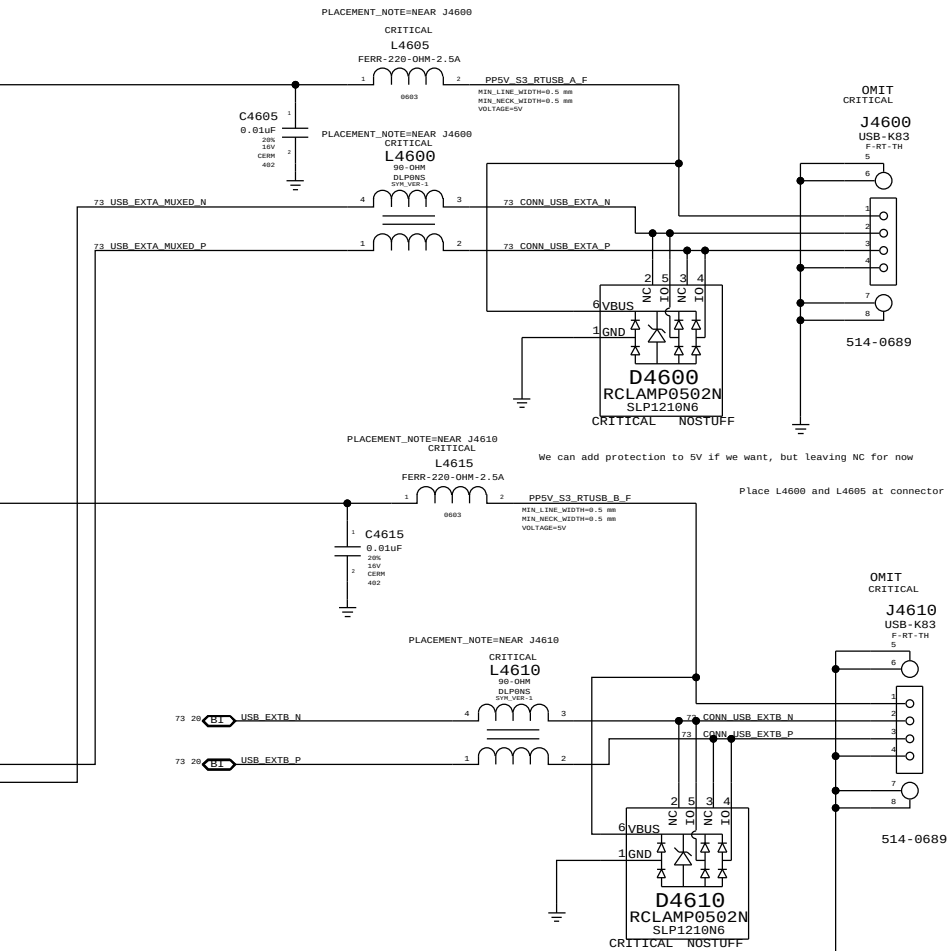


POR IS PLASTIC USB CONNECTOR PARTS BUT METAL PART'S SCHEMATIC AND CAD SYMBOLS HAVE BEEN USED AS ITS LAND PATTERN CAN ACCOMODATE BOTH TYPES

Port Power Switch



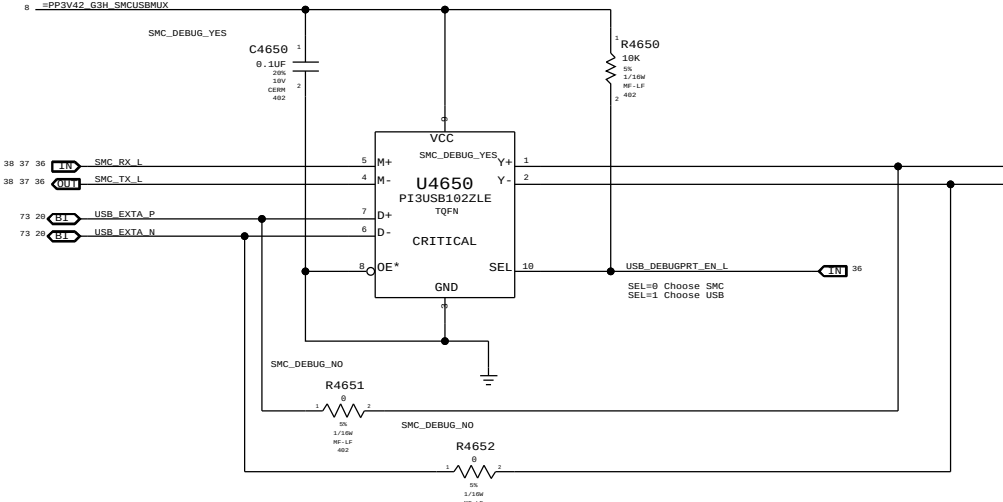
USB PORT A (FRONT PORT)

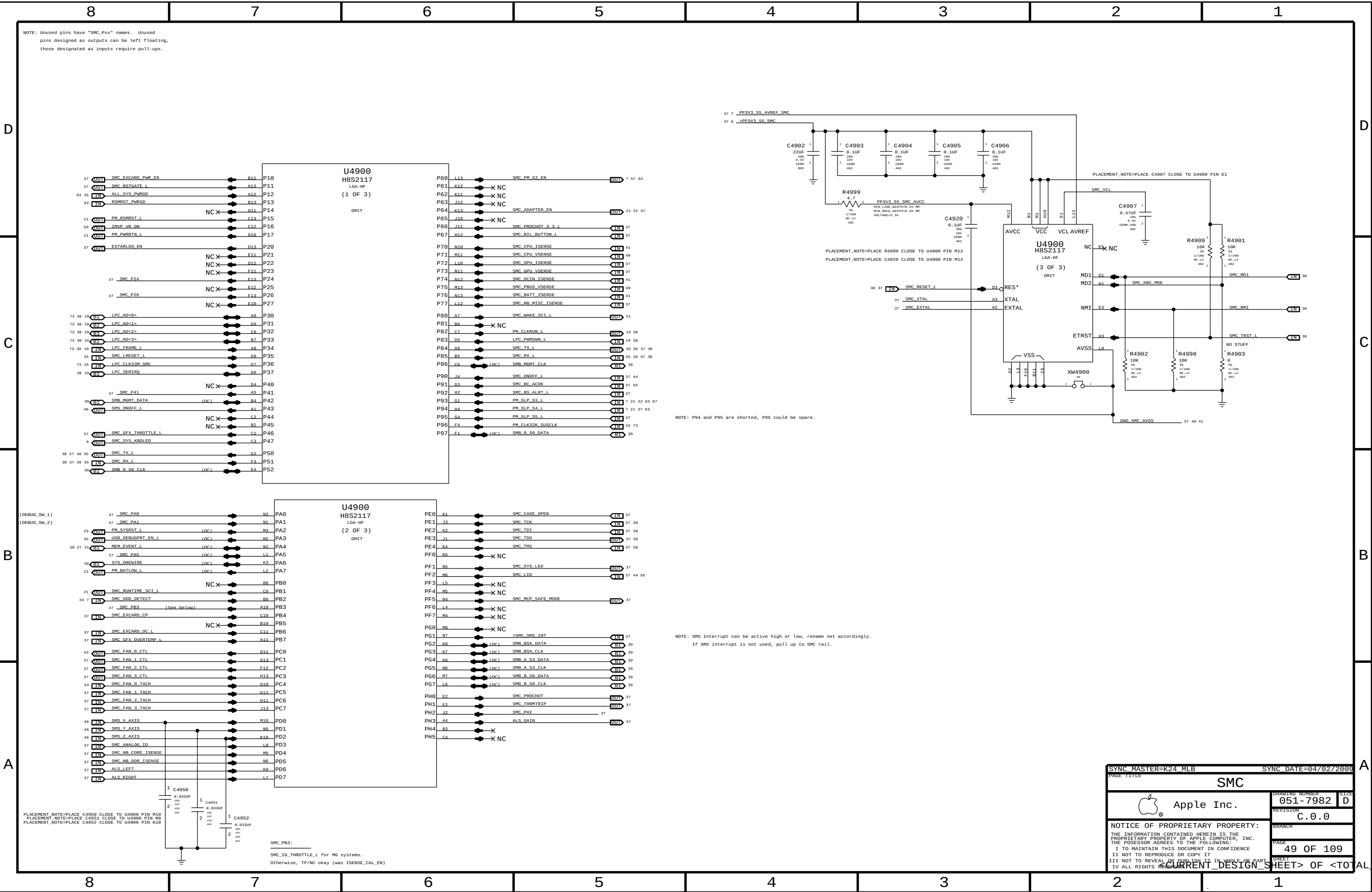


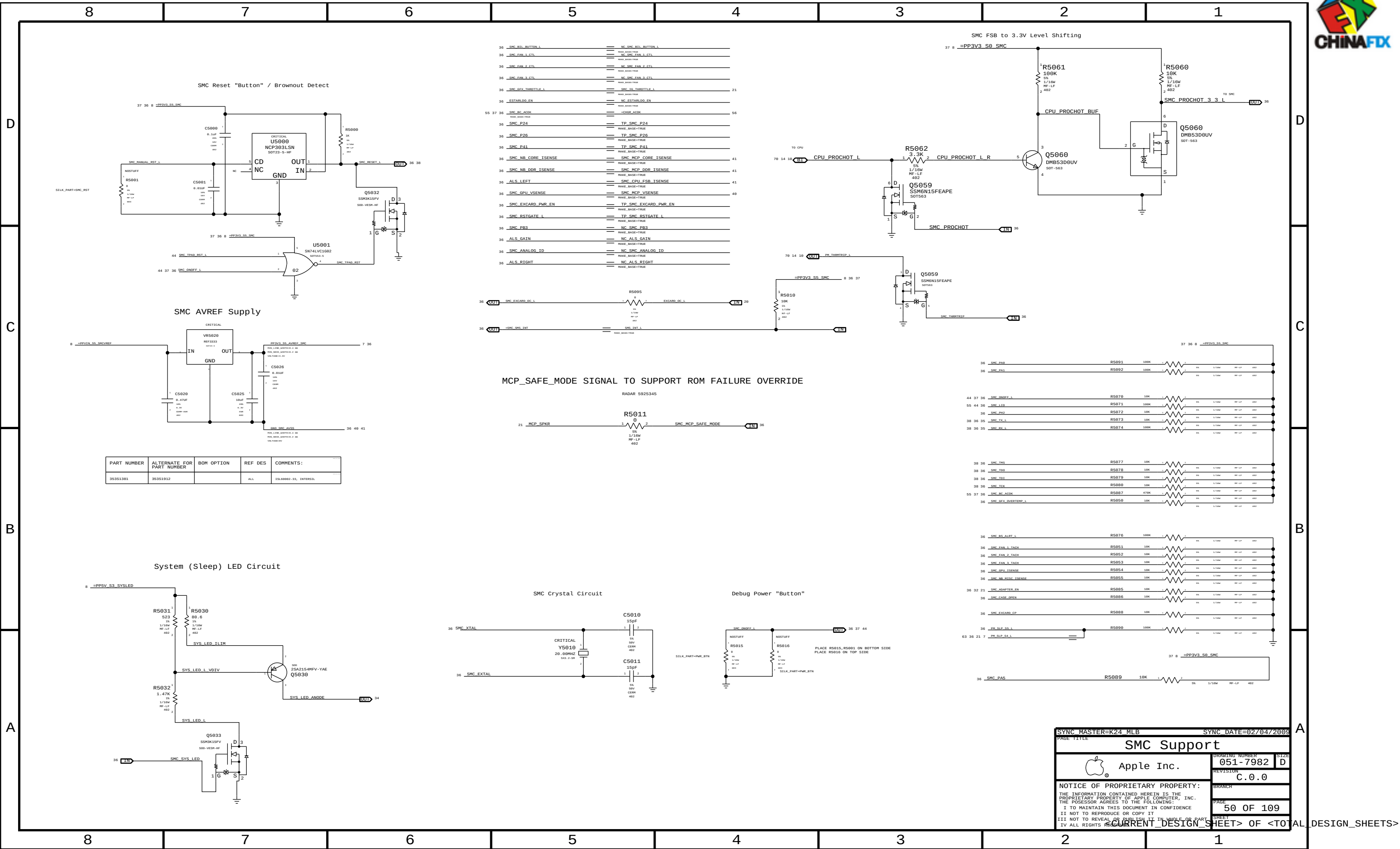
USB PORT B (BACK PORT)

SYNC MASTER=K24 MLB		SYNC DATE=02/05/2009	
PAGE TITLE			
External USB Connectors			
		DRAWING NUMBER	SIZE
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USB/SMC Debug Mux

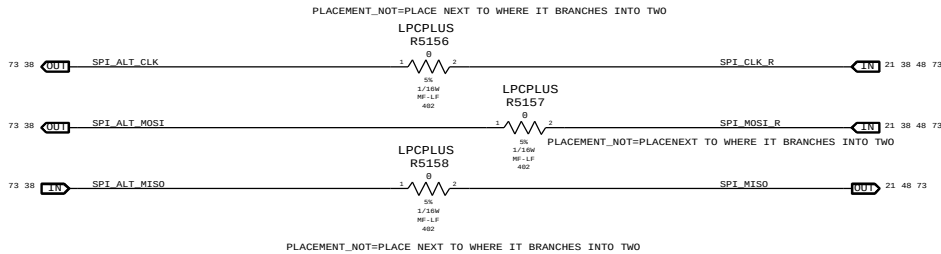
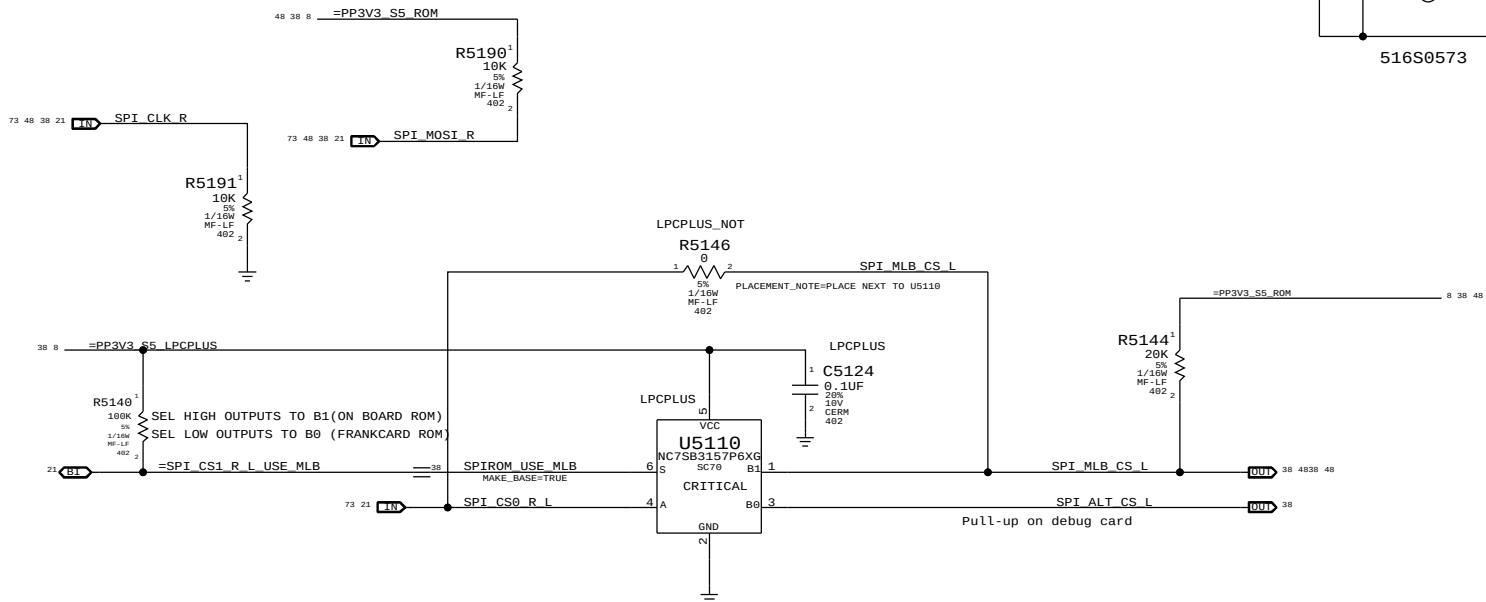
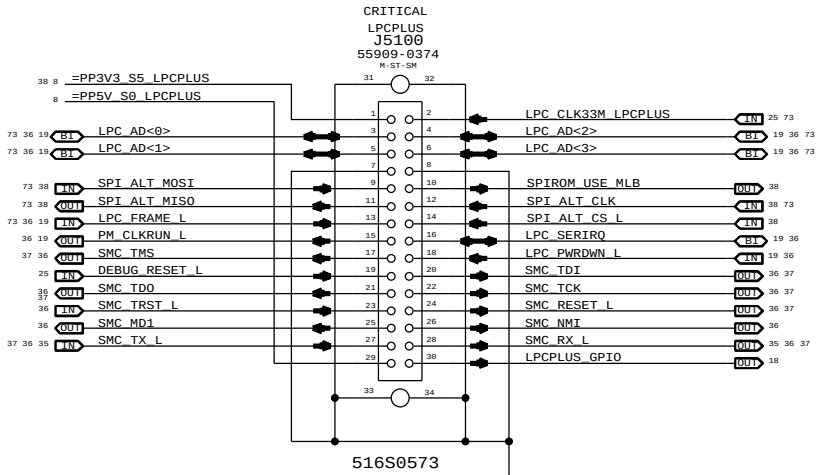




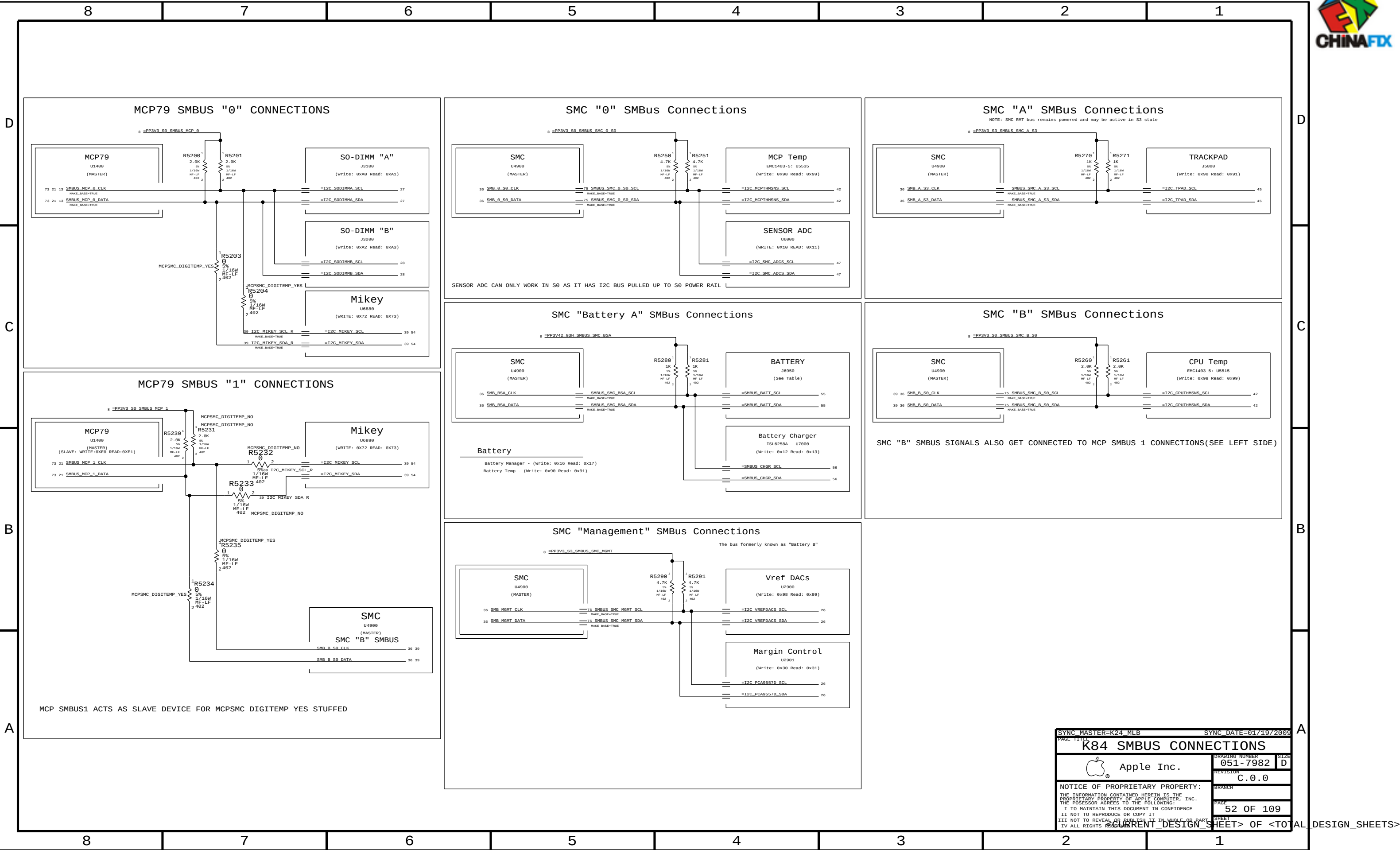


LPC+SPI Connector

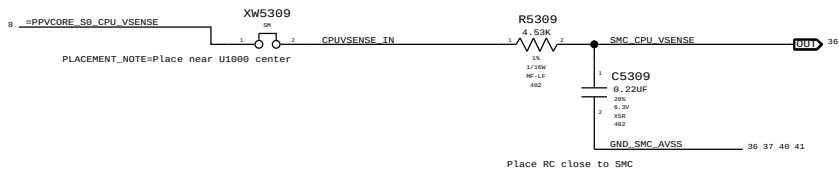
Alternate SPI ROM Support



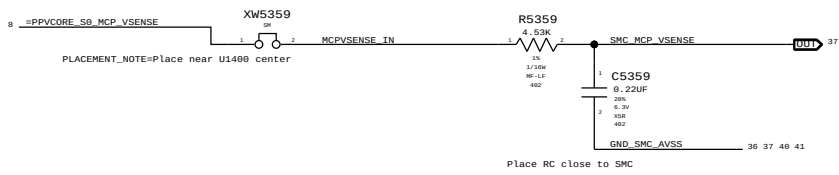
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PAGE TITLE			
LPC+SPI Debug Connector			
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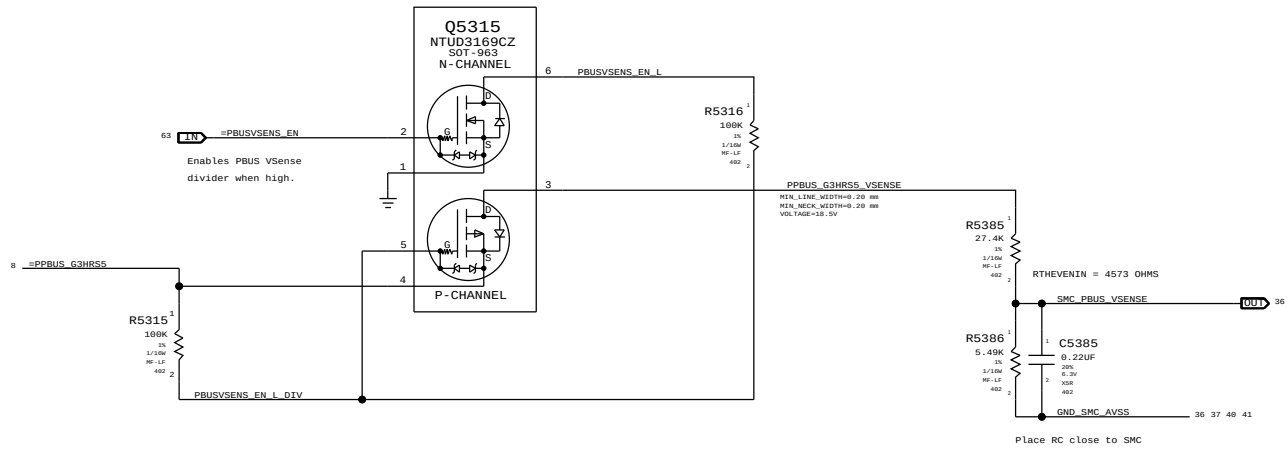
CPU Voltage Sense / Filter



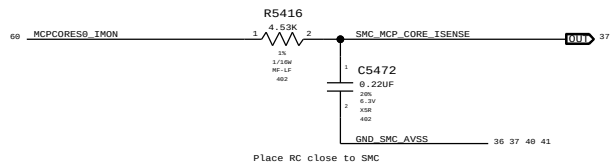
MCP Voltage Sense / Filter



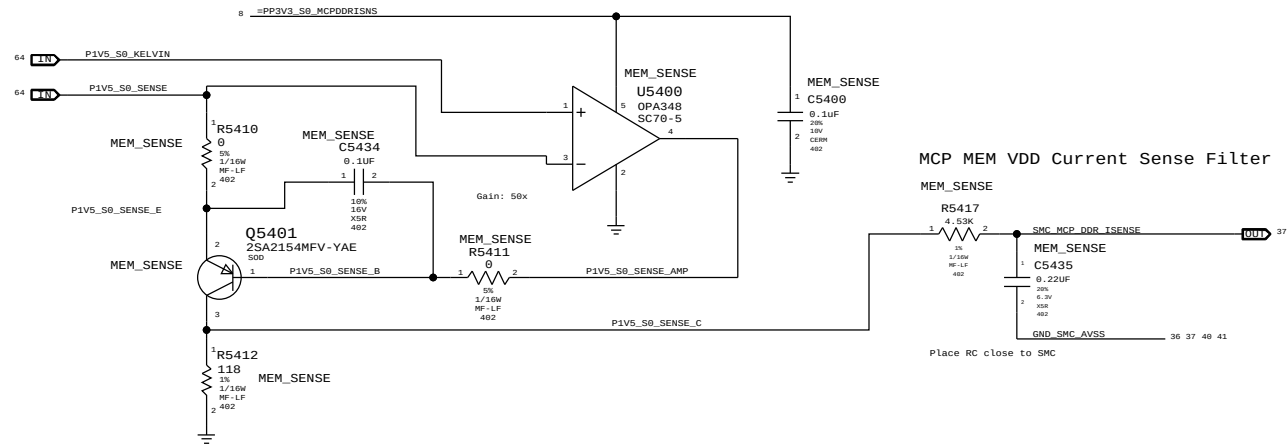
PBUS VOLTAGE SENSE ENABLE & FILTER



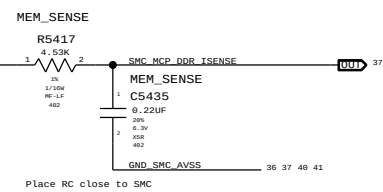
MCP VCore Current Sense Filter



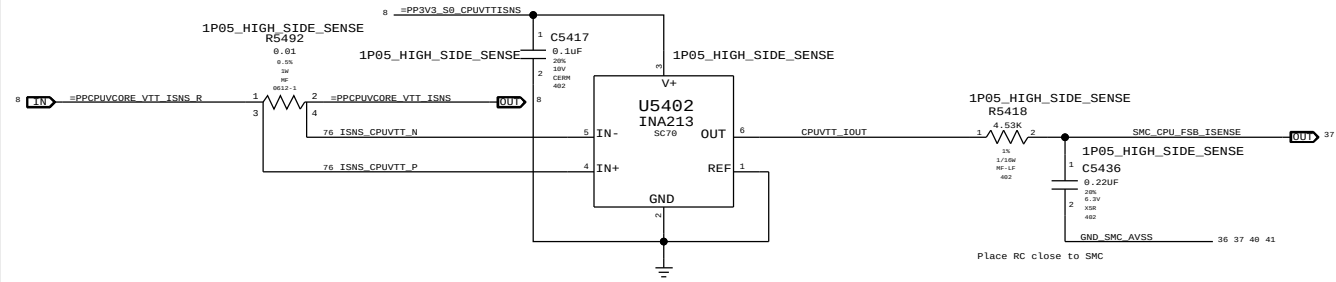
MCP MEM VDD Current Sense



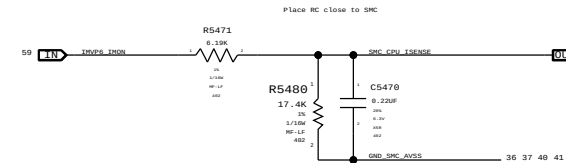
MCP MEM VDD Current Sense Filter



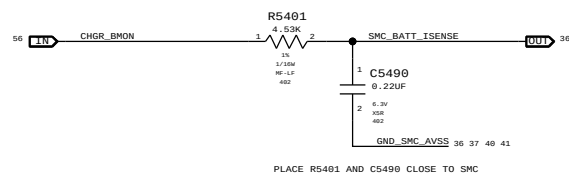
CPU 1.05V AND CPU VCore HIGH SIDE CURRENT SENSE



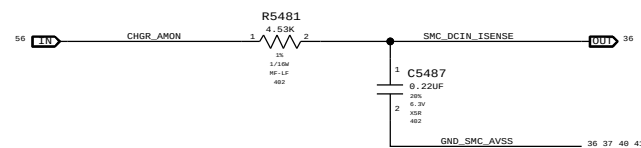
CPU VCore Load Side Current Sense / Filter



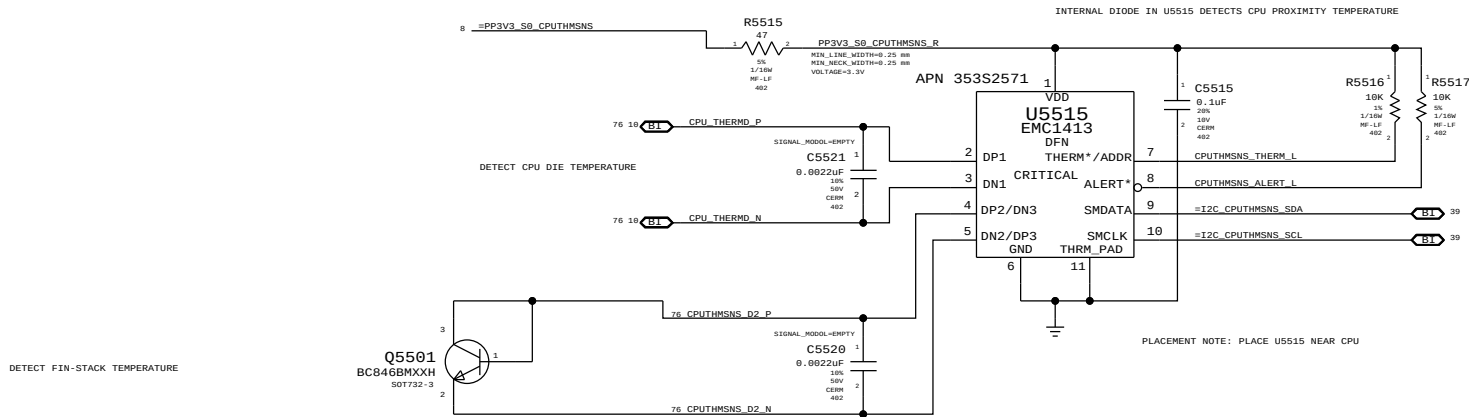
DC-IN (BMON) CURRENT SENSE



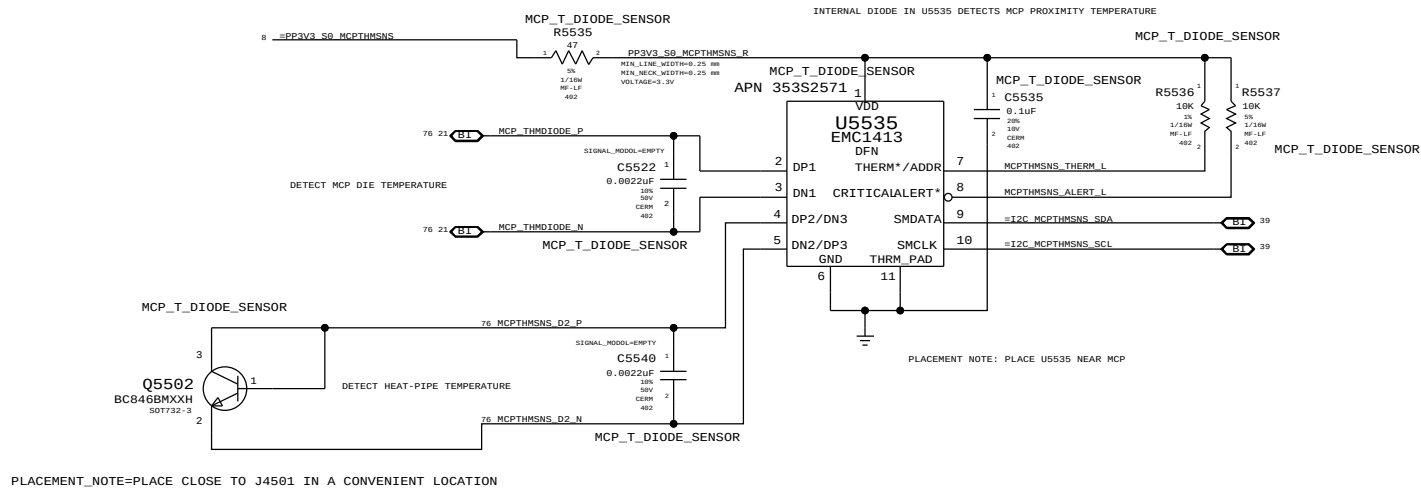
DC-IN (AMON) CURRENT SENSE



CPU T-Diode Thermal Sensor



MCP T-Diode Thermal Sensor





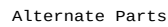
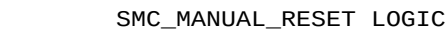
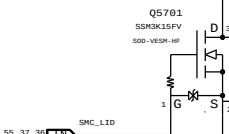
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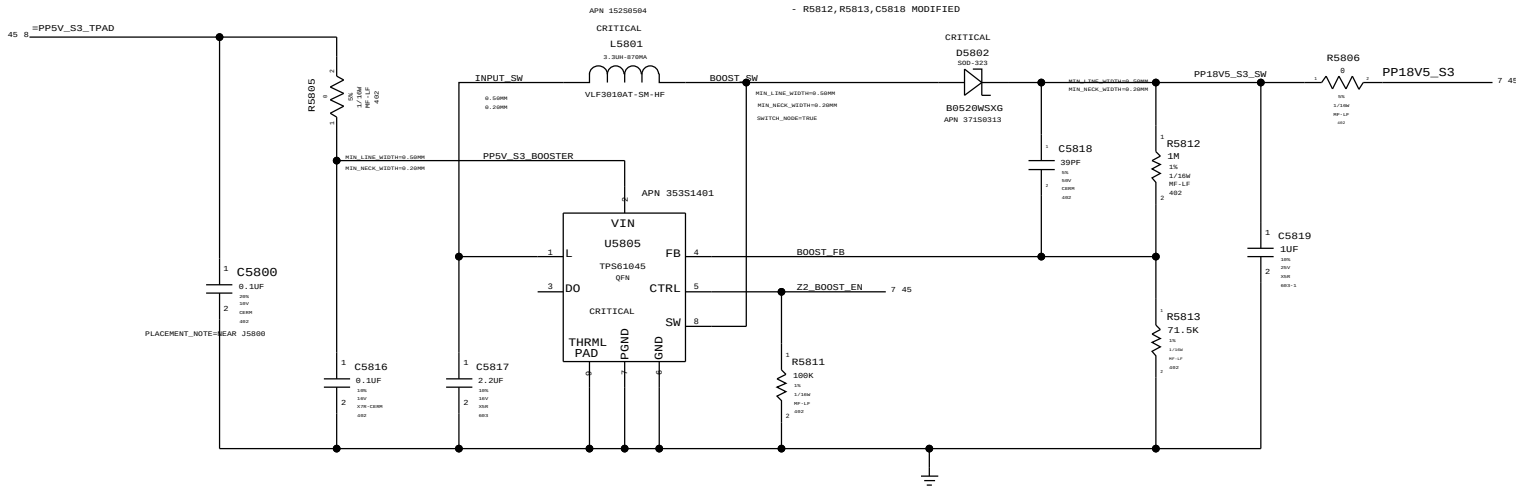
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ISOLATION CIRCUIT

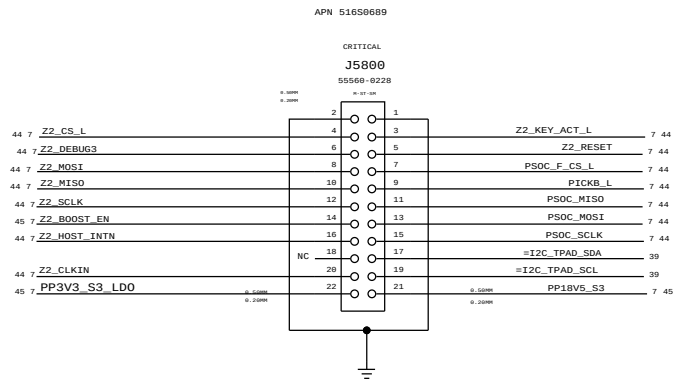
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BOOSTER +18.5VDC FOR SENSORS

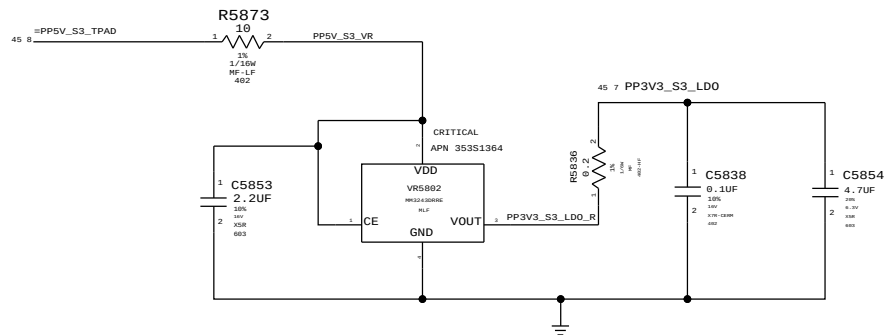
- BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
 - DROOP LINE REGULATION
 - RIPPLE TO MEET ERS
 - 100-300 KHZ CLEAN SPECTRUM
 - STARTUP TIME LESS THAN 2MS
 - RS812, RS813, C5818 MODIFIED

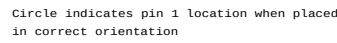


IPD FLEX CONNECTOR

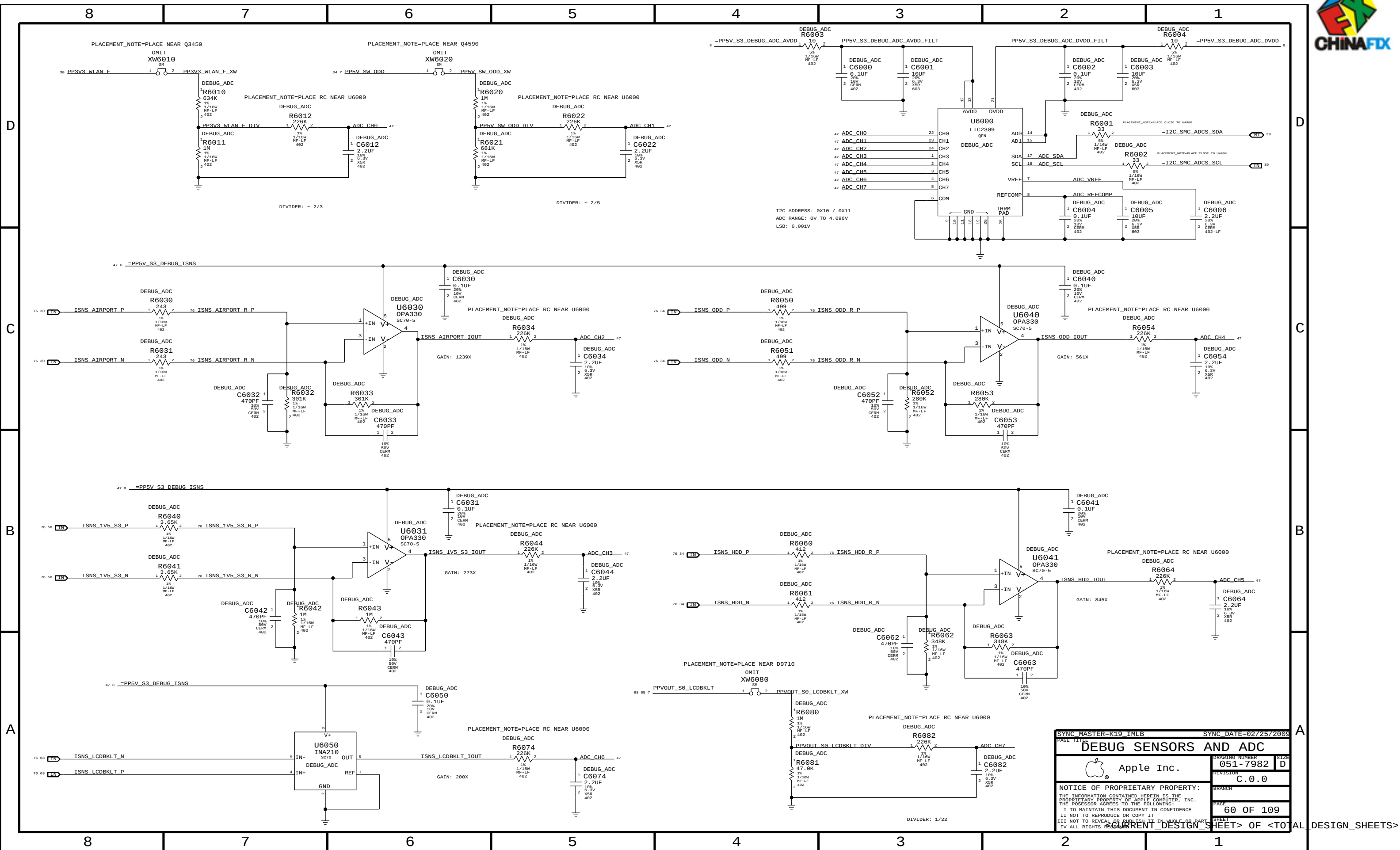


3V3 LDO FOR IPD

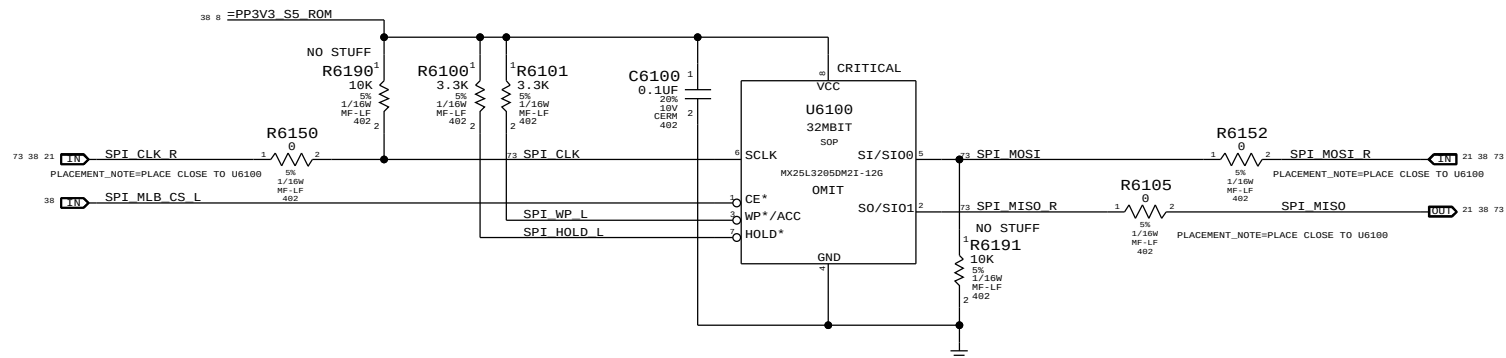




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DEBUG SENSORS AND ADC		051-7982	
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MCP79 SPI Frequency Select		
Frequency	SPI_MOSI	SPI_CLK
31 MHz	0	0
42 MHz	0	1
25 MHz	1	0
1 MHz	1	1

25MHz is selected with R5190 and R5191
Any of the 4 frequencies can be selected
with R6190, R6191, R5190 and R5191

SYNC MASTER=K24 MLB

SYNC DATE=02/16/2009

SPI ROM

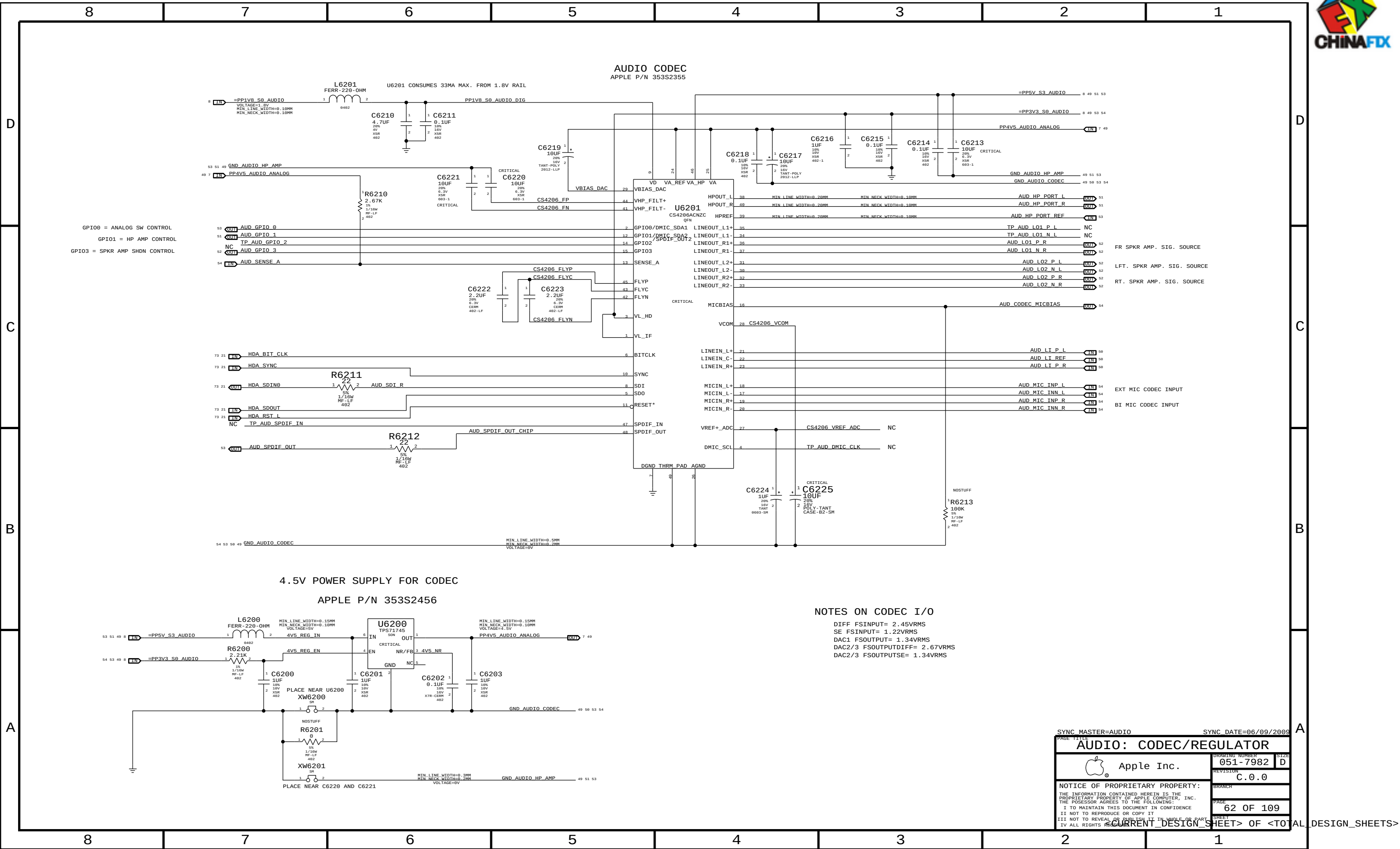
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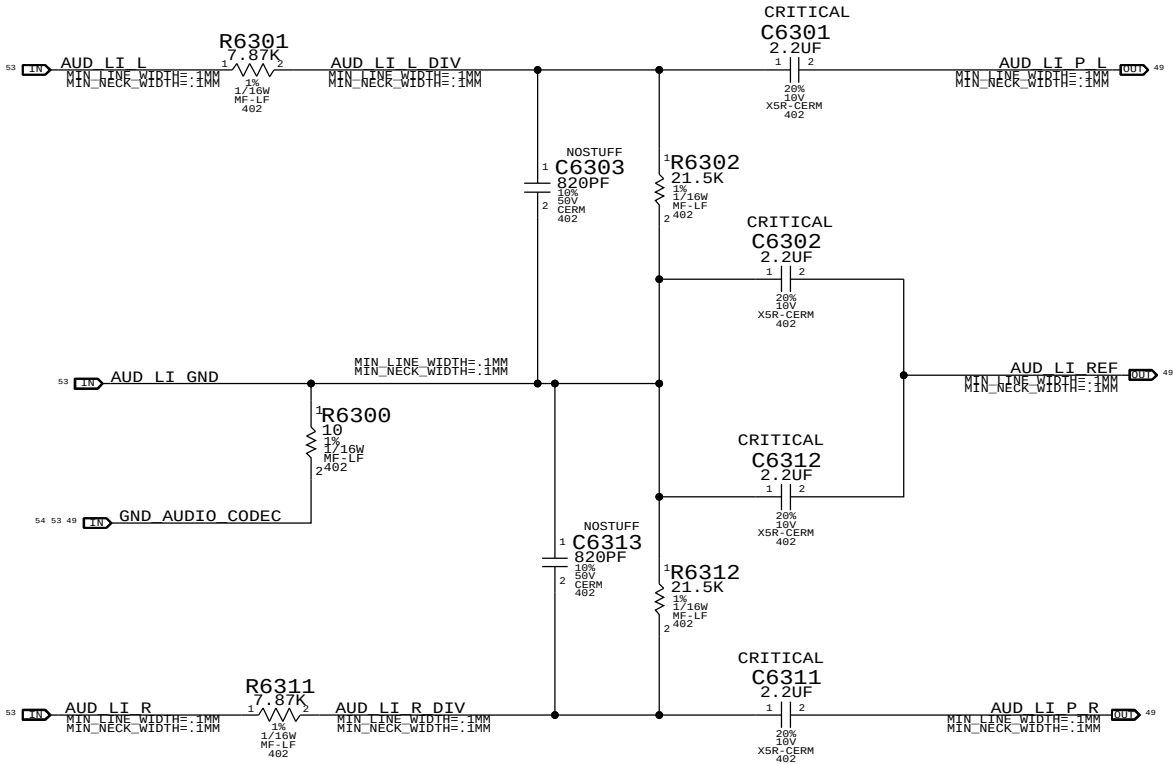
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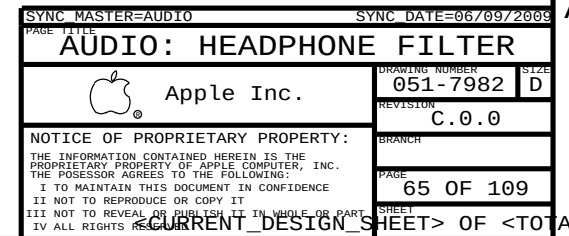
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LINE INPUT VOLTAGE DIVIDER

CODEC RIN = 20K OHMS
NET RIN = 10.36K OHMS (INCLUDING PULL-DOWNS AT ANALOG SWITCH COM PINS)
FC_HP = 3.6 HZ
FC_LP = 43KHZ
VIN = 2VRMS, CODEC VIN = 1.14 VRMS





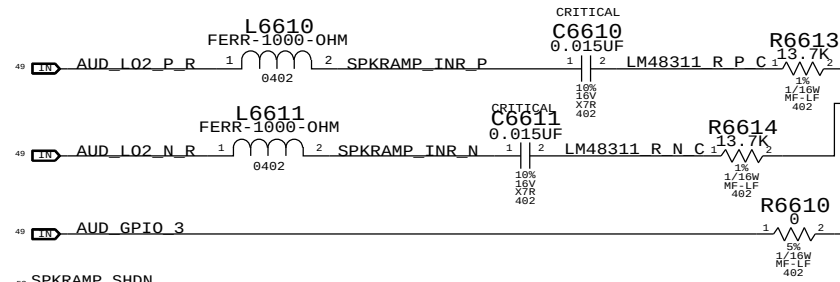
DYNAMIC (SUB) AND PIEZO (SATELLITE) SPKR AMPLIFIERS

SATELLITE HPF FC = 775 HZ
 SUB 80 HZ < HPF FC < 132 HZ
 SUB GAIN 6DB (2V/V)
 SAT GAIN 5.6DB (1.91V/V)

ALIAS OF PP5V_S3_REG, MIN_LINE_WIDTH=0.60MM, MIN_NECK_WIDTH=0.20MM

52 8 =PP5V_S3_AUDIO_AMP

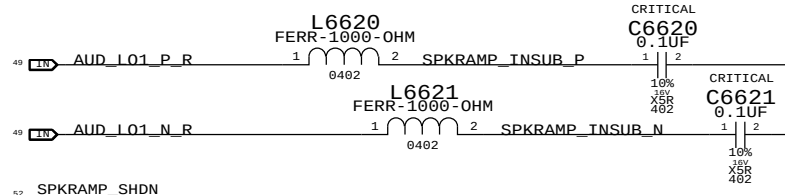
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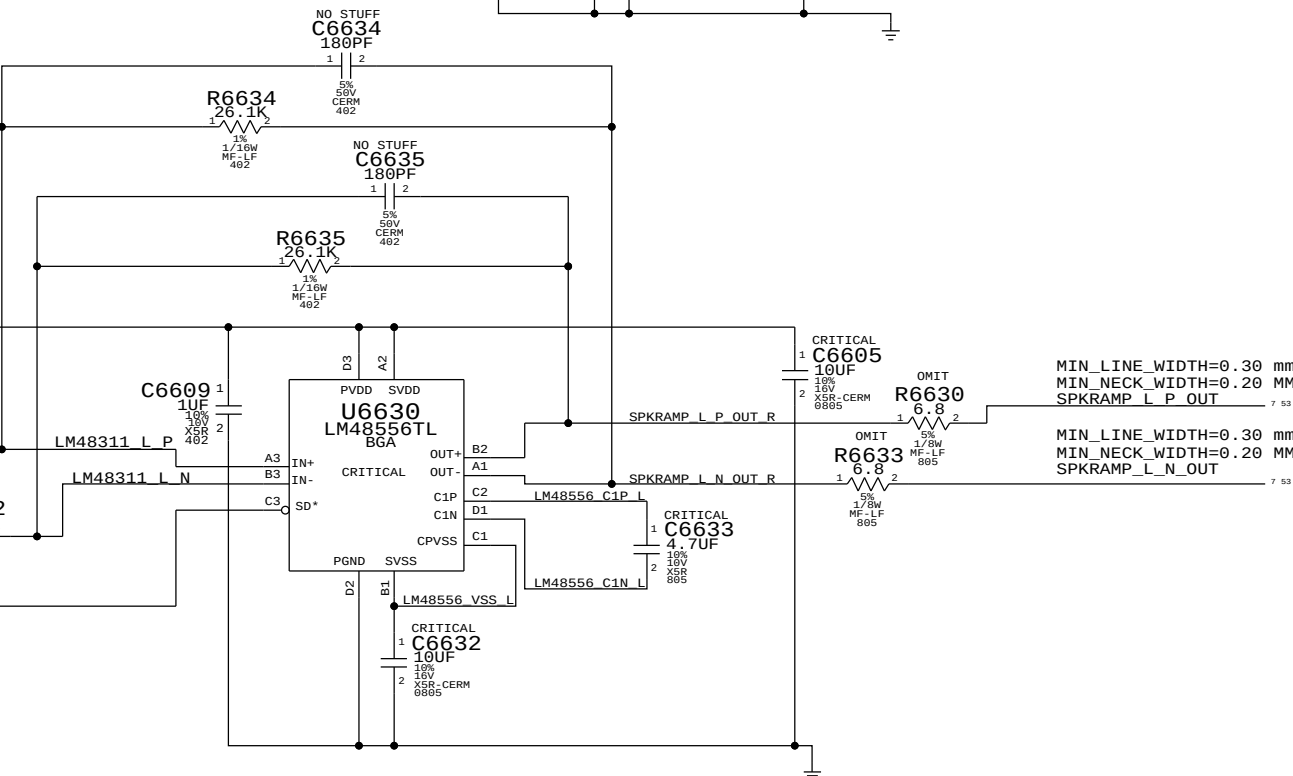
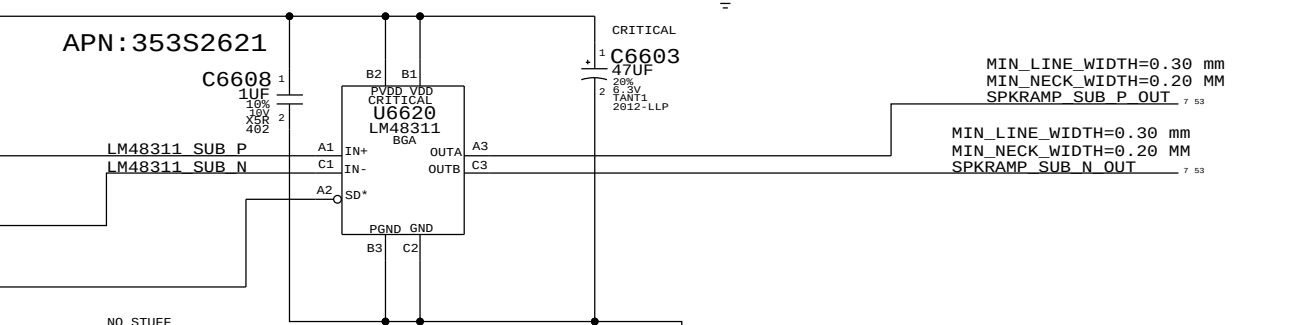
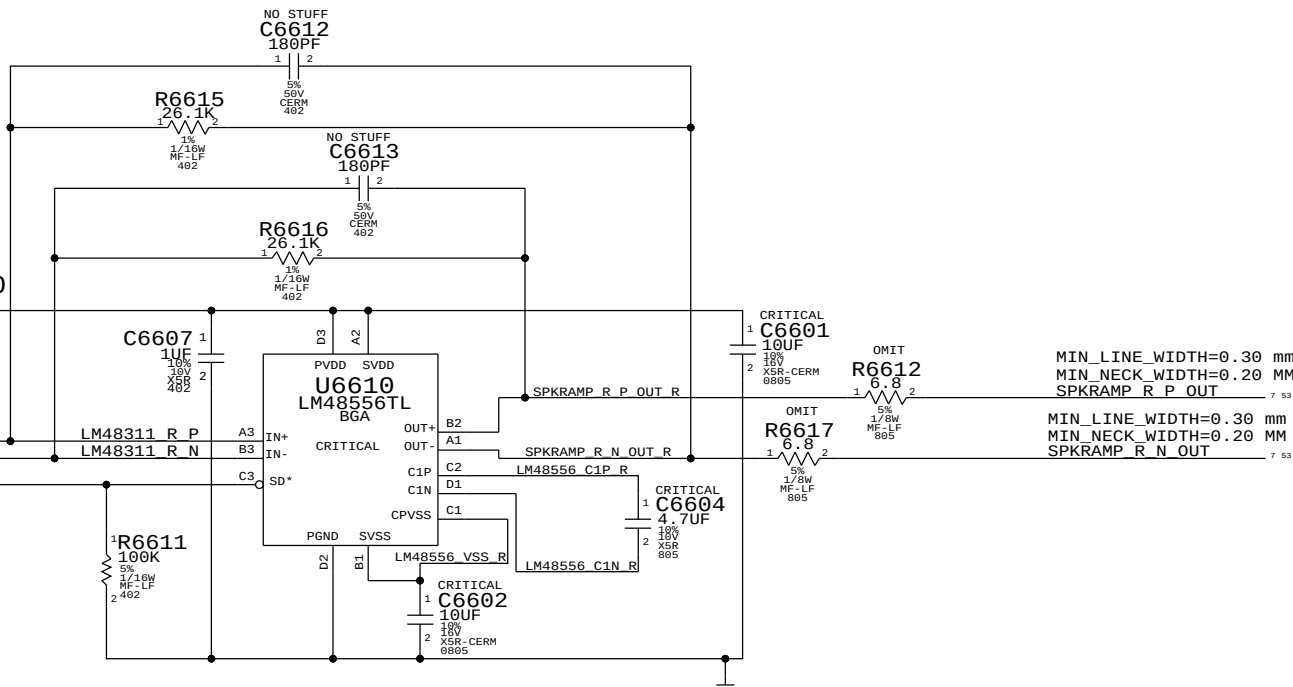
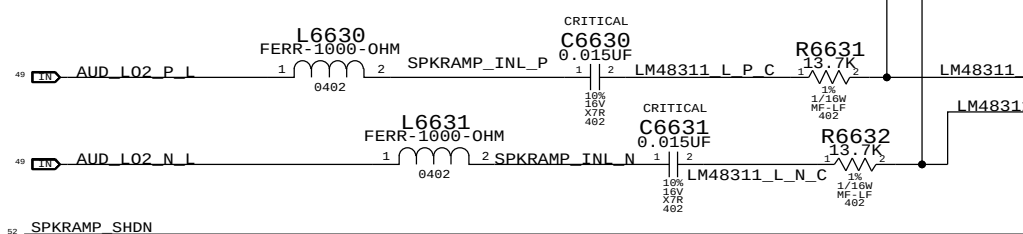
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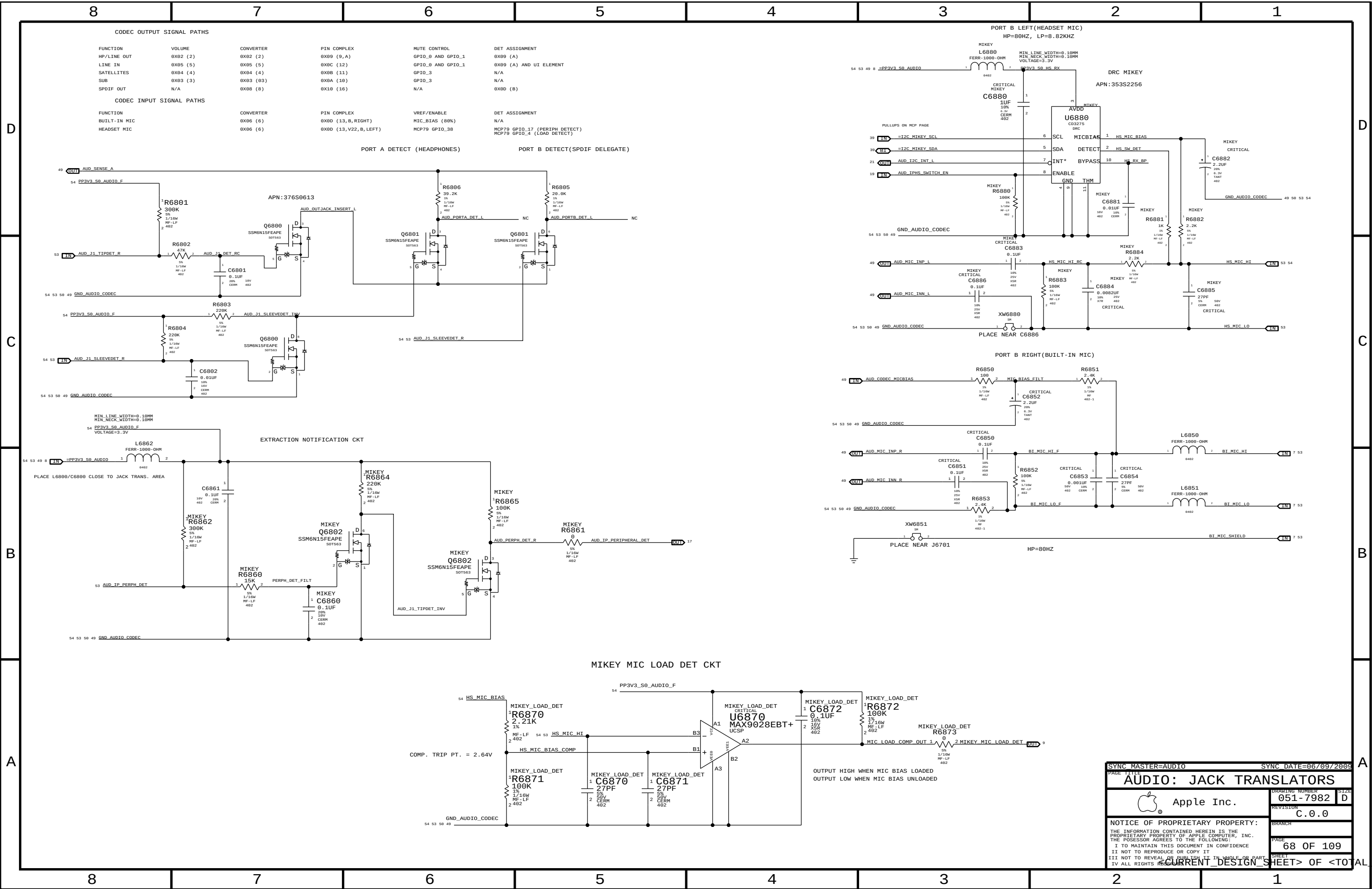
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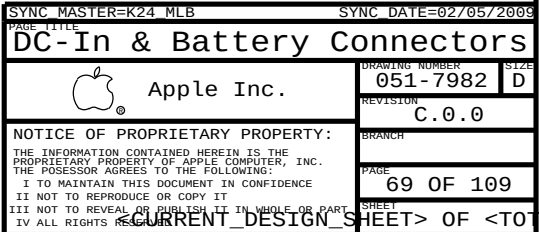
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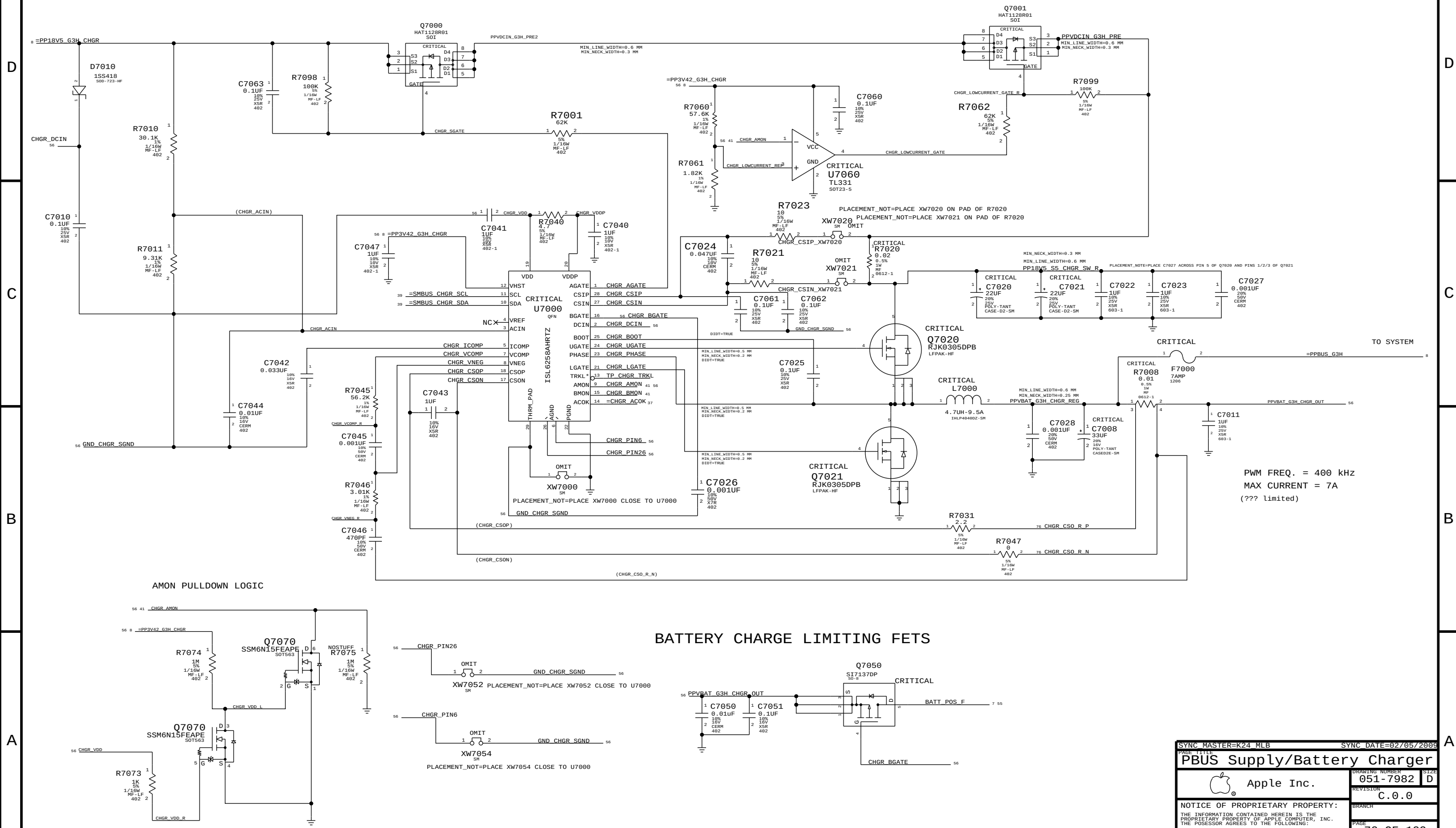
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PBUS SUPPLY / BATTERY CHARGER

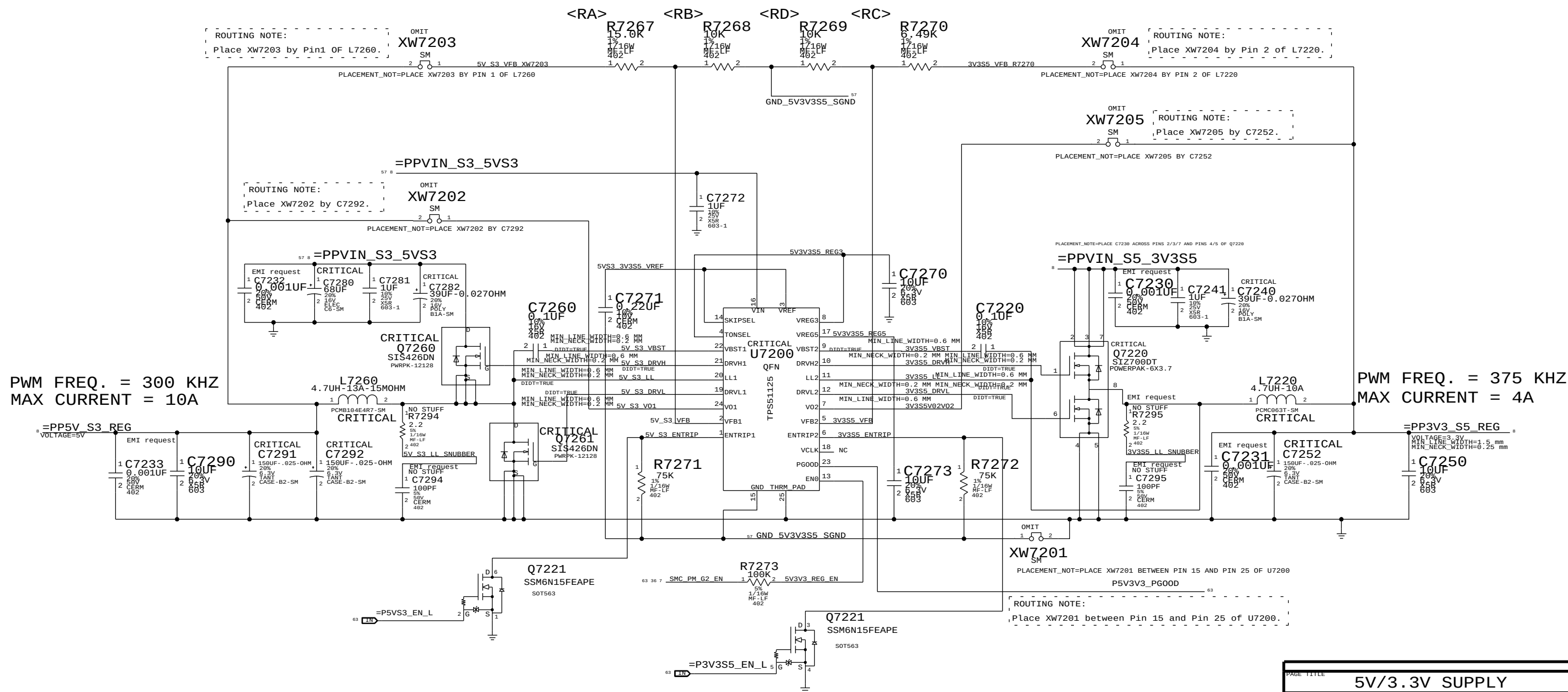


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PBUS Supply/Battery Charger			
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5V S3/3.3V S5 POWER SUPPLY

$$V_{OUT} = (2 * RA / RB) + 2$$

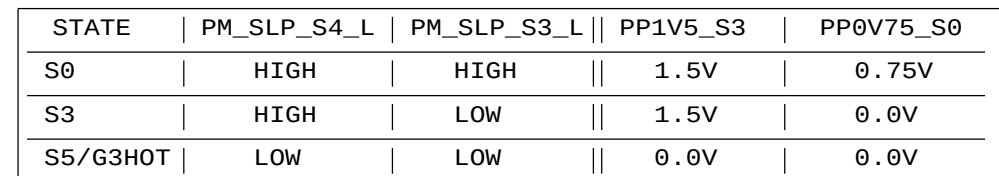
$$V_{OUT} = (2 * RC / RD) + 2$$



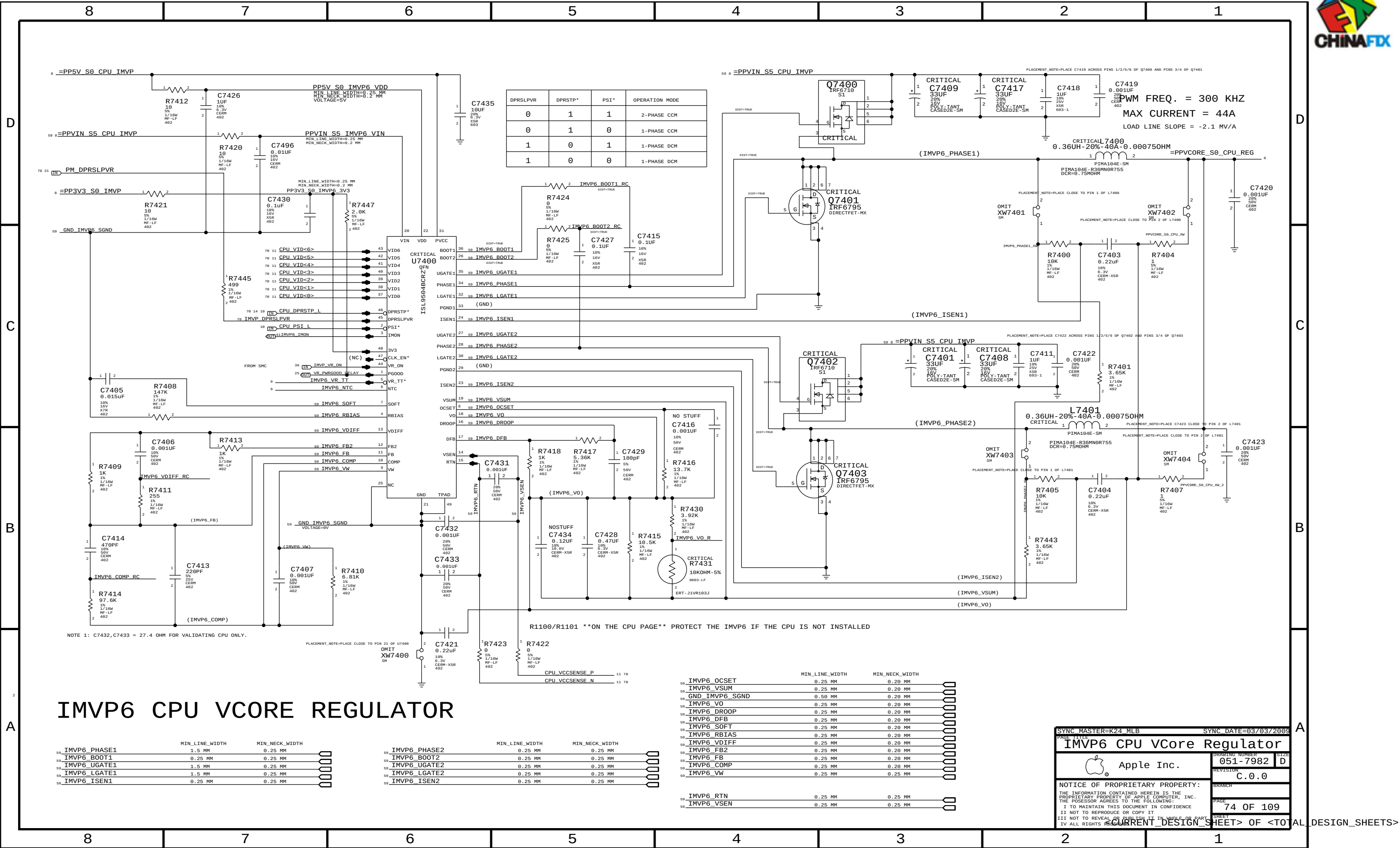
PWM FREQ. = 300 KHZ
MAX CURRENT = 10A

PWM FREQ. = 375 KHZ
MAX CURRENT = 4A

PAGE TITLE			
5V/3.3V SUPPLY			
 Apple Inc.	DRAWING NUMBER	051-7982	SIZE
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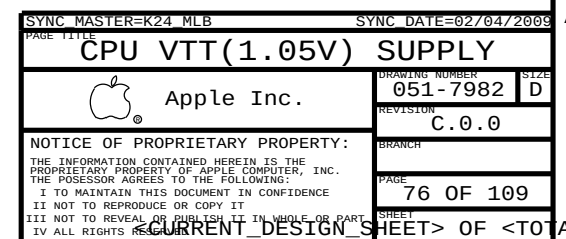
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1.5V/0.75V DDR3 SUPPLY		051-7982		D	
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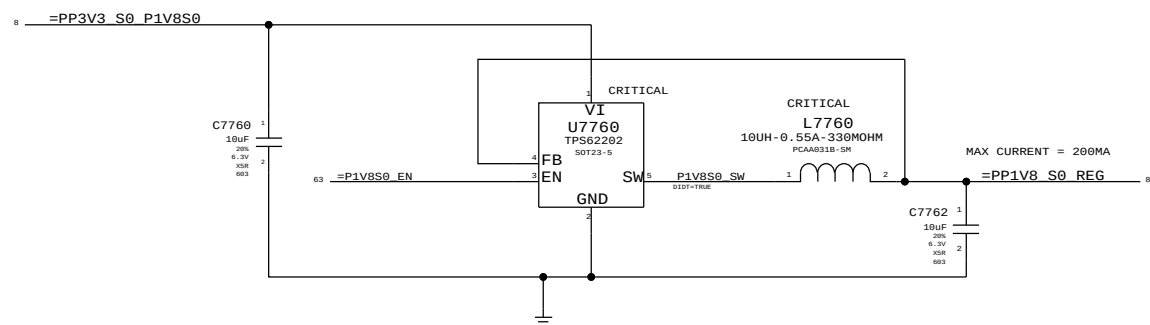


VID<2:0>	MCP TARGET
000	+1.05V
001	+1.00V
010	+0.95V
011	+0.90V
100	+0.85V
101	+0.80V
110	+0.75V

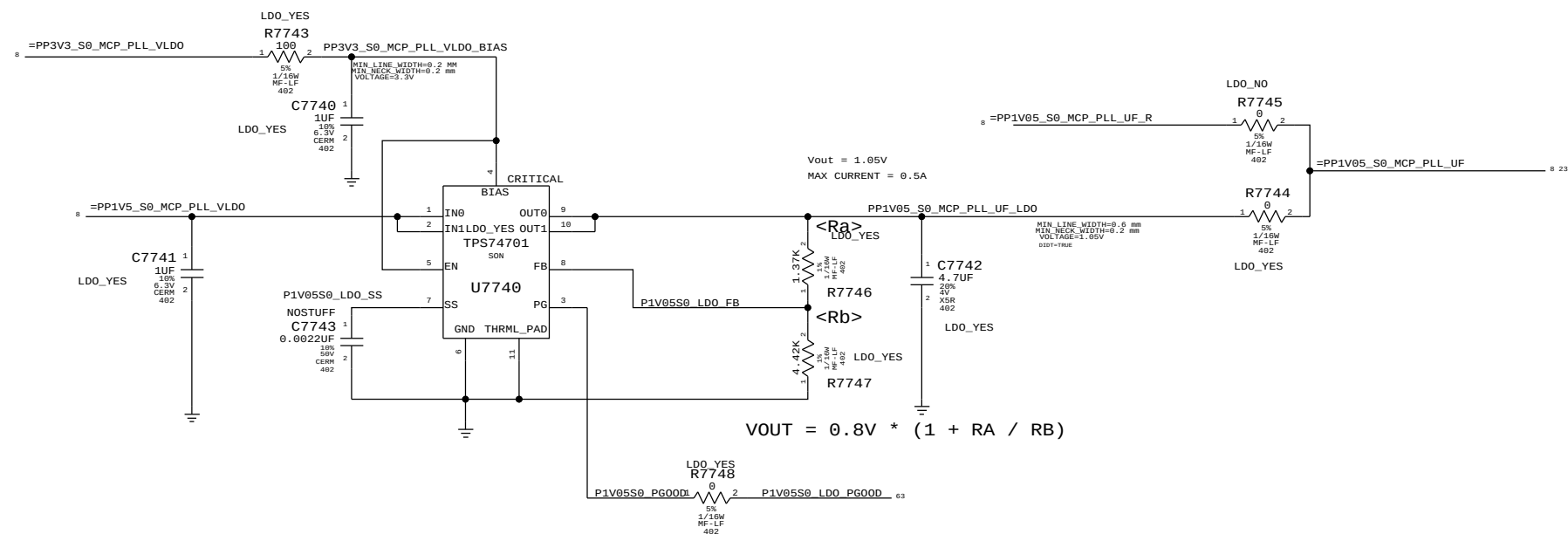
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MCP CORE REGULATOR		D	
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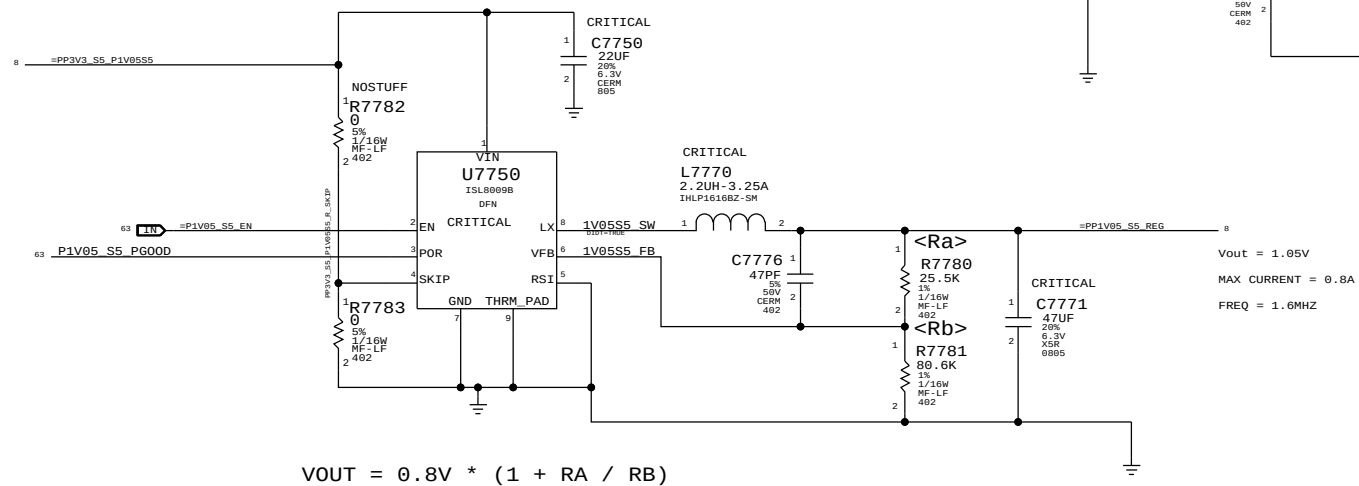
1.8V S0 SWITCHER

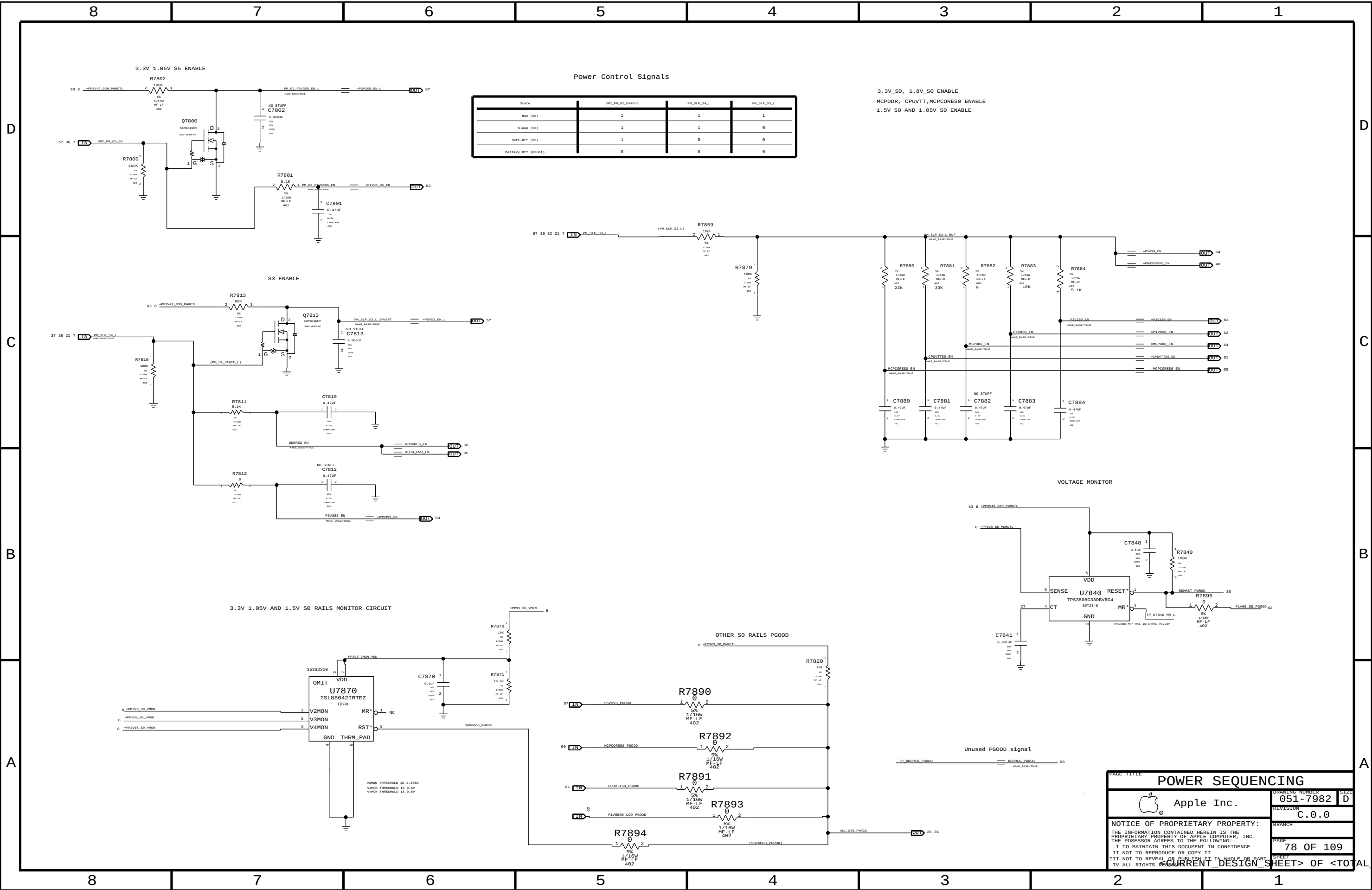


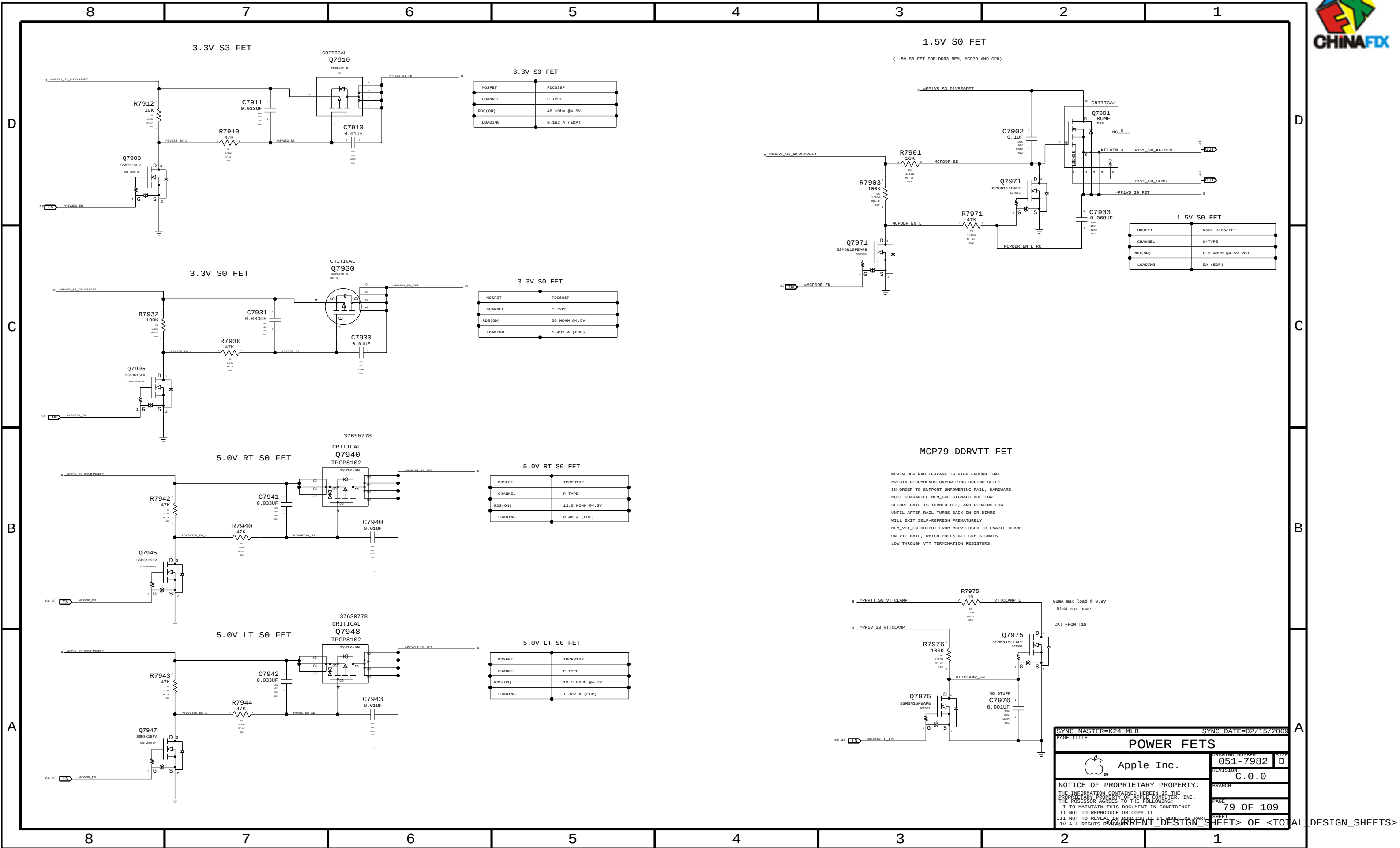
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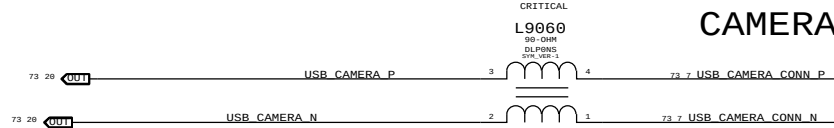
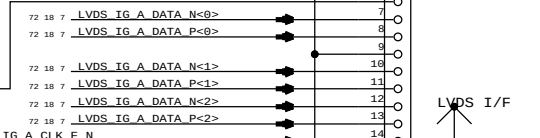
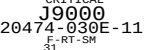
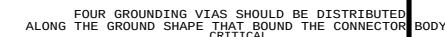
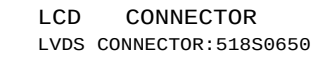
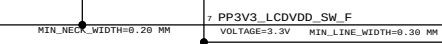
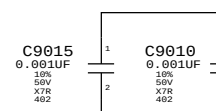
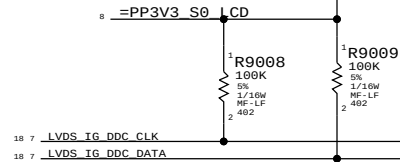
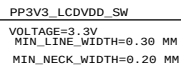
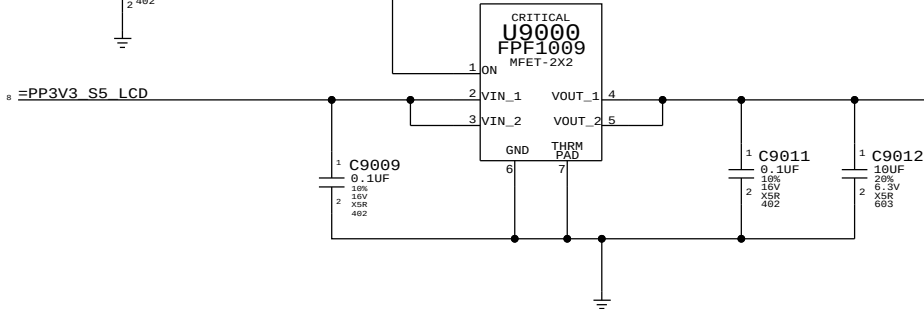
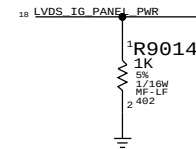


MCP 1.05V S5 (AUXC) SUPPLY







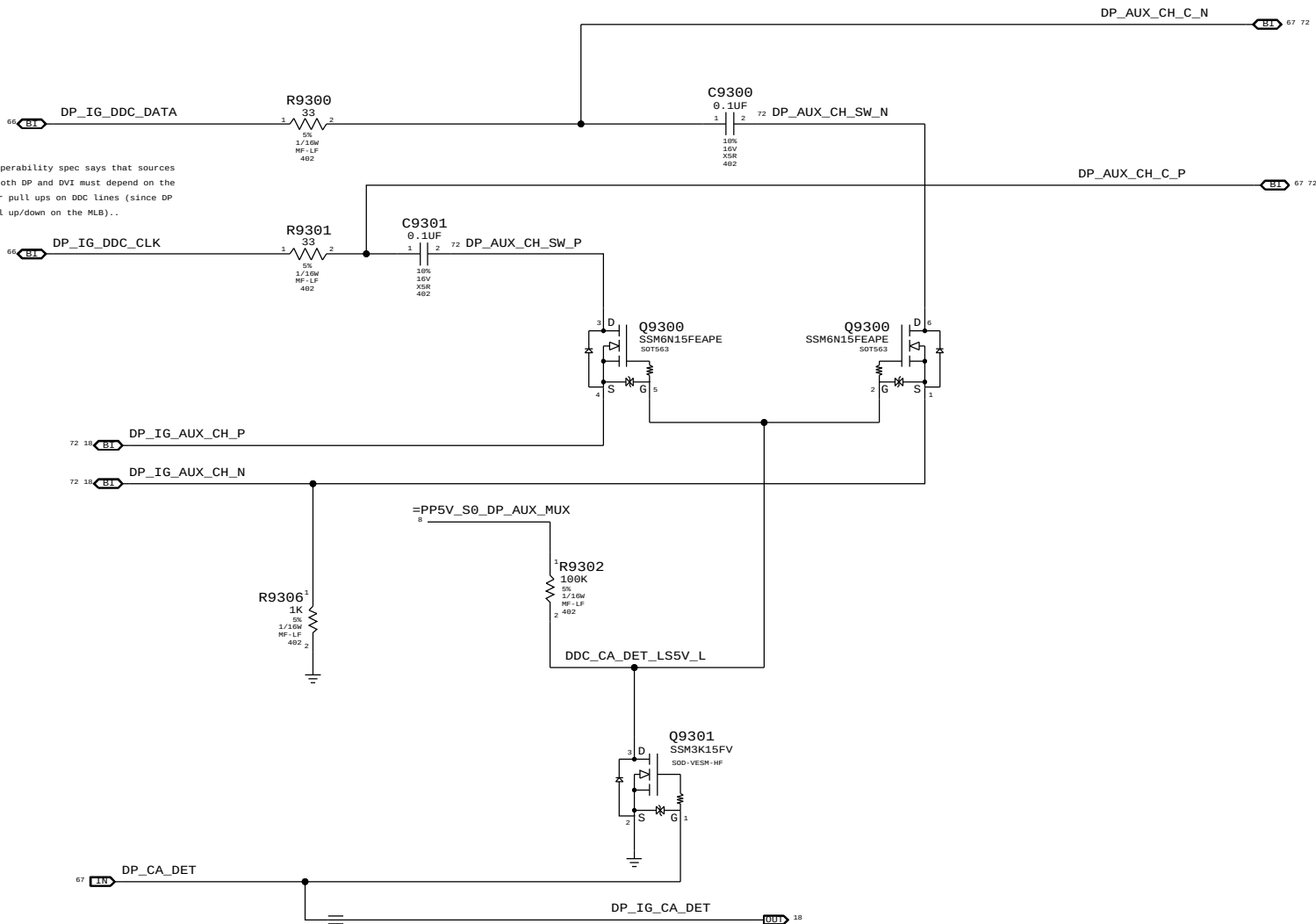


SYNCR MASTER=K24 MLB		SYNCR DATE=02/15/2009	
PAGE TITLE			
LVDS CONNECTOR			
 Apple Inc.		DRAWING NUMBER 051-7982	
		SIZE D	
		REVISION C.0.0	
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=MCP_HDMI_OOC_CLK	DP_IG_DDC_CLK	MAKE_BASE=TRUE	66
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Display Port Interoperability spec says that sources or sinks which do both DP and DVI must depend on the external adapter for pull ups on DDC lines (since DP AUX CH has 100K pull up/down on the MLB)..



SYNC MASTER=K24 MLB

SYNC DATE=04/06/2009

DISPLAYPORT SUPPORT

 Apple Inc.

DRAWING NUMBER
051-7982

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93 OF 109

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93 OF 109

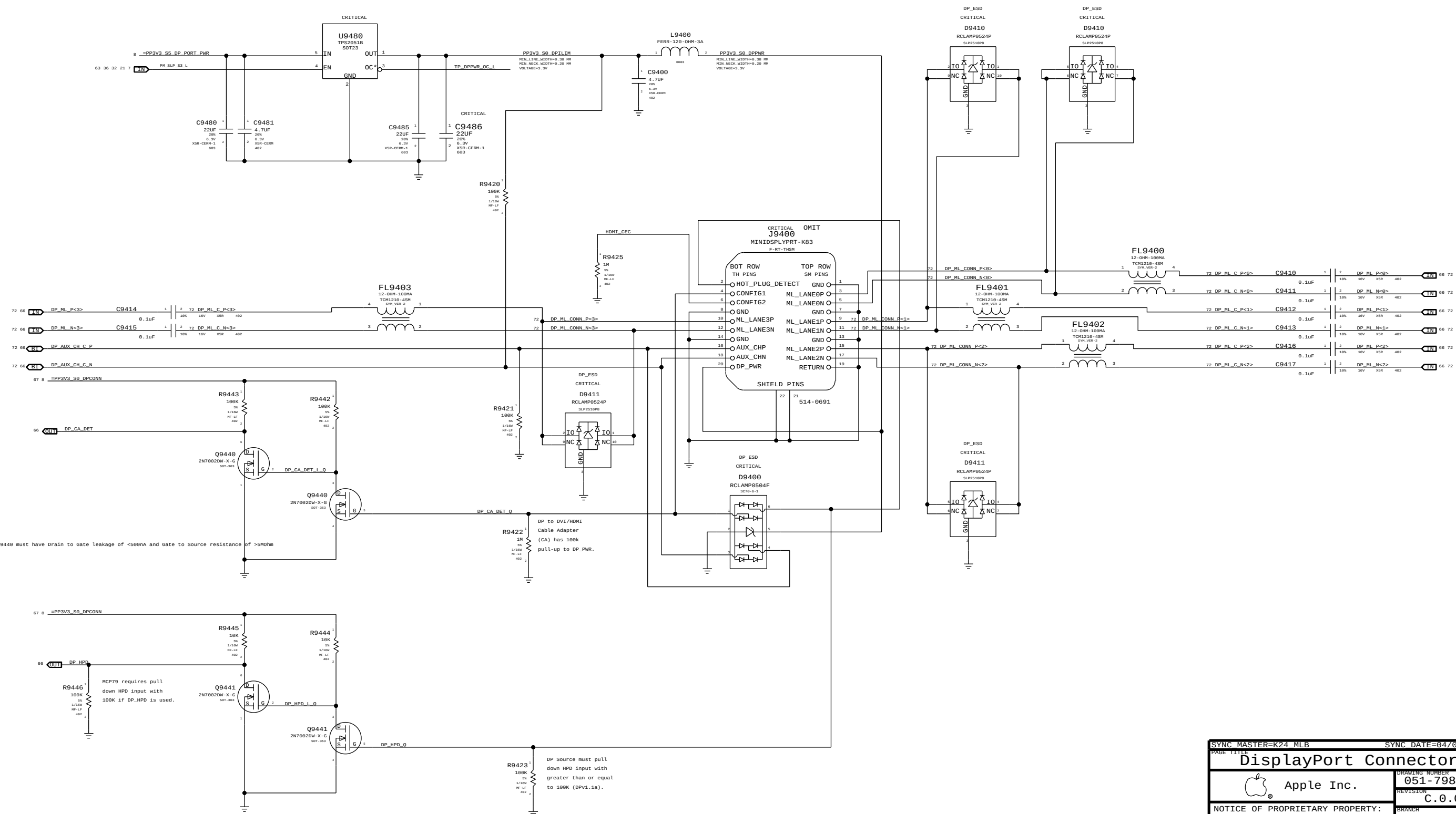
SHEET
CURRENT DESIGN SHEET

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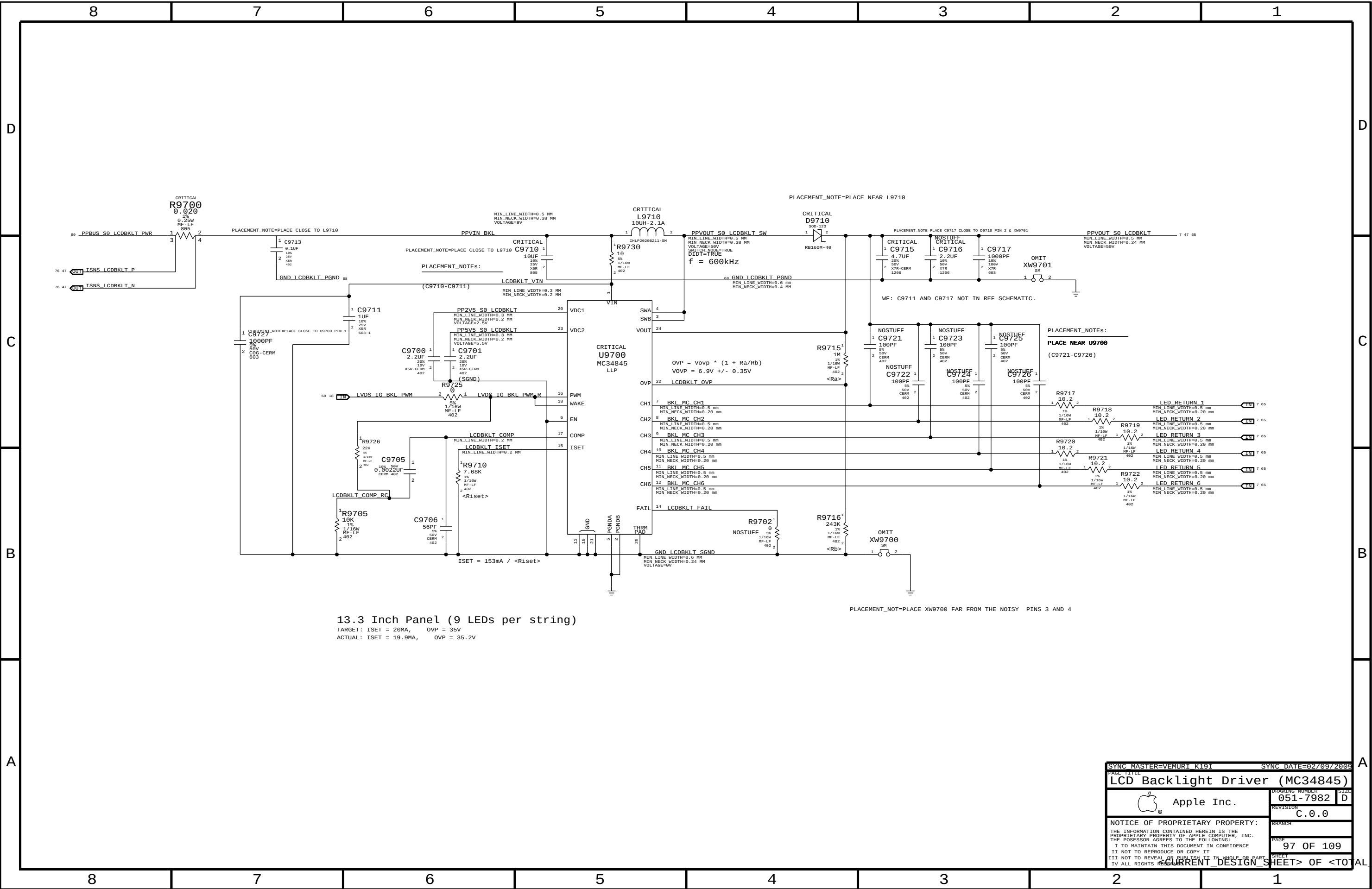
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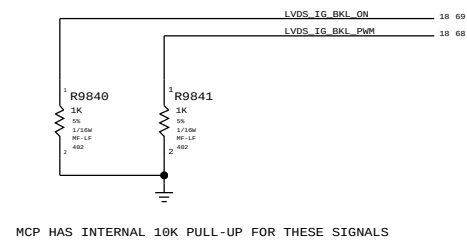
POR IS PLASTIC MINI DP CONNECTOR BUT METAL PART'S SCHEMATIC AND CAD SUMBOLS
HAVE BEEN USED BEACUSE ITS LAND PATTERN CAN ACCOMODATE BOTH TYPES

Port Power Switch



SYNC MASTER=K24 MLB		SYNC DATE=04/06/2009	
PAGE TITLE		DisplayPort Connector	
DRAWING NUMBER		051-7982	SIZE D
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PPBUS S0 LCDBKLT FET	
MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.4 A (EDP)

SYNCH MASTER=K24 MLB		SYNCH DATE=04/06/2009	
PAGE TITLE			
LCD Backlight		Support	
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CPU / FSB Net Properties

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		PHYSICAL	SPACING		
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	FSB_DATA_00000	FSB_3002	FSB_DATA	FSB_D1NV_L<0>	10 14
	FSB_D0T0	FSB_0010_500	FSB_0010	FSB_D0TB_L_P<0>	10 14
	FSB_D0T0	FSB_0010_500	FSB_0010	FSB_D0TB_L_N<0>	10 14
	FSB_DATA_00001	FSB_3002	FSB_DATA	FSB_D_L<31..16>	10 14
	FSB_DATA_00001	FSB_3002	FSB_DATA	FSB_D1NV_L<1>	10 14
	FSB_00TB1	FSB_0010_500	FSB_0010	FSB_D0TB_L_P<1>	10 14
	FSB_00TB1	FSB_0010_500	FSB_0010	FSB_D0TB_L_N<1>	10 14
	FSB_DATA_00007	FSB_3002	FSB_DATA	FSB_D_L<47..32>	10 14
	FSB_DATA_00007	FSB_3002	FSB_DATA	FSB_D1NV_L<2>	10 14
	FSB_D0T0	FSB_0010_500	FSB_0010	FSB_D0TB_L_P<2>	10 14
	FSB_D0T0	FSB_0010_500	FSB_0010	FSB_D0TB_L_N<2>	10 14
	FSB_DATA_00001	FSB_3002	FSB_DATA	FSB_D_L<63..48>	10 14
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	FSB_AD0P_00000	FSB_3002	FSB_AD0P	FSB_REQ_L<5..0>	10 14
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	FSB_1X	FSB_3002	FSB_1X	FSB_BNR_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_BPRT_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_OBSV_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_DEFER_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_ORDY_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_HIT_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_HITM_L	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_LOCK_L	10 14
	FSB_CPURST_L	FSB_3002	FSB_1X	FSB_CPURST_L	10 13 14
	FSB_1X	FSB_3002	FSB_1X	FSB_RS_L<2..0>	10 14
	FSB_1X	FSB_3002	FSB_1X	FSB_TRDY_L	10 14
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	CPU_B011	CPU_3002	CPU_A010	CPU_BSEL<2..0>	9 10
	CPU_F000_P	CPU_3002	CPU_B011	CPU_FERR_L	10 14
	CPU_A010C	CPU_3002	CPU_A010	CPU_I0GNE_L	10 14
	CPU_INIT_L	CPU_3002	CPU_A010	CPU_INIT_L	10 14
	CPU_A010C_0	CPU_3002	CPU_A010	CPU_INTR	10 14
	CPU_A010C_0	CPU_3002	CPU_A010	CPU_NMI	10 14
	CPU_PROCHOT_L	CPU_3002	CPU_A010	CPU_PROCHOT_L	10 14 37
	CPU_P0000	CPU_3002	CPU_A010	CPU_PWRGD	10 13 14
	CPU_A010C	CPU_3002	CPU_A010	CPU_SMI_L	10 14
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	CPU_DRPSTP_L	CPU_3002	CPU_A010	CPU_DRPSTP_L	10 14 59
	CPU_A010C	CPU_3002	CPU_A010	FSB_DPWR_L	10 14
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	MCP_CPU_COMP	MCP_3002	MCP_F000_COMP	MCP_BCLK_VML_COMP_GND	14
	MCP_CPU_COMP	MCP_3002	MCP_F000_COMP_VCC	MCP_CPU_COMP_VCC	14
	MCP_CPU_COMP	MCP_3002	MCP_F000_COMP	MCP_CPU_COMP_GND	14
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	FSB_CLK_CPU0	CLK_F000_1000	CLK_F000	FSB_CLK_CPU_N	10 14
	FSB_CLK_I1P	CLK_F000_1000	CLK_F000	FSB_CLK_I1P_P	13 14
	FSB_CLK_I1P	CLK_F000_1000	CLK_F000	FSB_CLK_I1P_N	13 14
	FSB_CLK_MCP	CLK_F000_1000	CLK_F000	FSB_CLK_MCP_P	14
	FSB_CLK_MCP	CLK_F000_1000	CLK_F000	FSB_CLK_MCP_N	14
	CPU_IERR_L	CPU_3002		CPU_IERR_L	10
	PM DPRSLPVR	CPU_3002	CPU_A010	PM DPRSLPVR	21 59
(see above)		CPU_3002	CPU_A010	IMVP DPRSLPVR	59
	CPU_GTLREF	CPU_3002	CPU_GTLREF	CPU_GTLREF	10 26
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	CPU_COMP	CPU_3002	CPU_COMP	CPU_COMP<1>	10
	CPU_COMP	CPU_27045	CPU_COMP	CPU_COMP<0>	10
	XDP_TDI	CPU_3002	CPU_I1P	XDP_TDI	10 13
	XDP_TDO	CPU_3002	CPU_I1P	XDP_TDO	10 13
	XDP_TMS	CPU_3002	CPU_I1P	XDP_TMS	10 13
	XDP_TCK	CPU_3002	CPU_I1P	XDP_TCK	10 13
	XDP_TRST_L	CPU_3002	CPU_I1P	XDP_TRST_L	10 13
	XDP_BPM_L	CPU_3002	CPU_I1P	XDP_BPM_L<4..0>	10 13
	XDP_BPM_L	CPU_I1P	CPU_I1P	XDP_BPM_L<5>	10 13
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	CPU_VID<6..0>	CPU_3002	CPU_B011	CPU_VID<6..0>	11 59
		CPU_3002	CPU_B011	IMVP_VID<6..0>	
	CPU_VCCSENSE	CPU_27045	CPU_VCCSENSE	CPU_VCCSENSE_P	11 59
	CPU_VCCSENSE	CPU_27045	CPU_VCCSENSE	CPU_VCCSENSE_N	11 59
(CPU_VCCSENSE)		CPU_27045	CPU_VCCSENSE	IMVP_VSEN_P	
(CPU_VCCSENSE)		CPU_27045	CPU_VCCSENSE	IMVP_VSEN_N	

SYNCH MASTER=K24 MLB		SYNCH DATE=04/06/2009	
PAGE TITLE			
CPU/FSB Constraints			
 Apple Inc.	DRAWING NUMBER		SIZE
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Memory Bus Constraints																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
<table><tr><th>PHYSICAL_RULE_SET</th><th>LAYER</th><th>ALLOW_ROUTE_ON_LAYER?</th><th>MINIMUM LINE WIDTH</th><th>MINIMUM NECK WIDTH</th><th>MAXIMUM NECK LENGTH</th><th>DIFFPAIR PRIMARY GAP</th><th>DIFFPAIR NECK GAP</th></tr><tr><td>MEM_4B5</td><td>*</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>MEM_4B5_VDD</td><td>*</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=4B_OHR_SE</td><td>=STANDARD</td><td>=STANDARD</td></tr><tr><td>MEM_7B0</td><td>*</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td></tr><tr><td>MEM_7B0_VDD</td><td>*</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td><td>=7B_OHR_DIFF</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	MEM_4B5	*	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=STANDARD	=STANDARD	MEM_4B5_VDD	*	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=STANDARD	=STANDARD	MEM_7B0	*	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	MEM_7B0_VDD	*	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
MEM_4B5	*	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
MEM_4B5_VDD	*	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=4B_OHR_SE	=STANDARD	=STANDARD																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
MEM_7B0	*	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF	=7B_OHR_DIFF																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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DDR2: DQ signals should be matched within 20 ps of associated DQS pair. DQS intra-pair matching should be within 1 ps, no inter-pair matching requirement. All DQS pairs should be matched within 100 ps of clocks. CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 140 ps. A/BA/cmd signals should be matched within 75 ps, no CLK matching requirement. All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate). DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
DDR3: DQ signals should be matched within 5 ps of associated DQS pair. DQS intra-pair matching should be within 1 ps, inter-pair matching should be within 180 ps No DQS to clock matching requirement. CLK intra-pair matching should be within 1 ps, inter-pair matching should be within 2 ps. A/BA/cmd signals should be matched within 5 ps of CLK pairs. All memory signals maximum length is 1.005 ps. CLK minimum length is 594 ps (lengths include substrate). DQ/A/BA/cmd signal spacing is 3x dielectric, DQS/CLK is 4x dielectric.																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
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colspan="2">NET_TYPE</th></tr><tr><th>PHYSICAL</th><th>SPACING</th></tr><tr><td>MEM_A_CLK</td><td>MEM_200_VDD</td><td>MEM_CLK</td><td>MEM_A_CLK_P<5..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CLK</td><td>MEM_200_VDD</td><td>MEM_CLK</td><td>MEM_A_CLK_N<5..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CTRL</td><td>MEM_4B5_VDD</td><td>MEM_CTRL</td><td>MEM_A_CKE<3..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CTRL</td><td>MEM_4B5_VDD</td><td>MEM_CTRL</td><td>MEM_A_CS_L<3..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CTRL</td><td>MEM_4B5_VDD</td><td>MEM_CTRL</td><td>MEM_A_ODT<3..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CMD</td><td>MEM_4B5_VDD</td><td>MEM_CMD</td><td>MEM_A_A<14..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CMD</td><td>MEM_4B5_VDD</td><td>MEM_CMD</td><td>MEM_A_BA<2..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_CMD</td><td>MEM_4B5_VDD</td><td>MEM_CMD</td><td>MEM_A_RAS_L</td><td>15</td><td>27</td></tr><tr><td>MEM_A_CMD</td><td>MEM_4B5_VDD</td><td>MEM_CMD</td><td>MEM_A_CAS_L</td><td>15</td><td>27</td></tr><tr><td>MEM_A_CMD</td><td>MEM_4B5_VDD</td><td>MEM_CMD</td><td>MEM_A_WE_L</td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<7..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<15..0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<23..16></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<31..24></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<39..32></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<47..48></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<55..48></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQ_BUFFER</td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DQ<63..56></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<0></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<1></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<1></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<2></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<2></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<3></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<3></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<4></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<4></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<5></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<5></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<6></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<6></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DM<7></td><td>MEM_4B5</td><td>MEM_DATA</td><td>MEM_A_DM<7></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<0></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<1></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<1></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<2></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<2></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<3></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<3></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<4></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<4></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<5></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<5></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<6></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<6></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_P<7></td><td>15</td><td>27</td></tr><tr><td>MEM_A_DQS</td><td>MEM_200</td><td>MEM_DQS</td><td>MEM_A_DQS_N<7></td><td>15</td><td>27</td></tr><tr><td>MEM_B_CLK</td><td>MEM_200_VDD</td><td>MEM_CLK</td><td>MEM_B_CLK_P<5..0></td><td>15</td><td>28</td></tr><tr><td>MEM_B_CLK</td><td>MEM_200_VDD</td><td>MEM_CLK</td><td>MEM_B_CLK_N<5..0></td><td>15</td><td>28</td></tr><tr><td>MEM_B_CTRL</td><td>MEM_4B5_VDD</td><td>MEM_CTRL</td><td>MEM_B_CKE<3..0></td><td>15</td><td>28</td></tr><tr><td>MEM_B_CTRL</td><td>MEM_4B5_VDD</td><td>MEM_CTRL</td><td>MEM_B_CS_L<3..0></td><td>15</td><td>28</td></tr>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EM_A_DQ<23..16>	15	27	MEM_A_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_A_DQ<31..24>	15	27	MEM_A_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_A_DQ<39..32>	15	27	MEM_A_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_A_DQ<47..48>	15	27	MEM_A_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_A_DQ<55..48>	15	27	MEM_A_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_A_DQ<63..56>	15	27	MEM_A_DM<0>	MEM_4B5	MEM_DATA	MEM_A_DM<0>	15	27	MEM_A_DM<1>	MEM_4B5	MEM_DATA	MEM_A_DM<1>	15	27	MEM_A_DM<2>	MEM_4B5	MEM_DATA	MEM_A_DM<2>	15	27	MEM_A_DM<3>	MEM_4B5	MEM_DATA	MEM_A_DM<3>	15	27	MEM_A_DM<4>	MEM_4B5	MEM_DATA	MEM_A_DM<4>	15	27	MEM_A_DM<5>	MEM_4B5	MEM_DATA	MEM_A_DM<5>	15	27	MEM_A_DM<6>	MEM_4B5	MEM_DATA	MEM_A_DM<6>	15	27	MEM_A_DM<7>	MEM_4B5	MEM_DATA	MEM_A_DM<7>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<0>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<0>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<1>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<1>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<2>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<2>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<3>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<3>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<4>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<4>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<5>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<5>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<6>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<6>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_P<7>	15	27	MEM_A_DQS	MEM_200	MEM_DQS	MEM_A_DQS_N<7>	15	27	MEM_B_CLK	MEM_200_VDD	MEM_CLK	MEM_B_CLK_P<5..0>	15	28	MEM_B_CLK	MEM_200_VDD	MEM_CLK	MEM_B_CLK_N<5..0>	15	28	MEM_B_CTRL	MEM_4B5_VDD	MEM_CTRL	MEM_B_CKE<3..0>	15	28	MEM_B_CTRL	MEM_4B5_VDD	MEM_CTRL	MEM_B_CS_L<3..0>	15	28	MEM_B_CTRL	MEM_4B5_VDD	MEM_CTRL	MEM_B_ODT<3..0>	15	28	MEM_B_CMD	MEM_4B5_VDD	MEM_CMD	MEM_B_A<14..0>	15	28	MEM_B_CMD	MEM_4B5_VDD	MEM_CMD	MEM_B_BA<2..0>	15	28	MEM_B_CMD	MEM_4B5_VDD	MEM_CMD	MEM_B_RAS_L	15	28	MEM_B_CMD	MEM_4B5_VDD	MEM_CMD	MEM_B_CAS_L	15	28	MEM_B_CMD	MEM_4B5_VDD	MEM_CMD	MEM_B_WE_L	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<7..0>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<15..0>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<23..16>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<31..24>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<39..32>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<47..48>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<55..48>	15	28	MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<63..56>	15	28	MEM_B_DM<0>	MEM_4B5	MEM_DATA	MEM_B_DM<0>	15	28	MEM_B_DM<1>	MEM_4B5	MEM_DATA	MEM_B_DM<1>	15	28	MEM_B_DM<2>	MEM_4B5	MEM_DATA	MEM_B_DM<2>	15	28	MEM_B_DM<3>	MEM_4B5	MEM_DATA	MEM_B_DM<3>	15	28	MEM_B_DM<4>	MEM_4B5	MEM_DATA	MEM_B_DM<4>	15	28	MEM_B_DM<5>	MEM_4B5	MEM_DATA	MEM_B_DM<5>	15	28	MEM_B_DM<6>	MEM_4B5	MEM_DATA	MEM_B_DM<6>	15	28	MEM_B_DM<7>	MEM_4B5	MEM_DATA	MEM_B_DM<7>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<0>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<0>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<1>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<1>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<2>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<2>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<3>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<3>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<4>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<4>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<5>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<5>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<6>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<6>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<7>	15	28	MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<7>	15	28	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_VDD	16		MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_GND	16	
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MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<39..32>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<47..48>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<55..48>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQ_BUFFER	MEM_4B5	MEM_DATA	MEM_B_DQ<63..56>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<0>	MEM_4B5	MEM_DATA	MEM_B_DM<0>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<1>	MEM_4B5	MEM_DATA	MEM_B_DM<1>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<2>	MEM_4B5	MEM_DATA	MEM_B_DM<2>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<3>	MEM_4B5	MEM_DATA	MEM_B_DM<3>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<4>	MEM_4B5	MEM_DATA	MEM_B_DM<4>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<5>	MEM_4B5	MEM_DATA	MEM_B_DM<5>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<6>	MEM_4B5	MEM_DATA	MEM_B_DM<6>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DM<7>	MEM_4B5	MEM_DATA	MEM_B_DM<7>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<0>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<0>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<1>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<1>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<2>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<3>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<3>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<4>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<4>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<5>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<5>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<6>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<6>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_P<7>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MEM_B_DQS	MEM_200	MEM_DQS	MEM_B_DQS_N<7>	15	28																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_VDD	16																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP	MCP_MEM_COMP_GND	16																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												

CHINAFTX

Memory Constraints

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D	PCI Bus Constraints								D						
	PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP							
	PCI_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD							
	CLK_PCI_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD							
	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT											
	PCI	*	=STANDARD	?											
	CLK_PCI	*	8 MIL	?											
	SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.8.														
	LPC Bus Constraints														
	PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP							
LPC_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
CLK_LPC_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT												
LPC	*	6 MIL	?												
CLK_LPC	*	8 MIL	?												
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.9.1.															
USB 2.0 Interface Constraints															
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP								
MCP_USB_RBIA5	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD								
USB_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT								
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?								
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.10.1.															
SMBus Interface Constraints															
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP								
SMB_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT												
SMB	*	=2x_DIELECTRIC	?												
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.11.1.															
HD Audio Interface Constraints															
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP								
HDA_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT												
HDA	*	=2x_DIELECTRIC	?												
MCP_HDA_COMP	*	8 MIL	?												
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.12.1.															
SIO Signal Constraints															
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP								
CLK_SLOW_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT												
CLK_SLOW	*	8 MIL	?												
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.13.															
SPI Interface Constraints															
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP								
SPI_S55	*	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=S5_OHM_SE	=STANDARD	=STANDARD								
SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT												
SPI	*	8 MIL	?												
SOURCE: MCP79 Interface DG (DG-03328-001_v00), Section 2.14.															



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MCP RGMII (Ethernet) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MCP_M11_COMP	*	=STANDARD	7.5 MIL	7.5 MIL	=STANDARD	=STANDARD	=STANDARD
ENET_M11_S5S	*	=S5_OHR_SE	=S5_OHR_SE	=S5_OHR_SE	=S5_OHR_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MCP_BUF8_CLK	*	=3:1_SPACING	?
ENET_M11	*	12 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Sections 2.7.2 & 2.7.4

88E1116R (Ethernet PHY) Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_MD1_1000	*	=100_OHR_DIFF	=100_OHR_DIFF	=100_OHR_DIFF	=100_OHR_DIFF	=100_OHR_DIFF	=100_OHR_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_MD1	*	25 MIL	?

SOURCE: MCP73 Interface DG (DG-02974-001_v01), Section 2.7.4

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
	PHYSICAL	SPACING	
	MCP_M11_COMP	MCP_M11_COMP	MCP_M11_COMP_VDD 18
	MCP_M11_COMP	MCP_M11_COMP	MCP_M11_COMP_GND 18
	MCP_CLK25M_BUF8	ENET_M11_S5S	MCP_BUF8_CLK 18
		MCP_BUF8_CLK	RTL8211_CLK25M_CKXTAL1 21
	ENET_INTR_L	ENET_M11_S5S	ENET_INTR_L 18
	ENET_MD10	ENET_M11_S5S	ENET_MD10 18
	ENET_MDC	ENET_M11_S5S	ENET_MDC 18
	ENET_PWDOWN_L	ENET_M11_S5S	ENET_PWDOWN_L 18
		ENET_M11_S5S	ENET_M11 31
	ENET_RXCLK	ENET_M11_S5S	ENET_CLK125M_RXCLK_R 31
		ENET_M11_S5S	ENET_CLK125M_RXCLK 18
	ENET_RXD	ENET_M11_S5S	ENET_RXD_R<3..0> 31
		ENET_M11_S5S	ENET_RXD<8> 18
	ENET_RXD_STRAP	ENET_M11_S5S	ENET_RXD<3..1> 18
	ENET_RXD	ENET_M11_S5S	ENET_RX_CTRL 18
		ENET_M11_S5S	ENET_RXCTL_R 21
		ENET_M11_S5S	ENET_CLK125M_TXCLK_R 31
	ENET_TXCLK	ENET_M11_S5S	ENET_CLK125M_TXCLK 18
	ENET_TXD	ENET_M11_S5S	ENET_TXD<8> 18
	ENET_TXD	ENET_M11_S5S	ENET_TXD<3..1> 18
	ENET_TXD	ENET_M11_S5S	ENET_TX_CTRL 18
		ENET_M11_S5S	ENET_RESET_L 18
	ENET_MD1	ENET_MD1_1000	ENET_MD1_P<3..0> 31
		ENET_MD1_1000	ENET_MD1_N<3..0> 31
		ENET_MD1_1000	ENET_MD1_TRAN_P<3..0> 33
		ENET_MD1_1000	ENET_MD1_TRAN_N<3..0> 33

SYNC MASTER=K24 MLB SYNC DATE=04/06/2009

Ethernet Constraints	
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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
1T01_DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

SMC SMBus Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
	SMBUS_SMC_A_S3_SCL	SMB_S05	SMB
	SMBUS_SMC_A_S3_SDA	SMB_S05	SMB
	SMBUS_SMC_B_S9_SCL	SMB_S05	SMB
	SMBUS_SMC_B_S9_SDA	SMB_S05	SMB
	SMBUS_SMC_C_S9_SCL	SMB_S05	SMB
	SMBUS_SMC_C_S9_SDA	SMB_S05	SMB
	SMBUS_SMC_BSA_SCL	SMB_S05	SMB
	SMBUS_SMC_BSA_SDA	SMB_S05	SMB
	SMBUS_SMC_MGMT_SCL	SMB_S05	SMB
	SMBUS_SMC_MGMT_SDA	SMB_S05	SMB

SMBus Charger Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
	CHGR_CS1	1T01_DIFFPAIR	
		1T01_DIFFPAIR	
	CHGR_CS0	1T01_DIFFPAIR	
		1T01_DIFFPAIR	

SMC Constraints

Apple Inc.

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	<table><tr><td>PHYSICAL_RULE_SET</td><td>LAYER</td><td>ALLOW_ROUTE_ON_LAYER?</td><td>MINIMUM_LINE_WIDTH</td><td>MINIMUM_NECK_WIDTH</td><td>MAXIMUM_NECK_LENGTH</td><td>DIFFPAIR_PRIMARY_GAP</td><td>DIFFPAIR_NECK_GAP</td></tr><tr><td>DIFFPAIR</td><td>*</td><td>=STANDARD</td><td>=STANDARD</td><td>=STANDARD</td><td>=STANDARD</td><td>0.1 MM</td><td>0.1 MM</td></tr></table>								PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP	DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM																																																																																			
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP																																																																																																				
DIFFPAIR	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM																																																																																																				
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CHGR_C50_R_N

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CPUTHMSNS_D2_P

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CPUTHMSNS_D2_N

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CPU_THERMD_P

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ISNS_CPUVTT_P

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ISNS_1V5_S3_P

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
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